



MAXIMO ENTERPRISE ASSET MANAGEMENT AT OTORITA IBU KOTA NEGARA (OIKN)

Training Material – Work Management (part 1)

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1 Person, Labor, User, & Craft

1.1 Overview

Person represent an individual that related to EMS system. Person record is created for any individual whose name appears anywhere on any data. Person record is required when creating a user or a labor. A person data contain basic information about an individual's name, address, contact information, and other generic information.

Labor represent employees and contractors who perform work on work orders. Labor records contain information about an individual's skills and qualifications. These records are used to plan and schedule work, and to track labor costs for work orders. A labor record must have a person record associated with it.

User represent authentication to access the system. User records contain user names, passwords, and security profiles that determine which applications, options, and data a user can access. A user must have a person record. A labor record and user record can be associated to the same person record.

Crafts represents an occupation or trade, and typically the craft name reflects the type of work done by members of the trade. If pay rates differ based on expertise, multiple skill levels with different pay rates for each skill level can be assigned to the craft. For each craft user can specify standard rates, and add premium pay codes for premium pay rates.

1.2 People Application

People application is used to create, modify, view, and delete records person data. The **People** application stores personal information, employee information, and delegation configurations.

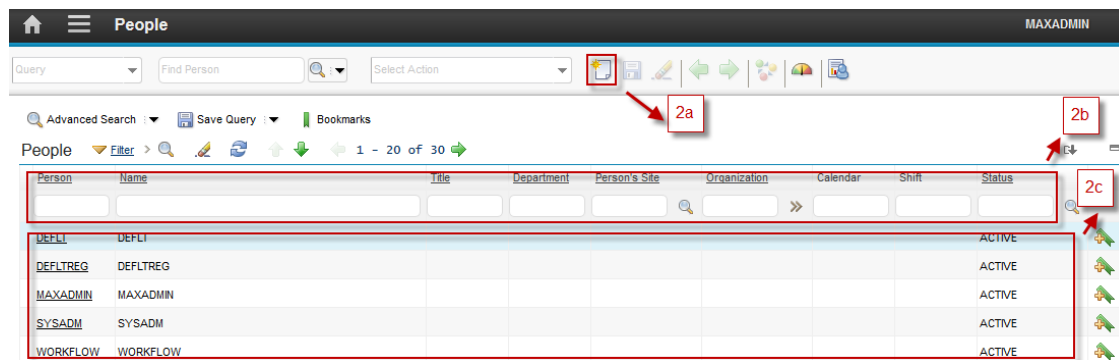
Followings are details about **People** applications accessibility:

1



From **Go To** menu, choose **Administration – Resources – People**

2



Header Bar will give information that user is in the **People** application.

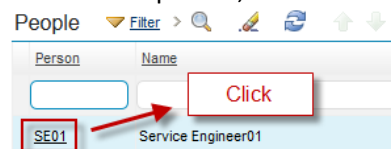
2.a **New Person** button, to add new person

2.b **Filter** person by **Person ID, Name, Title, Status**, etc.

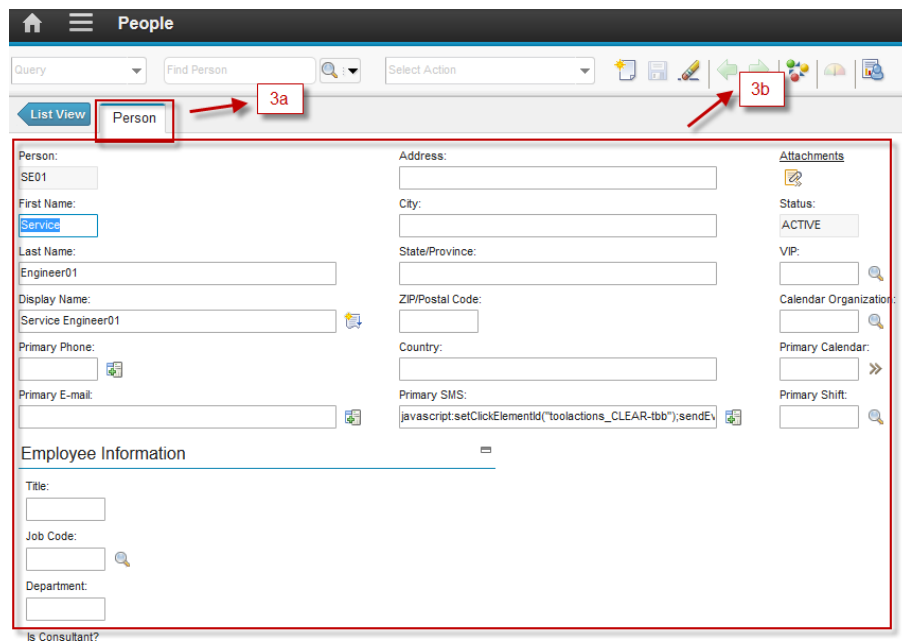
2.c List of all persons based of filtering result.

3

Choose one person, to show the details of the person



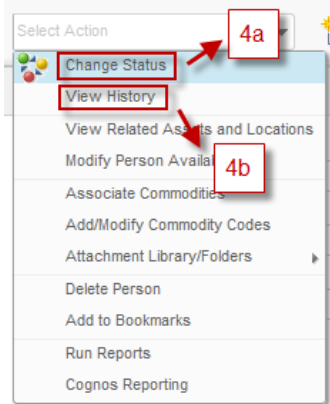
Will show display like below picture.



3.a **Person** tab, to view and update details of person

3.b Detail information of the person, such as first name, last name, display name, primary phone, primary email, etc.

- 4 User can perform some action related to the person from Select Action menu (on top of the form).

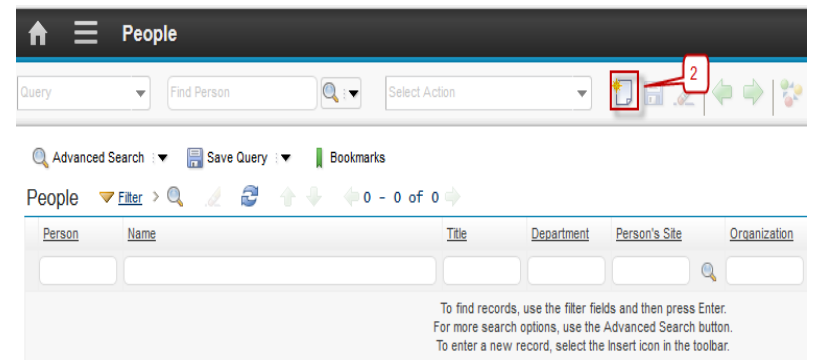


4.a **Change status** action to change status this person

4.b **View History**, to view the change status history of the person.

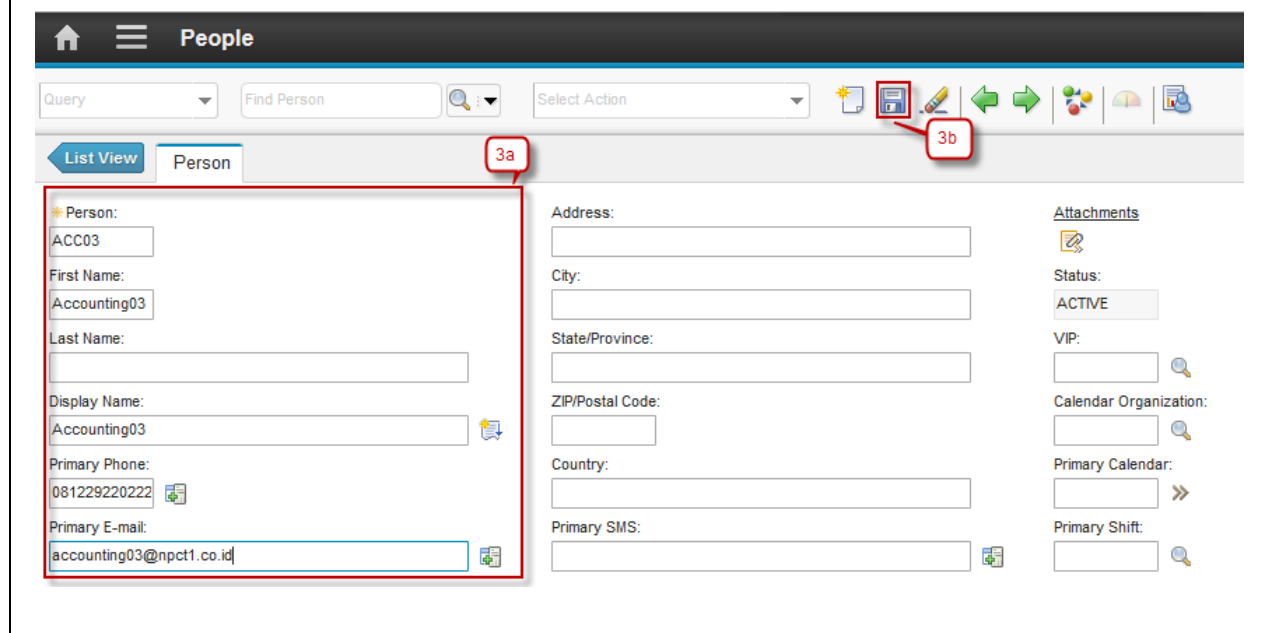
1.2.1 Create Persons

Person creation data can be created by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create new person:

Step	Action
1. Go to People application by click  (GoTo) – Administration Resources – People.	
2. Click  New Person.	

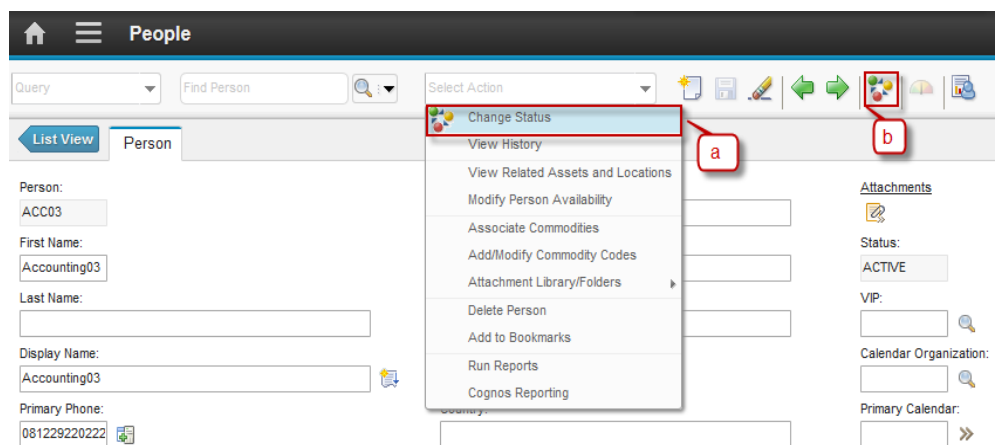
3a. Input the Person Code and any additional information about Person.

3b. Click **Save**.

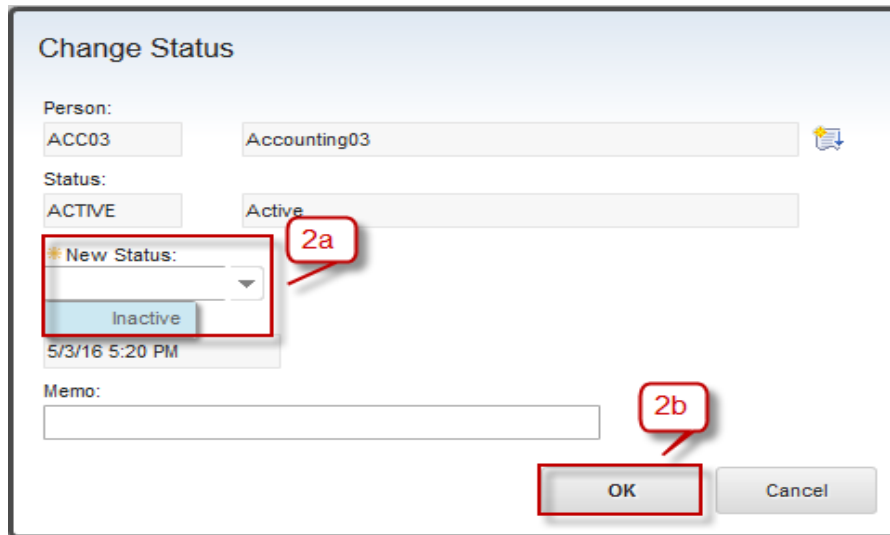


1.2.2 Change Status of Person

Person status change can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Change Status of a person data:

Step	Action
1. Change Status of Person have two ways:	
a. From Select Action → Change Status	
b. Click button 	
	

2a. Dialog box **Change Status** will be appeared, see the figure below :

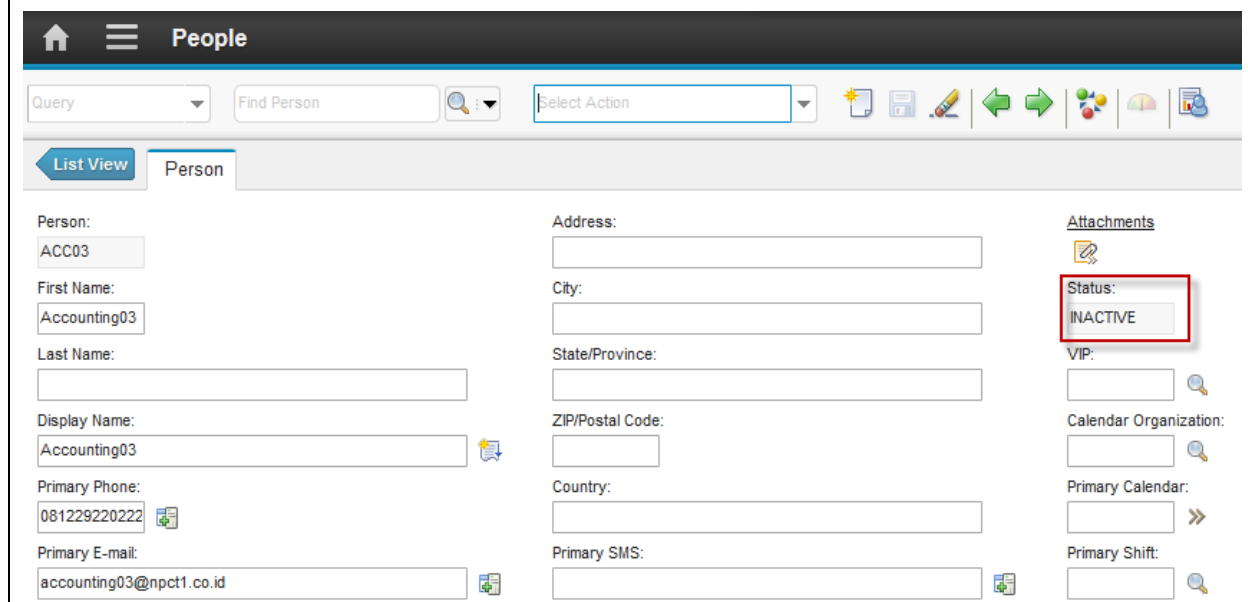


The 'Change Status' dialog box is shown. It contains the following fields and controls:

- Person:** ACC03 (selected in a dropdown), Accounting03 (text input)
- Status:** ACTIVE (selected in a dropdown), Active (text input)
- New Status:** A dropdown menu with 'Inactive' selected. A red callout '2a' points to this dropdown.
- Memo:** A text input field.
- Timestamp:** 5/3/16 5:20 PM
- Buttons:** OK and Cancel. A red callout '2b' points to the OK button.

2b. Click button **OK**

3. Status will be changed to **INACTIVE**

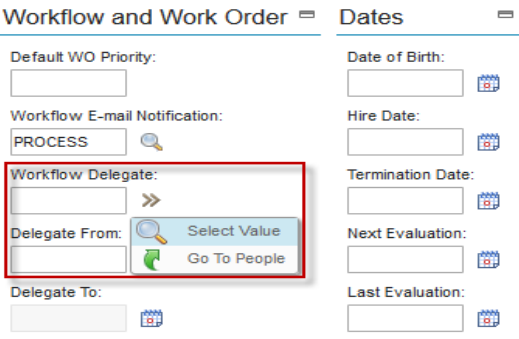
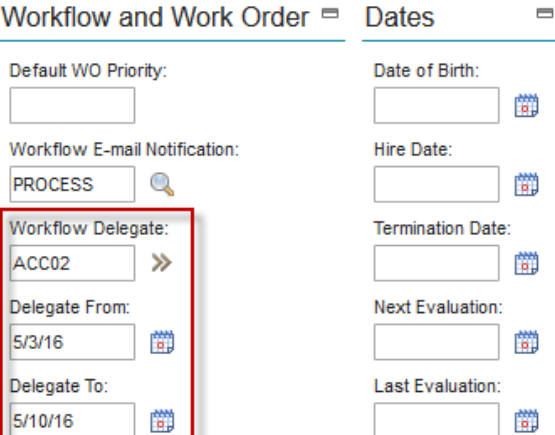


The 'People' management interface is shown. It includes a header with a home icon, a menu icon, and the title 'People'. Below the header is a search bar with 'Find Person' and a 'Select Action' dropdown. The main content area is divided into two tabs: 'List View' and 'Person'. The 'Person' tab is active, showing a form for a person with the following fields:

- Person:** ACC03
- First Name:** Accounting03
- Last Name:** (empty)
- Display Name:** Accounting03
- Primary Phone:** 081229220222
- Primary E-mail:** accounting03@npct1.co.id
- Address:** (empty)
- City:** (empty)
- State/Province:** (empty)
- ZIP/Postal Code:** (empty)
- Country:** (empty)
- Primary SMS:** (empty)
- Attachments:** A section with a red box around the 'Status' field, which is set to 'INACTIVE'.
- VIP:** (empty)
- Calendar Organization:** (empty)
- Primary Calendar:** (empty)
- Primary Shift:** (empty)

1.2.3 Set Person's Delegation

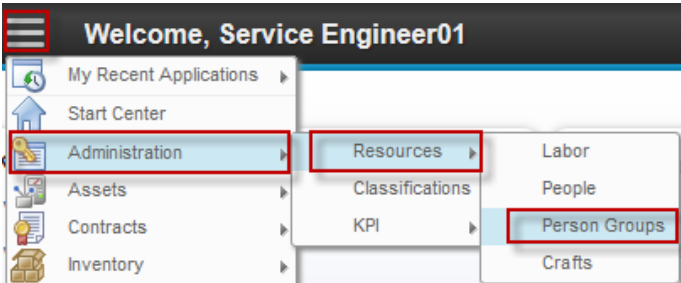
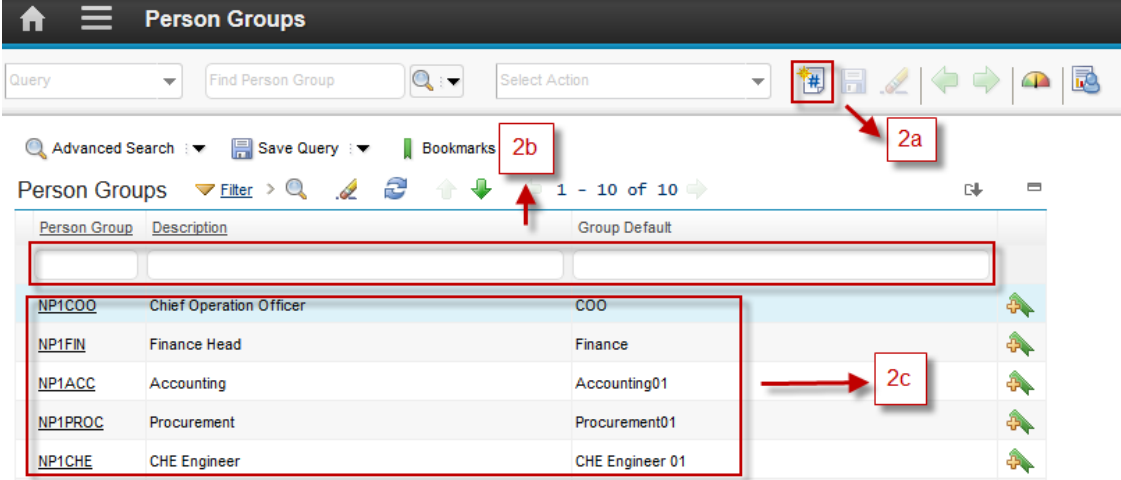
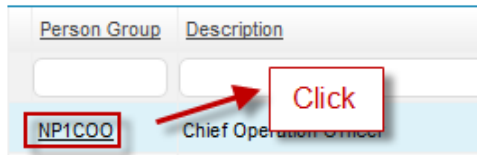
When a person is off, delegation can set toward those person, so any upcoming assignment can be redirected to the delegation. Person's delegation can be configured by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to set person's delegation:

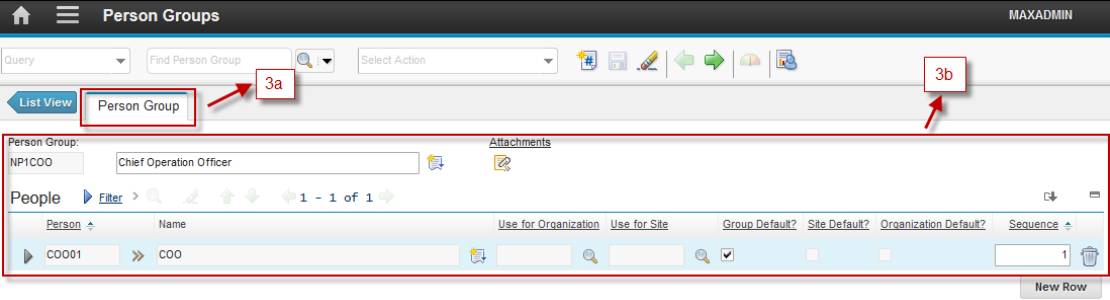
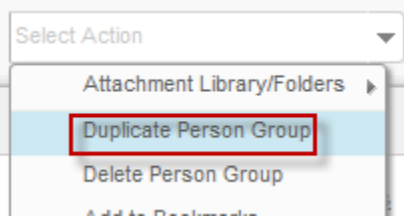
Step	Action
1. Go to section Workflow and Work Order → click button » → Select Value or Go To People to select the person who will be delegated	
2. When click button Select Value will be appeared list of all person and select one of record, see the figure below:	
3. Input field Delegate From and field Delegate To using button 	

1.3 Person Groups Application

Person Groups application is used to manage the groups of person within organization. A person group can be the recipient of a document that is routed by a workflow process. If a document is routed to a person group, everyone in the group receives the document, unless it is configured to send the document only to the person whose calendar indicates availability.

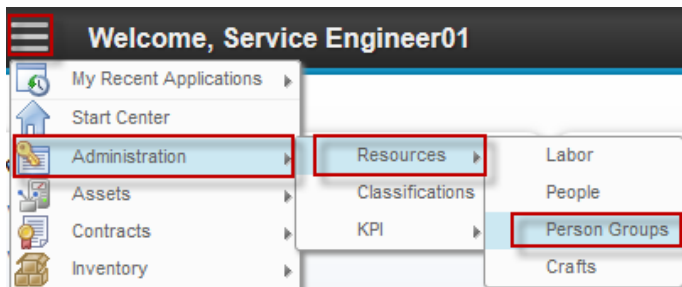
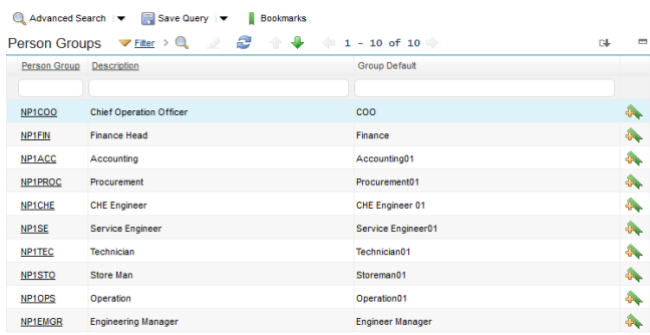
Followings are details about **Person Groups** applications accessibility:

1	 From Go To menu, choose Administration – Resources – Person Groups																		
2	 Header Bar will give information that user is in the Person Group application. 2.a New Person Group button, to add new person 2.b Filter person group by Person Group Code, Description, etc. 2.c List of all person group based of filtering result. <table border="1" data-bbox="272 1100 1284 1360"> <thead> <tr> <th>Person Group</th> <th>Description</th> <th>Group Default</th> </tr> </thead> <tbody> <tr> <td>NP1COO</td> <td>Chief Operation Officer</td> <td>COO</td> </tr> <tr> <td>NP1FIN</td> <td>Finance Head</td> <td>Finance</td> </tr> <tr> <td>NP1ACC</td> <td>Accounting</td> <td>Accounting01</td> </tr> <tr> <td>NP1PROC</td> <td>Procurement</td> <td>Procurement01</td> </tr> <tr> <td>NP1CHE</td> <td>CHE Engineer</td> <td>CHE Engineer 01</td> </tr> </tbody> </table>	Person Group	Description	Group Default	NP1COO	Chief Operation Officer	COO	NP1FIN	Finance Head	Finance	NP1ACC	Accounting	Accounting01	NP1PROC	Procurement	Procurement01	NP1CHE	CHE Engineer	CHE Engineer 01
Person Group	Description	Group Default																	
NP1COO	Chief Operation Officer	COO																	
NP1FIN	Finance Head	Finance																	
NP1ACC	Accounting	Accounting01																	
NP1PROC	Procurement	Procurement01																	
NP1CHE	CHE Engineer	CHE Engineer 01																	
3	Choose one person group, to show the details of the person group  Will show display like below picture.																		

	 3.a Person Group tab, to view and update details of person group 3.b Detail information of the person group and all person listed in the person group.
4	User can perform some action related to the person from Select Action menu (on top of the form).  Duplicate Person group action to create new person group based on opened person group.

1.3.1 Add/Remove Person in a Person Group

Person group's member modification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to add/remove person groups:

Step	Action																																	
1. Go to Person Groups application by click  (GoTo) – Administration – Resources – Person Groups.																																		
2. Select one of Person Group to be added or removed	 <table><thead><tr><th>Person Group</th><th>Description</th><th>Group Default</th></tr></thead><tbody><tr><td>NP1COO</td><td>Chief Operation Officer</td><td>COO</td></tr><tr><td>NP1FIN</td><td>Finance Head</td><td>Finance</td></tr><tr><td>NP1ACC</td><td>Accounting</td><td>Accounting01</td></tr><tr><td>NP1PROC</td><td>Procurement</td><td>Procurement01</td></tr><tr><td>NP1CHE</td><td>CHE Engineer</td><td>CHE Engineer 01</td></tr><tr><td>NP1SE</td><td>Service Engineer</td><td>Service Engineer01</td></tr><tr><td>NP1TEC</td><td>Technician</td><td>Technician01</td></tr><tr><td>NP1STO</td><td>Store Man</td><td>Storeman01</td></tr><tr><td>NP1OPS</td><td>Operation</td><td>Operation01</td></tr><tr><td>NP1EMGR</td><td>Engineering Manager</td><td>Engineer Manager</td></tr></tbody></table>	Person Group	Description	Group Default	NP1COO	Chief Operation Officer	COO	NP1FIN	Finance Head	Finance	NP1ACC	Accounting	Accounting01	NP1PROC	Procurement	Procurement01	NP1CHE	CHE Engineer	CHE Engineer 01	NP1SE	Service Engineer	Service Engineer01	NP1TEC	Technician	Technician01	NP1STO	Store Man	Storeman01	NP1OPS	Operation	Operation01	NP1EMGR	Engineering Manager	Engineer Manager
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NP1PROC	Procurement	Procurement01																																
NP1CHE	CHE Engineer	CHE Engineer 01																																
NP1SE	Service Engineer	Service Engineer01																																
NP1TEC	Technician	Technician01																																
NP1STO	Store Man	Storeman01																																
NP1OPS	Operation	Operation01																																
NP1EMGR	Engineering Manager	Engineer Manager																																

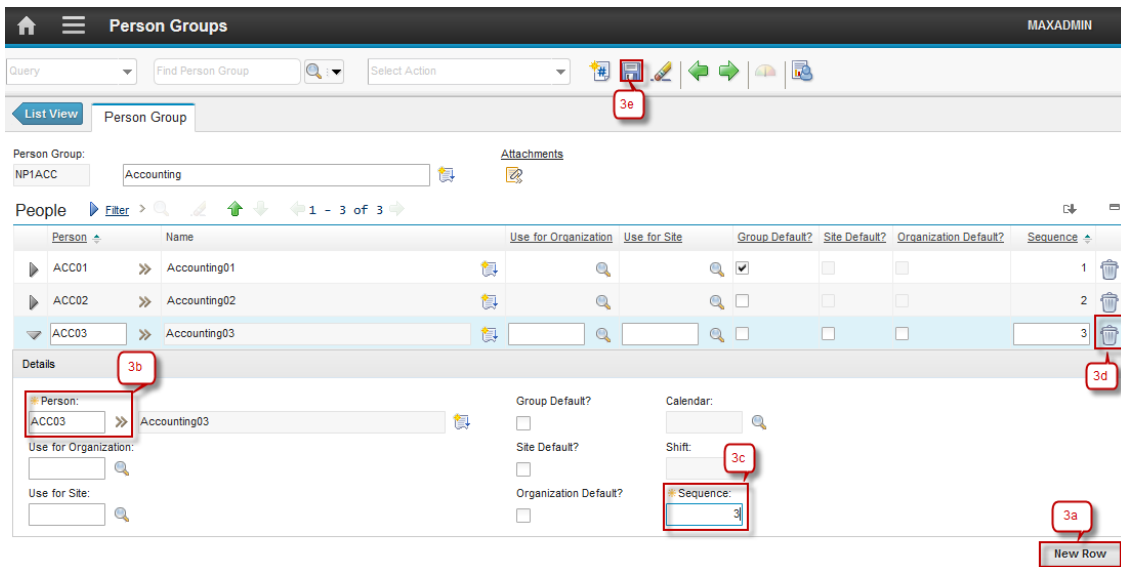
3a. Click **New Row** to add person as member of this person group.

3b. Select the person as member of this person group.

3c. Input the sequence.

3d. Click button  to remove a member of this person group

3e. Click **Save**.



The screenshot shows the 'Person Groups' application interface. At the top, there's a header with a home icon, a menu icon, the title 'Person Groups', and the user 'MAXADMIN'. Below the header is a toolbar with various icons. A 'List View' button is visible. The main content area shows a 'Person Group' section with 'NPIACC' and 'Accounting'. Below this is a 'People' section with a table of members. The table has columns for 'Person', 'Name', 'Use for Organization', 'Use for Site', 'Group Default?', 'Site Default?', 'Organization Default?', and 'Sequence'. Three rows are visible: 'ACC01' (Accounting01), 'ACC02' (Accounting02), and 'ACC03' (Accounting03). The 'ACC03' row is highlighted. To the right of the table is a 'Details' section for 'Accounting03'. It contains fields for 'Person' (set to 'ACC03'), 'Use for Organization', 'Use for Site', 'Group Default?', 'Site Default?', 'Organization Default?', 'Calendar', 'Shift', and 'Sequence'. The 'Sequence' field is highlighted. At the bottom right, there is a 'New Row' button. Red callouts are placed over the interface: '3a' points to the 'New Row' button, '3b' points to the 'Person' dropdown in the details section, '3c' points to the 'Sequence' input field, '3d' points to the trash icon in the table, and '3e' points to the 'Save' button in the toolbar.

1.4 Crafts Application

Crafts application is used to define craft records for a work plan, and to define skill levels, standard rates. The craft code reflects the type of work that employees and contractors perform.

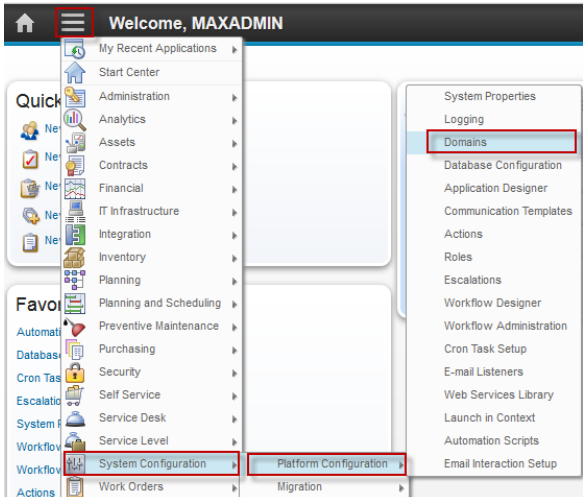
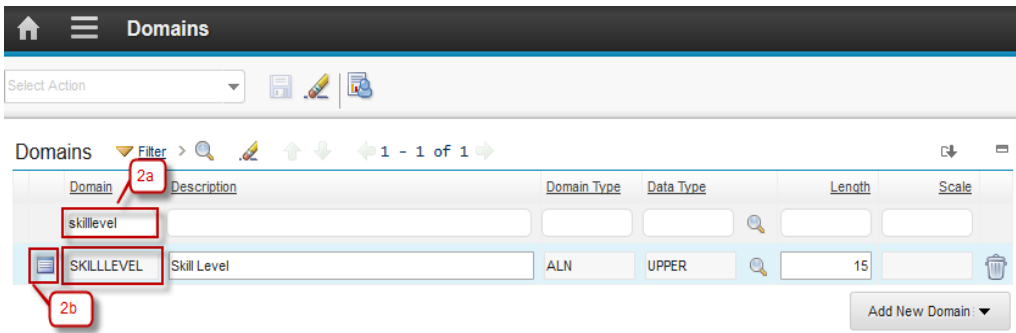
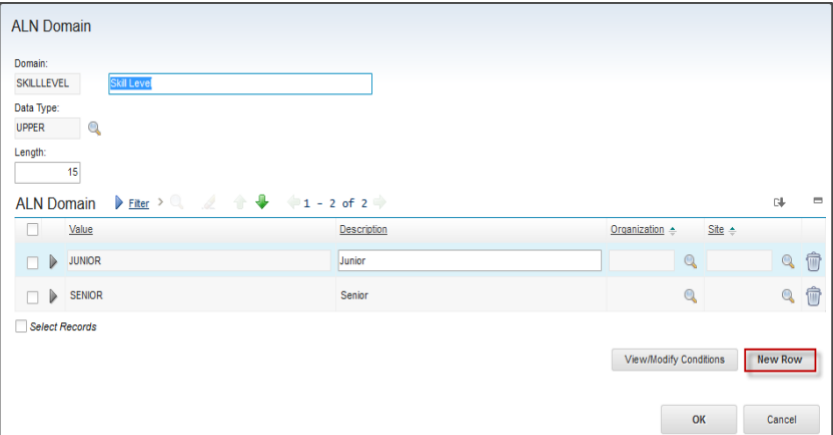
Besides, in **Crafts** application user also can:

- Associate labor records with crafts and skill levels
- Specify multiple skill levels for each craft

To distinguish between a junior level mechanic and a senior level mechanic, it is not required to create two separate craft records. Only one mechanic craft record is needed and set up junior and senior skill levels within the mechanic craft (skill level 1 is highest, and 5 is lowest). Each skill level within the mechanic craft can have different standard rates.

1.4.1 Define Skill Level

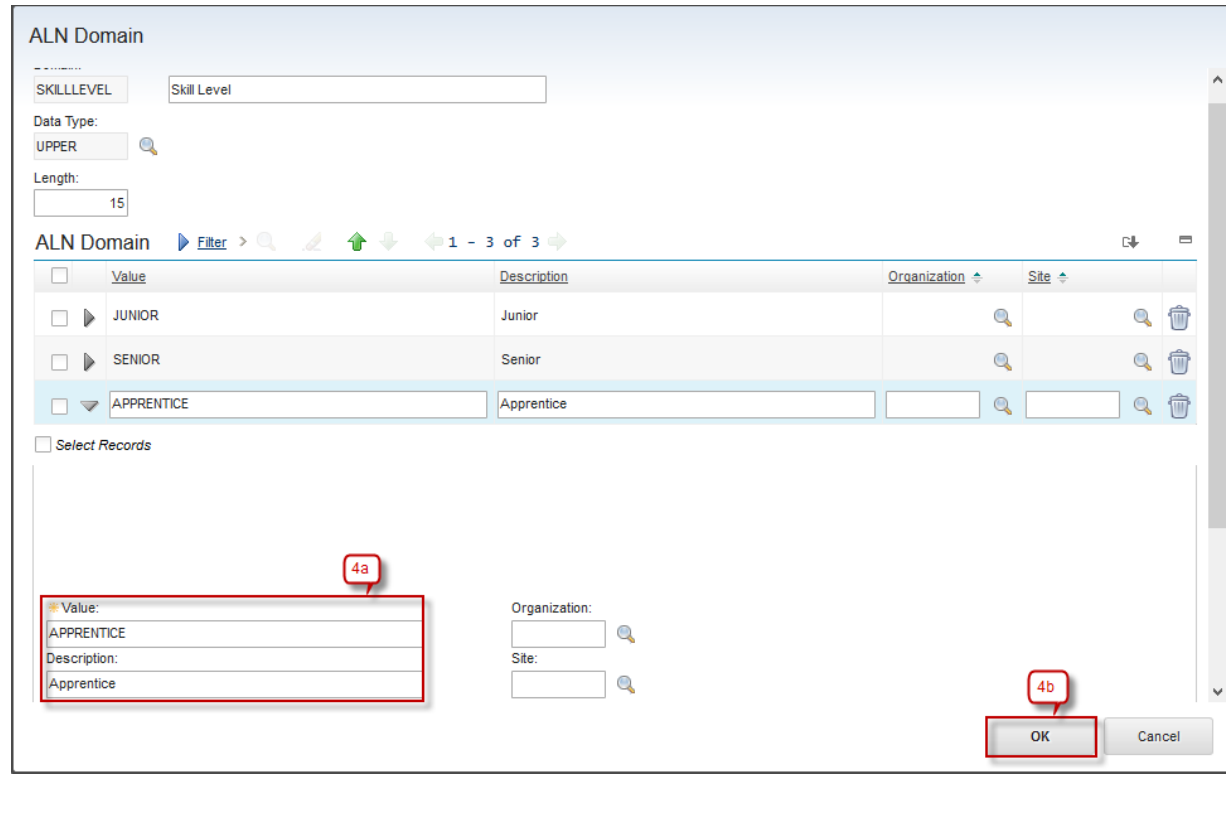
Skill level definition can only be done by *Administrator*. Below are step by step to define Skill Level:

Step	Action
1. Go to Domains application by click  (GoTo) – System Configuration – Platform Configuration – Domains.	
2a. Search Domains name in field Domains = 'skilllevel' 2b. Click button  to add skill level.	
3. Click button New Row	

4a. Input the value of Skill Level.

4b. Input the value description.

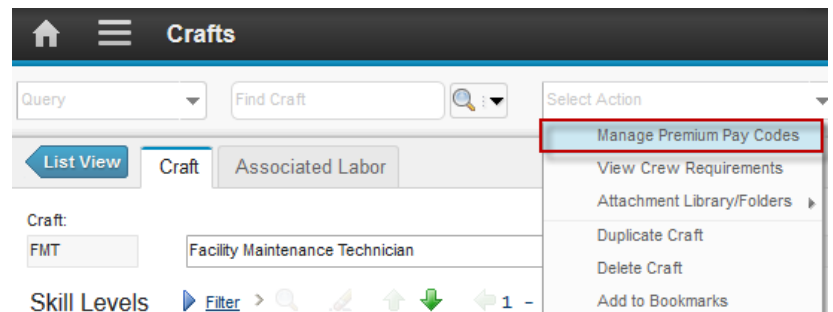
4c. Click button **OK**.



The screenshot shows the 'ALN Domain' form. At the top, there is a 'SKILLLEVEL' field with the value 'Skill Level'. Below it, 'Data Type' is set to 'UPPER' and 'Length' is '15'. A table lists three records: JUNIOR, SENIOR, and APPRENTICE. The 'APPRENTICE' record is selected. Below the table, there is a 'Select Records' checkbox. A red box labeled '4a' highlights the 'Value' field in the 'APPRENTICE' record, which contains 'APPRENTICE'. Another red box labeled '4b' highlights the 'OK' button at the bottom right of the form.

1.4.2 Define Premium Pay Codes

Managing premium pay codes can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to define Premium Pay Codes:

Step	Action
1. Click Craft Tab → Select Action → Manage Premium Pay Codes	 The screenshot shows the 'Crafts' tab interface. The 'Select Action' dropdown menu is open, and the 'Manage Premium Pay Codes' option is highlighted with a red box. Other options in the menu include 'View Crew Requirements', 'Attachment Library/Folders', 'Duplicate Craft', 'Delete Craft', and 'Add to Bookmarks'.

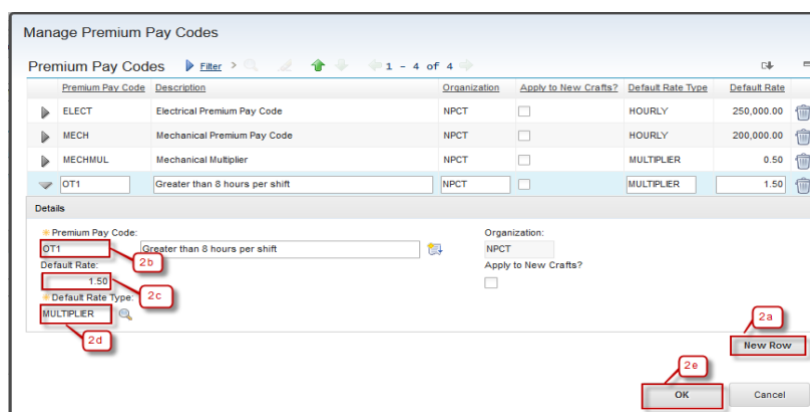
2a. Click **New Row**.

2b. Input the Premium Pay Code.

2c. Input Default Rate.

2d. Input Default Rate Type.

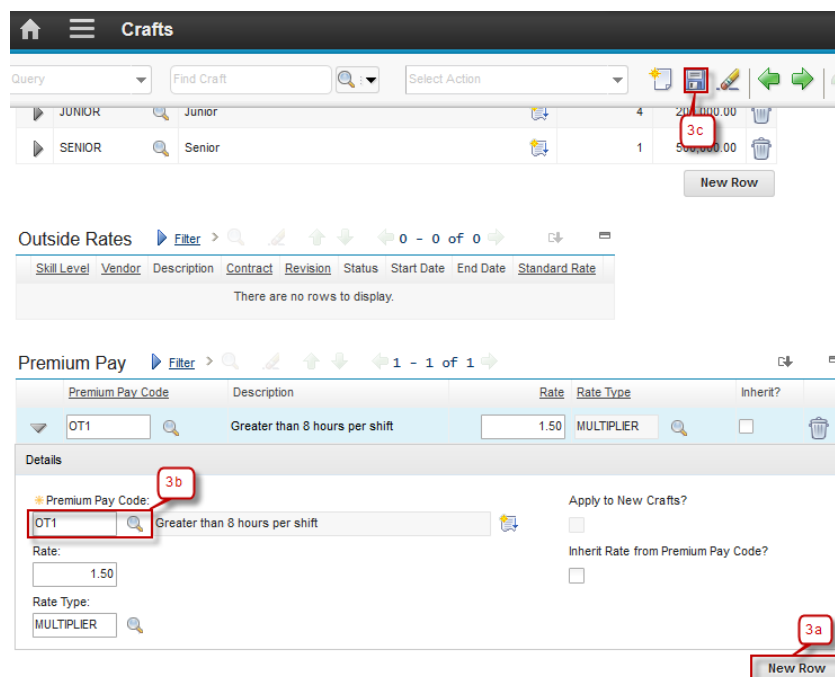
2e. Click **OK**.



3a. Click **New Row** to add Premium Pay in this Craft.

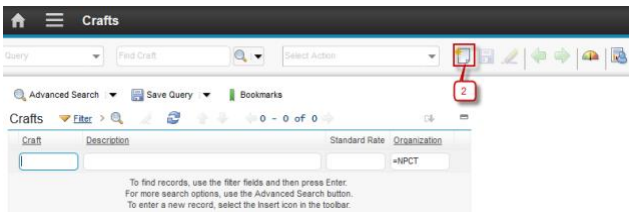
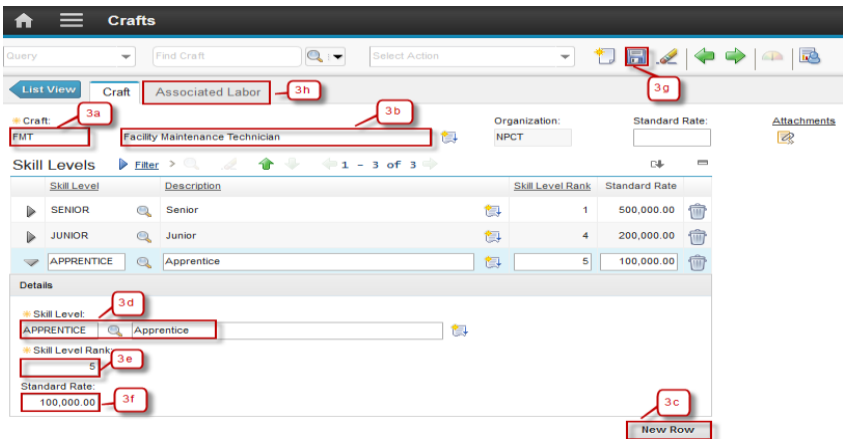
3b. Select Premium Pay Code using button 

3c. Click **Save**

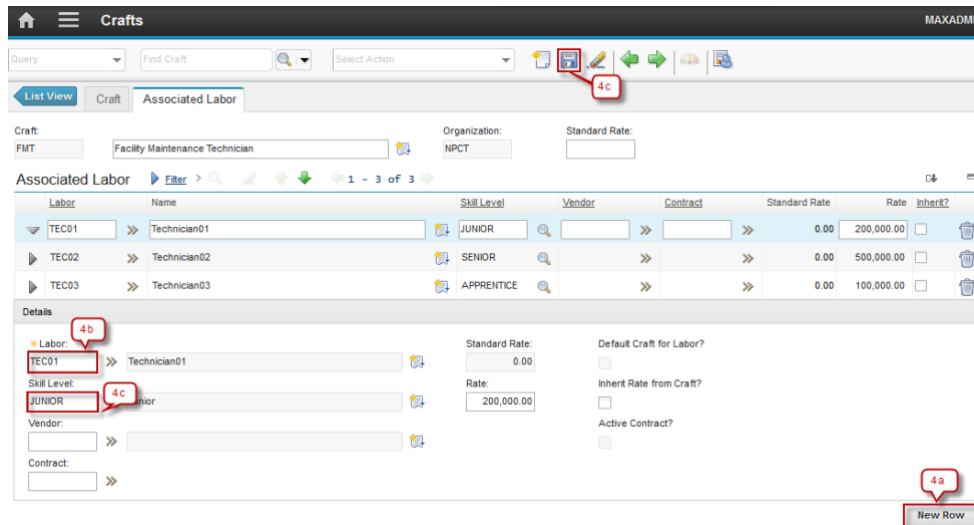


1.4.3 Create Crafts

Craft creation can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Crafts:

Step	Action
1. Go to Crafts application by click  (GoTo) – Administration – Resources – Crafts.	
2. Click  New Craft .	
3a. Input the Craft name. 3b. Input the Craft description. 3c. Click New Row to add Skill Level of this craft. 3d. Input the Skill Level name and the description. 3e. Input the Skill Level Rank (1 is highest, 5 is lowest). 3f. Input Standard Rate for that skill level of that craft. 3g. Click Save. 3h. Click Associated Labor tab to input labor that has this craft skill.	

- 4a. Click **New Row** to associate labor with this craft skill.
- 4b. Input the labor name that has this craft skill.
- 4c. Input the skill level of this craft that owned by the labor.
- 4d. Click **Save**.



1.5 Labor Application

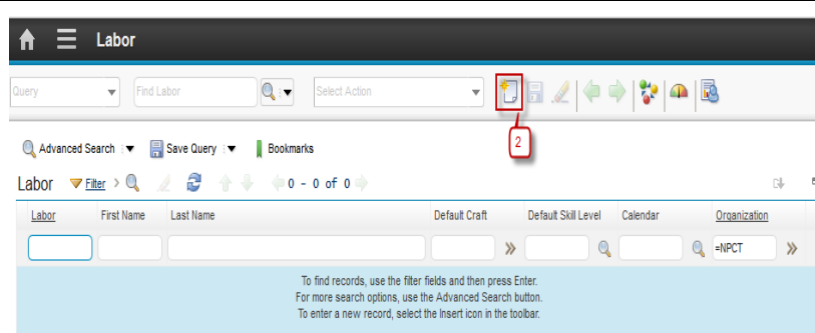
Labor application is used to create, change, view, and delete labor records for employees and contractors. Labor records can contain personal and work-related information, as well as the type and location of work. In addition, crafts, skill levels, qualifications, and certifications can be specified that are associated with labor records. Before labor record can be created, a person record corresponding to the labor must be created first.

1.5.1 Create Labors

Labor data creation can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Labor:

Step	Action
1. Go to Labor application by click  (GoTo) – Administration – Resources – Labor .	

2. Click  **New Labor.**



The screenshot shows the 'Labor' management interface. At the top, there's a header with a home icon, a menu icon, and the title 'Labor'. Below the header is a toolbar with various icons. The 'New Labor' icon, which is a document with a plus sign, is highlighted with a red box and labeled with the number '2'. Below the toolbar, there are search and filter options, including 'Query', 'Find Labor', and 'Select Action'. There are also links for 'Advanced Search', 'Save Query', and 'Bookmarks'. The main area shows a table with columns for 'Labor', 'First Name', 'Last Name', 'Default Craft', 'Default Skill Level', 'Calendar', and 'Organization'. The table is currently empty, and a message at the bottom states: 'To find records, use the filter fields and then press Enter. For more search options, use the Advanced Search button. To enter a new record, select the Insert icon in the toolbar.'

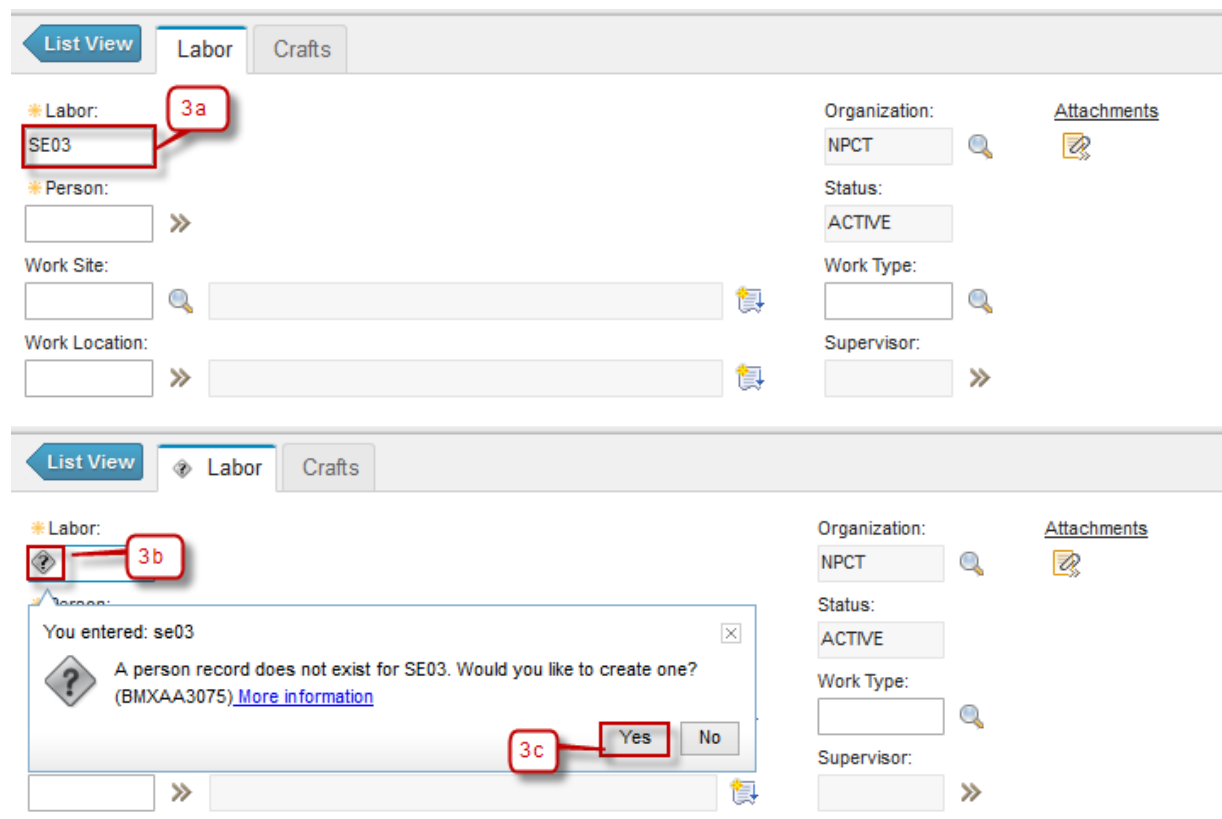
3a. Input the Labor Code.

3b. If the labor code that inputted above doesn't have a person data yet, the question mark icon will appear, Click the question mark.

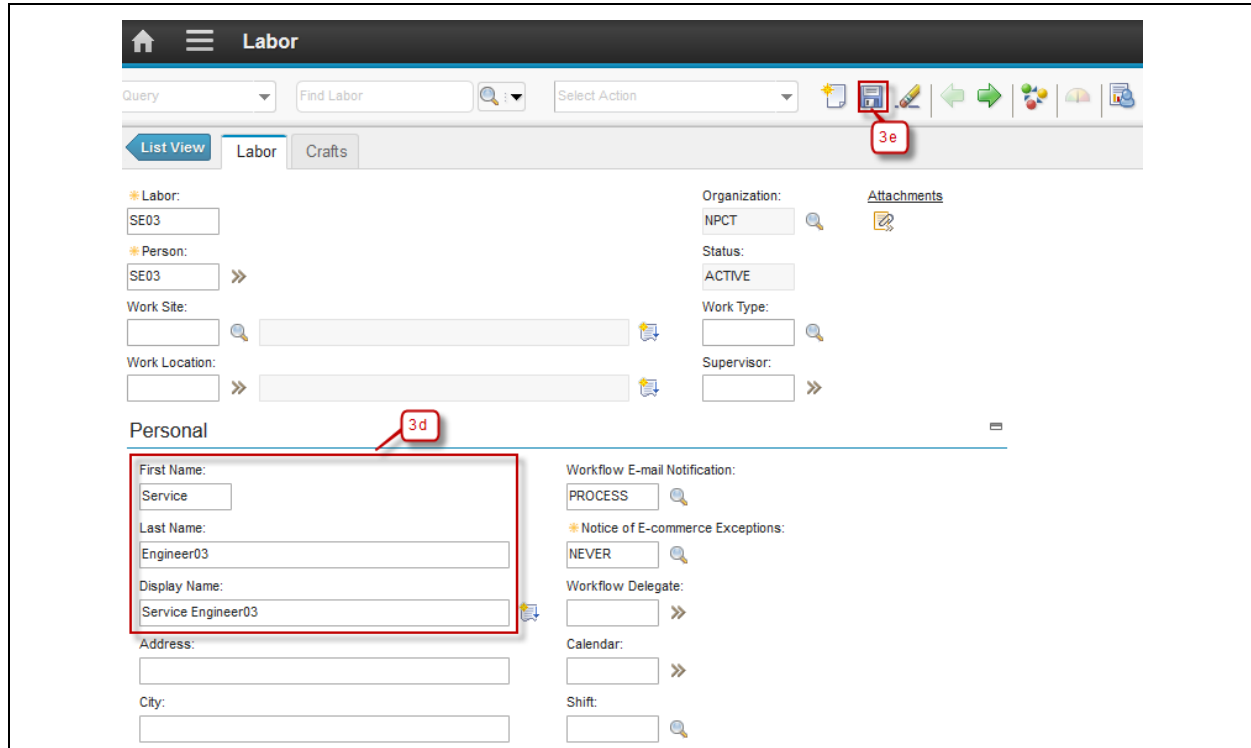
3c. Click **Yes** to create the person data based on the labor code that inputted above

3d. Input any additional information about Labor

3e. Click **Save.**



The first screenshot shows the 'New Labor' form. The 'Labor' field is set to 'SE03' and is highlighted with a red box labeled '3a'. Other fields include 'Person', 'Work Site', 'Work Location', 'Organization' (set to 'NPCT'), 'Status' (set to 'ACTIVE'), 'Work Type', and 'Supervisor'. The second screenshot shows the same form, but the 'Labor' field now has a question mark icon next to it, highlighted with a red box labeled '3b'. A dialog box is open, asking 'You entered: se03. A person record does not exist for SE03. Would you like to create one? (BMXAA3075) [More information](#)'. The 'Yes' button in the dialog box is highlighted with a red box labeled '3c'.

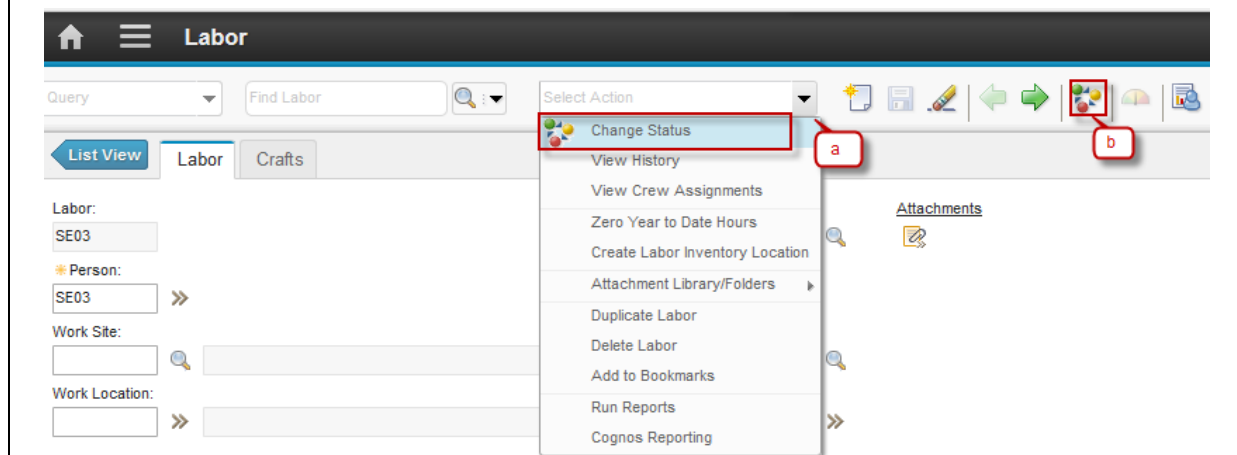


The screenshot shows the 'Labor' management interface. The 'Personal' section is highlighted with a red box and labeled '3d'. The 'Attachments' section is highlighted with a red box and labeled '3e'.

1.5.2 Change Status of Labor

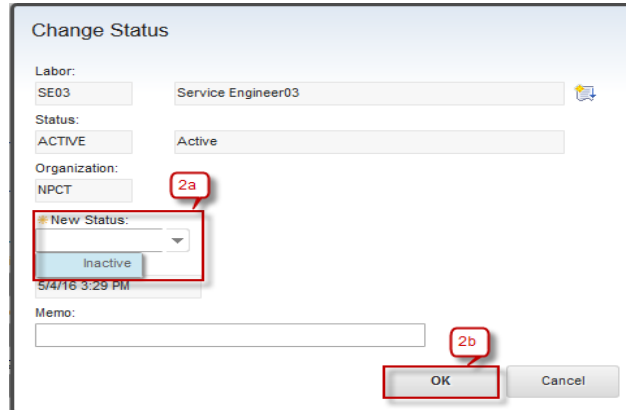
Labor status change can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Change Status of Person

Step	Action
1. Change Status of Person have two ways:	
a. From Select Action → Change Status	
b. Click button 	



The screenshot shows the 'Labor' management interface. The 'Select Action' dropdown menu is open, and the 'Change Status' option is highlighted with a red box and labeled 'a'. The 'Attachments' section is highlighted with a red box and labeled 'b'.

2a. Dialog box **Change Status** will be appeared, see the figure below :

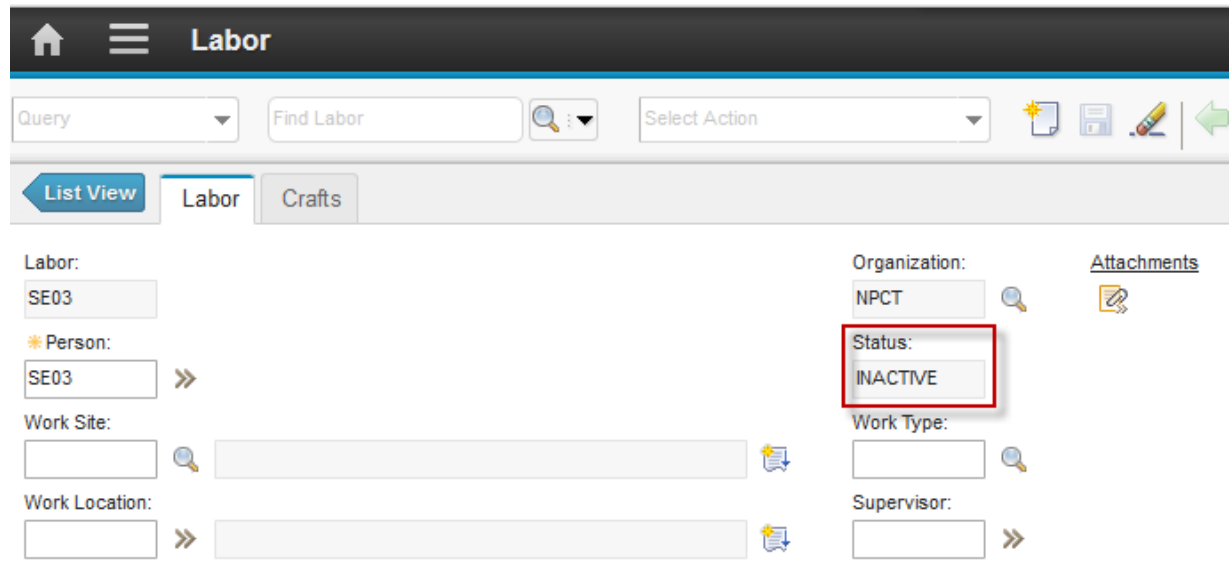


The 'Change Status' dialog box contains the following fields and controls:

- Labor:** SE03 (text field), Service Engineer03 (text field)
- Status:** ACTIVE (text field), Active (text field)
- Organization:** NPCT (text field)
- New Status:** A dropdown menu with 'Inactive' selected. A red box labeled '2a' highlights this dropdown.
- Memo:** A text area.
- Buttons:** OK and Cancel. A red box labeled '2b' highlights the OK button.

2b. Click button **OK**

4. Status will be changed to **INACTIVE**



The 'Labor' management interface shows the following details:

- Header:** Labor
- Query:** Find Labor (text field), Select Action (dropdown menu)
- Views:** List View, Labor, Crafts (tabs)
- Labor Details:**
 - Labor:** SE03
 - Person:** SE03 (with a double arrow icon)
 - Work Site:** (text field)
 - Work Location:** (text field)
- Organization:** NPCT
- Status:** INACTIVE (highlighted with a red box)
- Work Type:** (text field)
- Supervisor:** (text field)
- Attachments:** (link)

1.6 Exercise Phase

2 Asset & Location

2.1 Overview

Asset is an item of economic value that a company own, benefit from, or has use of, in generating income. In EMS, an item is treated as asset when user wants to track its moving history, and maintenance work can occurs toward it.

Asset can be either rotating or non-rotating. Rotating assets are assets that are interchangeable, such as motors, pumps, or fire extinguishers. In EMS, remarkable difference between rotating asset and non-rotating asset is that rotating assets have both an asset number and an inventory item number, while non-rotating just have an asset number, without an inventory item number. The item number lets user to track assets as a group as they are moved in and out of inventory and other types of locations, while the asset number is useful to track individual instances of asset as it is moved from one location to another and from one site to another.

Here is example to help understanding inventory item number and asset. A company might have four identical (same make, same model) centrifugal pumps, so all four pumps would have the same item number. To track the use, repair, and locations of each individual pump, each pump would have its own, unique asset number.

Location is generally defined as a place where assets are operated, stored, and repaired. Generally locations are defined as a means of tracking assets. Another consideration, a maintenance work can be done toward **Locations**.

2.2 Classification Application

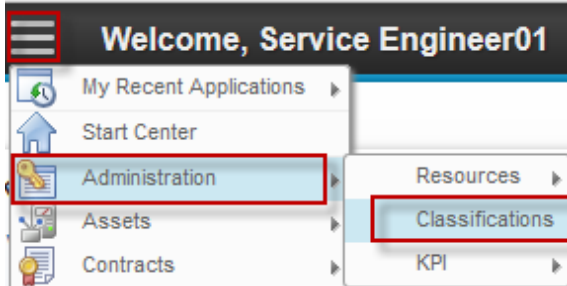
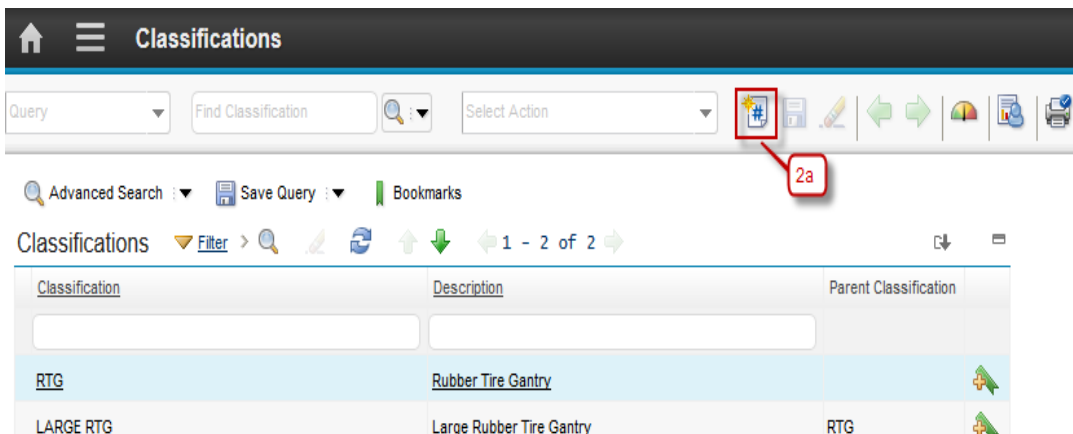
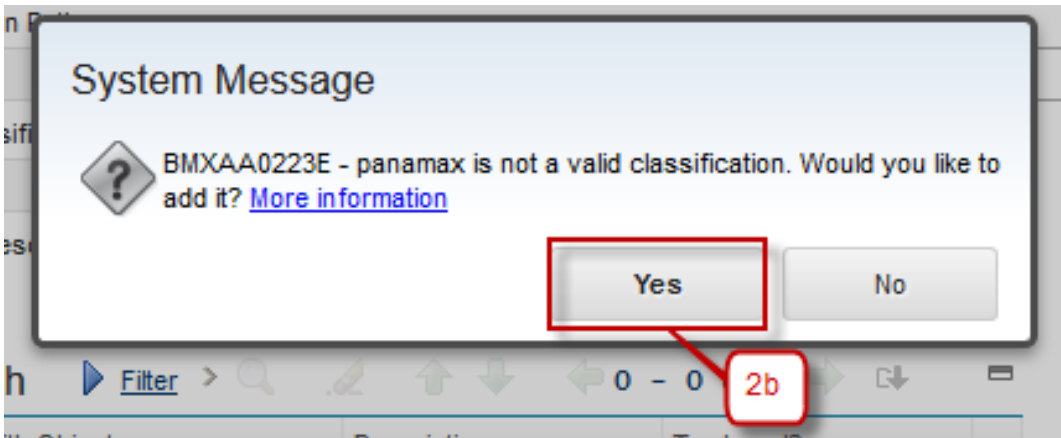
The Specifications tab on asset and location records can be used to add Classification information to asset and location records. Specification is available on item data too. Using Classifications lets user to structure assets and locations in organized hierarchies so that:

- Records can be easily located via Classifications.
- Records are not duplicated unintentionally.
- Asset and item lists are consistent with vendor's lists.

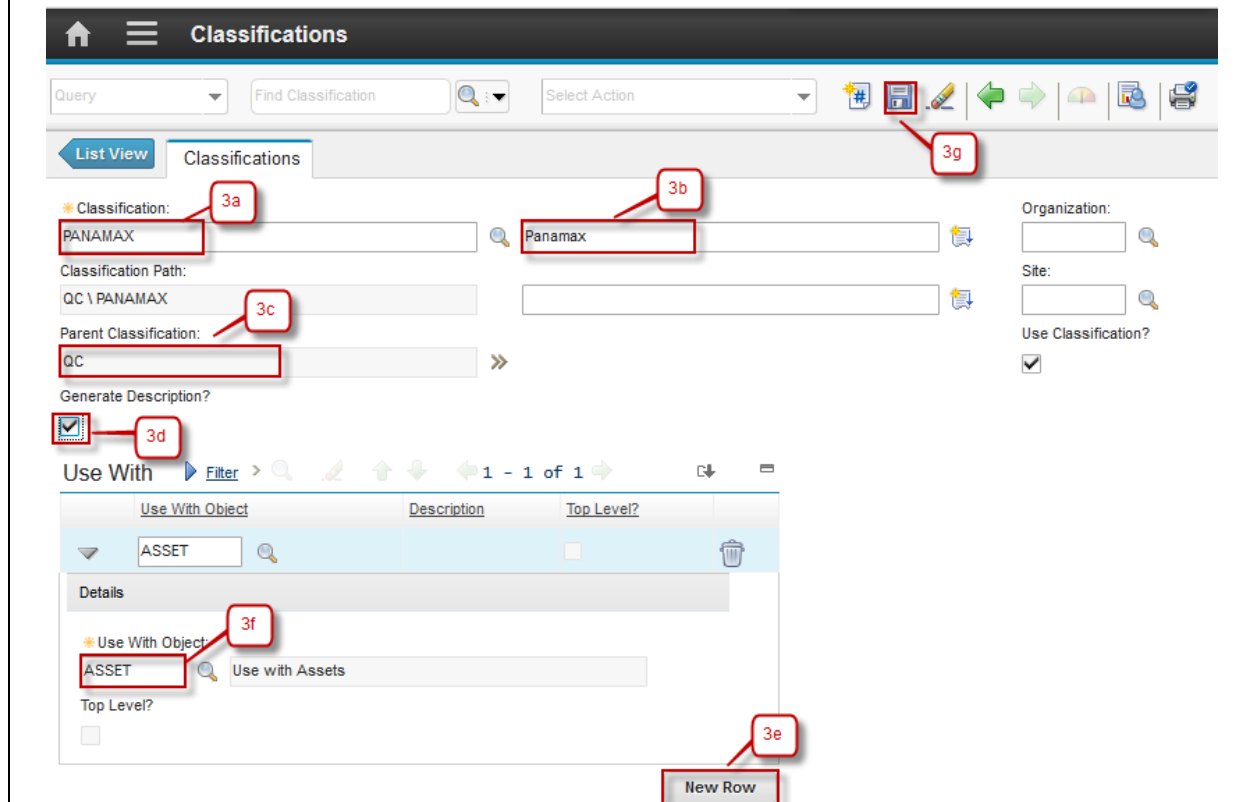
Classifications use attributes to define the classification category. When you list a rotating item that has a classification on an asset or location record, Maximo copies the Classification and Attributes to **Specifications** tab of the asset or location record.

2.2.1 Create Classifications

Classification creation can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Classification:

Step	Action
1. Go to Classification application by click  (GoTo) – Administration – Classifications.	
2a. Click New Classification .	 
2b. If the classification that inputted above doesn't valid, the message box to added the classification will appear. Click button Yes and then classification will be added.	

- 3a. Input the Classification name.
- 3b. Input the Classification description.
- 3c. Input the Parent Classification if this classification has a parent.
- 3d. Default from **Generate Description** is checked, if description classification don't want generated when the classification in use make sure the check box is unchecked.
- 3e. Click **New Row** to add the object that will use this classification.
- 3f. Input the object that will use this classification.
- 3g. Click **Save**.



The screenshot shows the 'Classifications' management interface. At the top, there's a header with a home icon, a menu icon, and the title 'Classifications'. Below this is a toolbar with a 'Query' dropdown, a 'Find Classification' search bar, a 'Select Action' dropdown, and several icons including a plus sign (3g), a save icon, a refresh icon, and a print icon. The main area is divided into a 'List View' tab and a 'Classifications' tab. In the 'Classifications' tab, there's a form with the following fields: 'Classification:' (3a) with the value 'PANAMAX', 'Classification Path:' with the value 'QC \ PANAMAX', 'Parent Classification:' (3c) with the value 'QC', and 'Generate Description?' (3d) which is checked. To the right of these fields are 'Organization:', 'Site:', and 'Use Classification?' (checked). Below the form is a 'Use With' section with a table. The table has columns 'Use With Object', 'Description', and 'Top Level?'. It contains one row with 'ASSET' in the 'Use With Object' column. Below the table is a 'Details' section with a 'Use With Object' field (3f) containing 'ASSET' and a 'Top Level?' checkbox. At the bottom right of the 'Details' section is a 'New Row' button (3e).

2.2.2 Define Attributes to a Classification

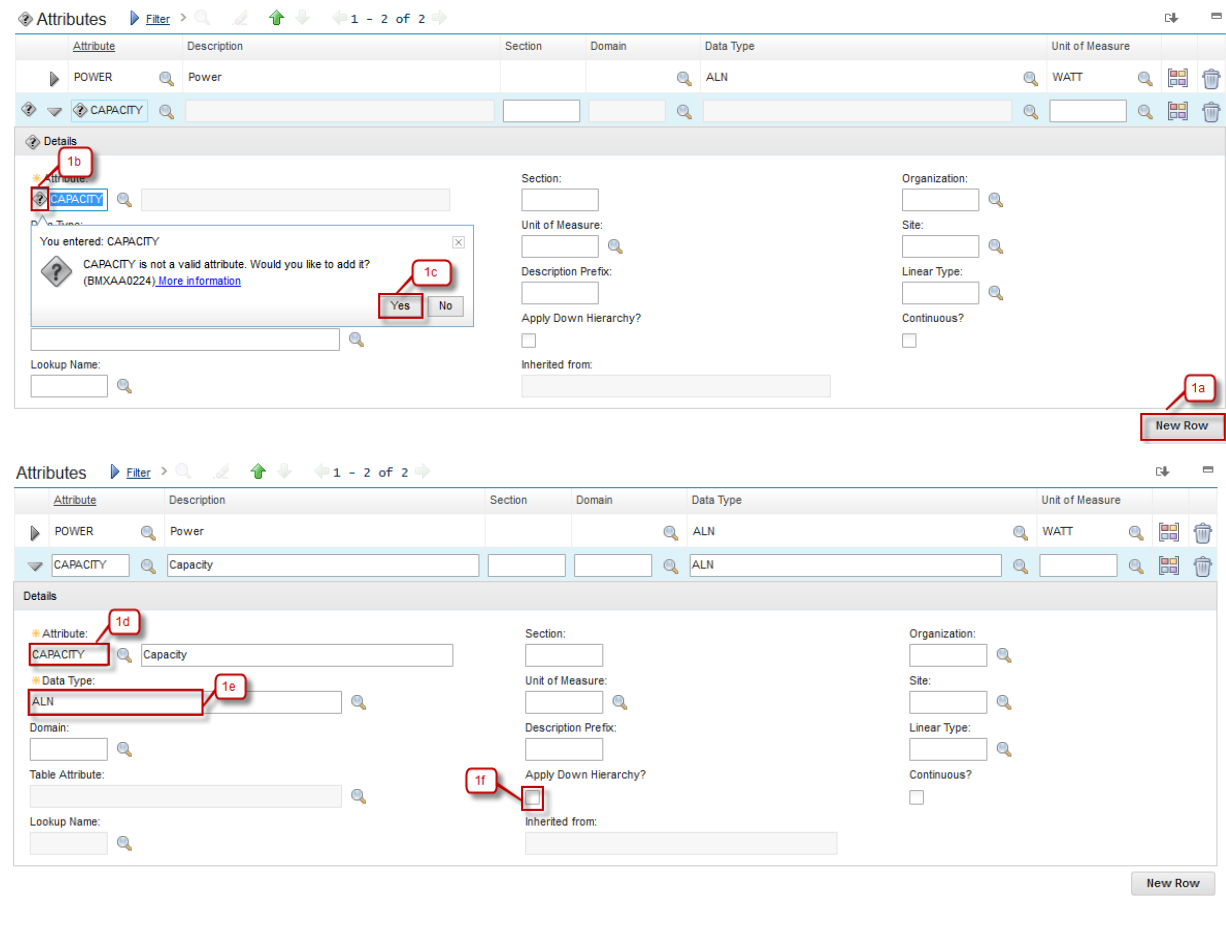
Defining attributes to a classification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Define Attribute to a Classification:

Step	Action
1a. Click New Row to add attribute on this classification.	
1b. If the attribute you added is a new attribute, question mark icon will appear.	
1c. Click Yes to add the attribute.	

1d. Input the Description of that attribute.

1e. Input the Data Type of that attribute.

1f. Check the **Apply Down Hierarchy** checkbox if you want this attribute will appear in any child classification of this classification and click **Save**.



The screenshot shows the 'Attributes' management interface. At the top, there is a table with columns: Attribute, Description, Section, Domain, Data Type, and Unit of Measure. The first row is 'POWER' with description 'Power', domain 'ALN', and unit 'WATT'. The second row is 'CAPACITY' with description 'Capacity', domain 'ALN', and unit 'WATT'. Below the table, there is a 'Details' section for the selected attribute. In the 'Details' section, the 'Attribute' field is labeled '1b' and contains 'CAPACITY'. The 'Data Type' field is labeled '1e' and contains 'ALN'. The 'Description' field is labeled '1d' and contains 'Capacity'. The 'Apply Down Hierarchy' checkbox is labeled '1f' and is checked. A message box is displayed: 'You entered: CAPACITY. CAPACITY is not a valid attribute. Would you like to add it? (BMXAA0224) More information'. The 'Yes' button is labeled '1c'. The 'New Row' button is labeled '1a'.

2.3 Locations Application

Locations application is used to add, view, modify, and delete location records for assets, and organize these location into systems. Systems are used to maintain location's hierarchical. When location is organized into systems, user can quickly find a location using the drilldown and then identify the assets at a specific location.

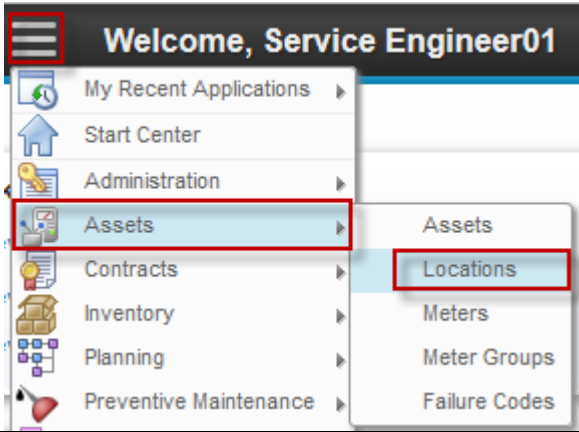
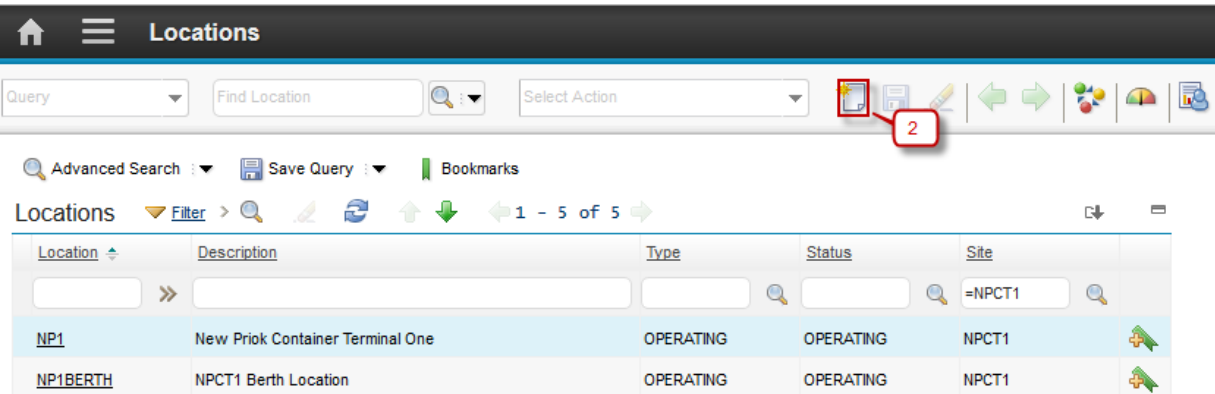
Locations are categorized into several types:

- **Courier** - This is an inventory location, used to indicate an individual who is responsible for inventory items while in transit from one inventory location to another.

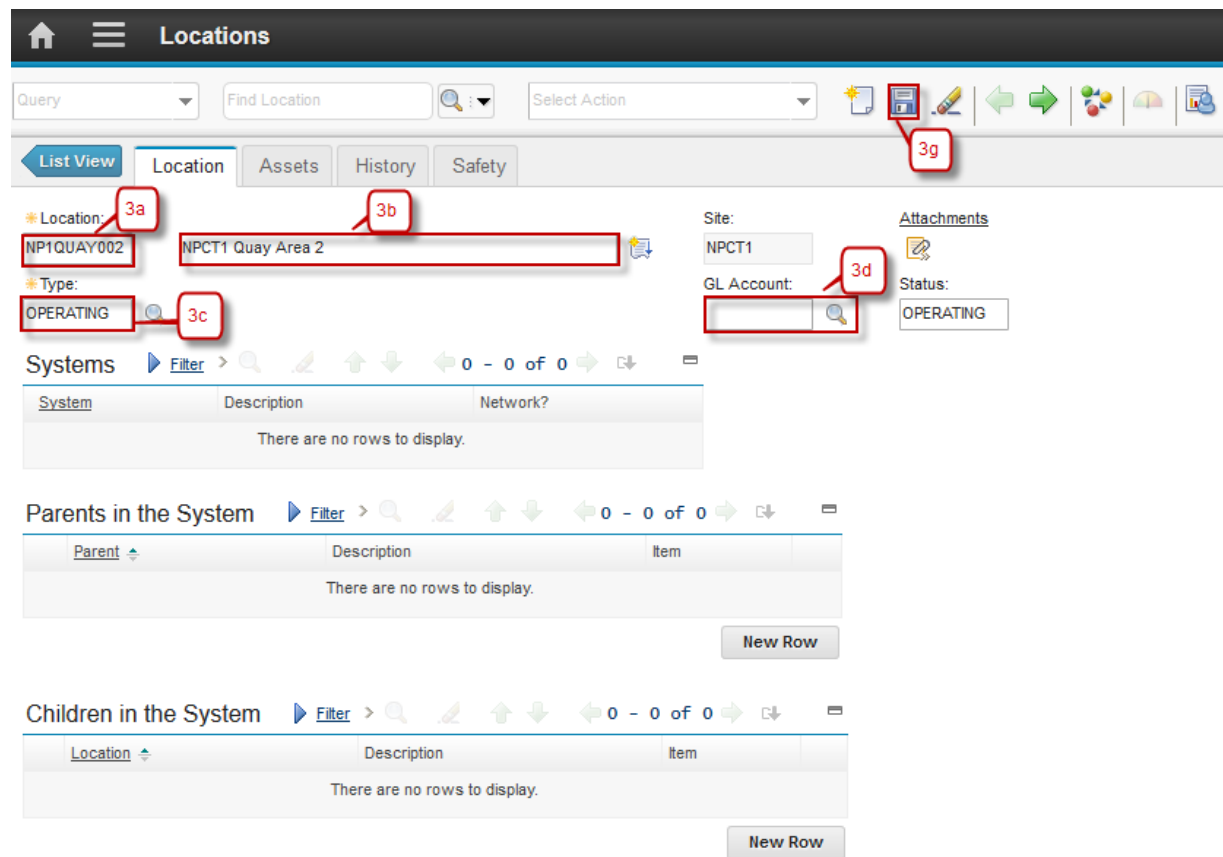
- **Holding** - This is an inventory location, used to hold items received with a status of WINSP (waiting for inspection). One site can only have one holding location.
- **Operating** - This is an asset location, used to indicate a location where assets are located or equipment is operated.
- **Repair** - This is an asset location, used to indicate a location where assets are repaired or refurbished.
- **Salvage** - This is an asset location, used to indicate a location used to store retired or decommissioned assets.
- **Vendor** - This is an inventory location, used to indicate the source of items, material, and rotating assets, and may be used to indicate the location of an asset returned to the vendor for repair or refurbishment.

2.3.1 Create Locations

Location creation can be performed by *Service Engineer*. Below are step by step to Create Locations:

Step	Action
1. Go to Locations application by click (GoTo) – Assets – Locations.	
2. Click  New Locations .	

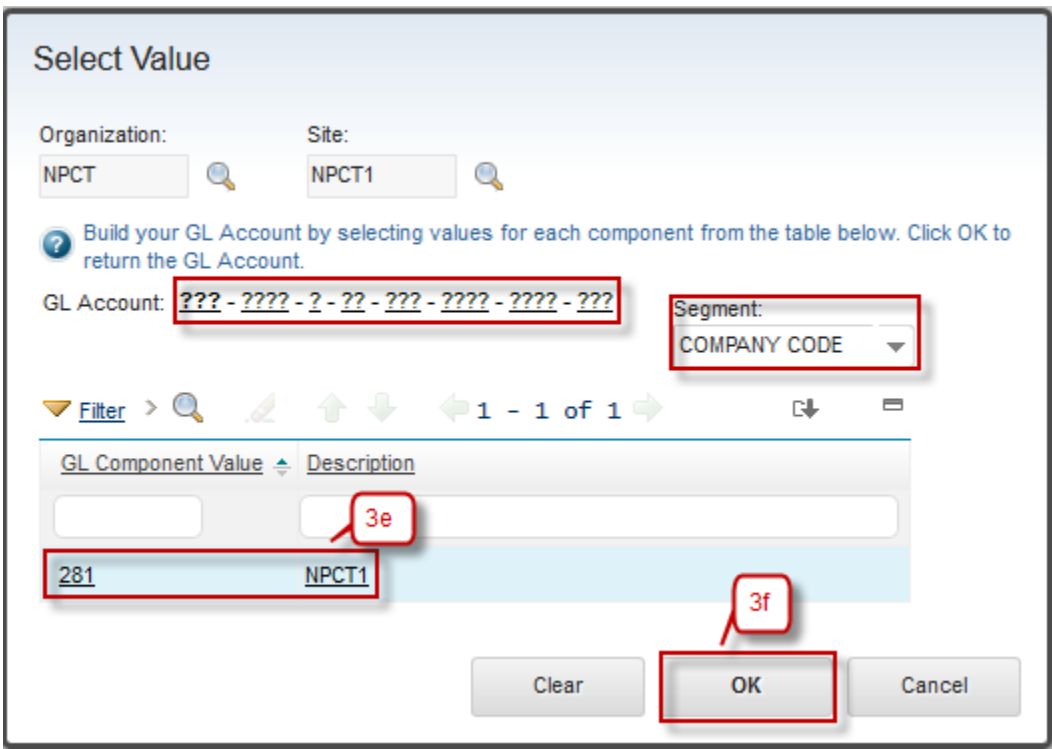
- 3a. Input the Locations name.
- 3b. Input the Locations description.
- 3c. Select the type of Locations, default type is 'OPERATING'.
- 3d. Define GL Account of Locations, click button 
- 3e. Select account value for every segment of GL Account.
- 3f. Click **OK**.
- 3g. Click **Save**.



The screenshot shows the 'Locations' form in the Agile system. The form is titled 'Locations' and has a navigation bar with tabs: 'List View', 'Location', 'Assets', 'History', and 'Safety'. The 'Location' tab is active. The form contains several fields and buttons:

- Location:** A text field containing 'NP1QUAY002' (annotated with 3a).
- Description:** A text field containing 'NPCT1 Quay Area 2' (annotated with 3b).
- Type:** A dropdown menu with 'OPERATING' selected (annotated with 3c).
- Site:** A text field containing 'NPCT1'.
- GL Account:** A text field with a magnifying glass icon (annotated with 3d).
- Status:** A dropdown menu with 'OPERATING' selected.
- Attachments:** A button with a plus icon.
- Buttons:** 'Save' (annotated with 3g), 'OK', and 'Cancel'.

Below the form, there are three sections: 'Systems', 'Parents in the System', and 'Children in the System'. Each section has a table with columns for 'System', 'Description', and 'Network?'. All three sections show 'There are no rows to display.' and a 'New Row' button.



Select Value

Organization: NPCT Site: NPCT1

Build your GL Account by selecting values for each component from the table below. Click OK to return the GL Account.

GL Account: ??? - ??? - ? - ?? - ??? - ??? - ??? Segment: COMPANY CODE

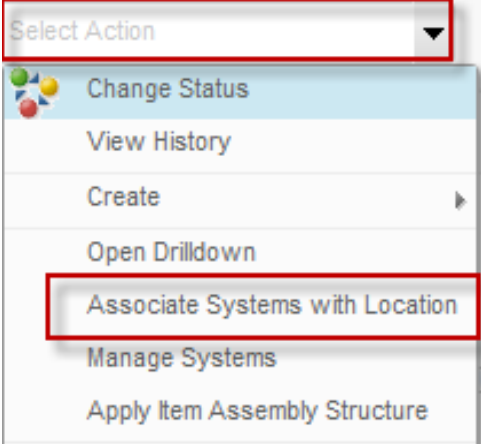
Filter > 1 - 1 of 1

GL Component Value	Description
281	NPCT1

Clear OK Cancel

2.3.2 Associate Systems with Locations

Below are step by step to Associate Systems with Locations by *Service Engineer*:

Step	Action
1. Go to Select Action – Associate Systems with Location.	

- 2a. Click **New Row**.
- 2b. Select System of Locations.
- 2c. Select Parent of Locations.
- 2d. Click button **OK**.

Associate Systems with Location

To associate a new system with the current location, click New Row.

Location: NP1QUAY002 NPCT1 Quay Area 2

Site: NPCT1

System Filter > 1 - 1 of 1

System	Description	Primary System?	Network?	Parent
PRIMARY	Primary System	☒	☐	NP1QUAY001

Details

* System: PRIMARY Primary System

Parent: NP1QUAY001 NPCT1 Quay Area

Primary System? ☒

Network? ☐

2a New Row

2b

2c

2d OK Cancel

3. Field System and Parent will defined

List View Location Assets History Safety

Location: NP1QUAY002 NPCT1 Quay Area 2

Site: NPCT1

Type: OPERATING

GL Account: 281-0091-0-01

Attachments

Status: OPERATING

Systems Filter > 1 - 1 of 1

System	Description	Network?
PRIMARY	Primary System	☐

Parent of NP1QUAY002 in the PRIMARY System Filter > 1 - 1 of 1

Parent	Description	Item
NP1QUAY001	NPCT1 Quay Area	

Children of NP1QUAY002 in the PRIMARY System Filter > 0 - 0 of 0

Location	Description	Item
There are no rows to display.		

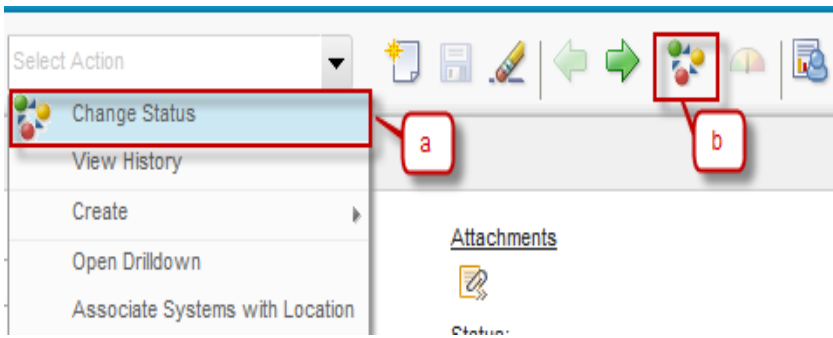
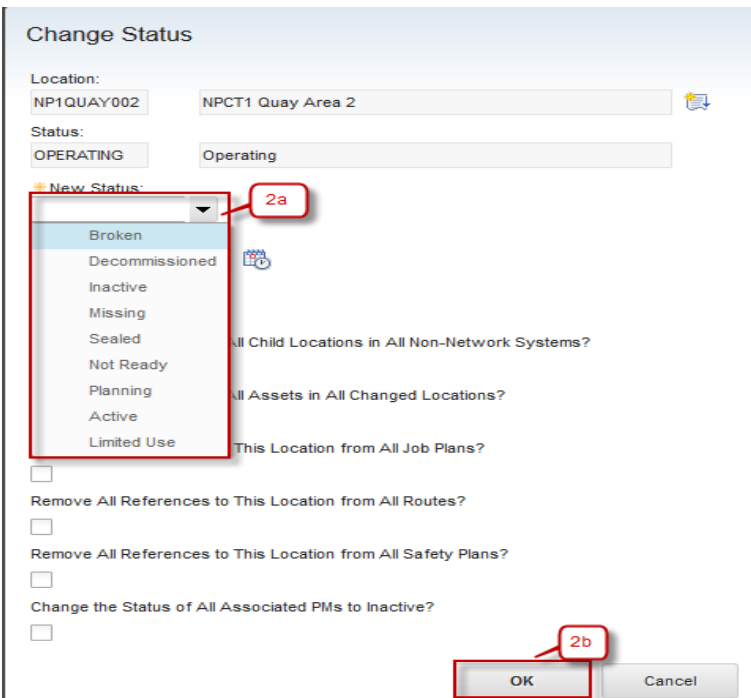
3

New Row

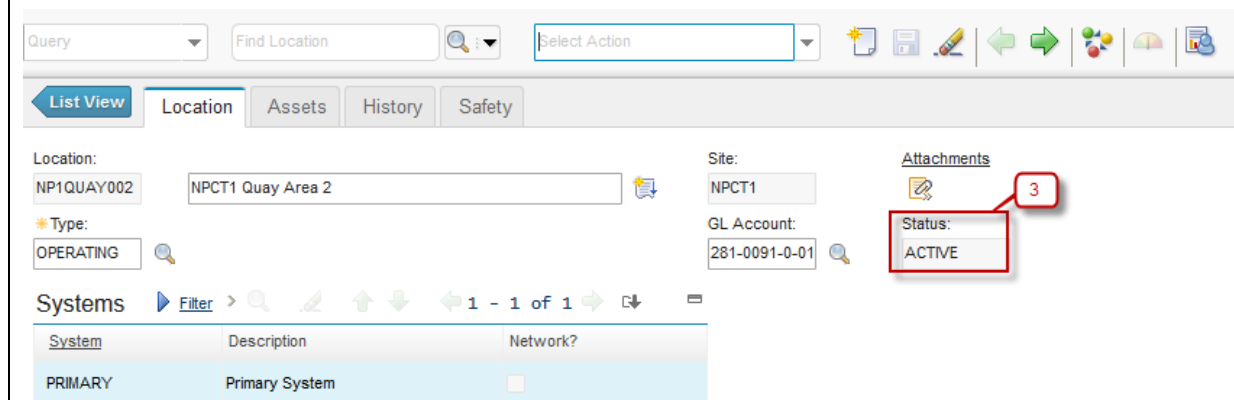
New Row

2.3.3 Change Status of a Location

Below are step by step to Change Status of a Location by *Service Engineer*:

Step	Action
1. Change Status of Person have two ways: a. From Select Action → Change Status b. Click button 	
2. Dialog box Change Status will be appeared, see the figure below : 2a. Select New Status of Locations. 2b. Click OK .	

3. Status will be change to **ACTIVE**



The screenshot shows the Agile Solution interface for managing assets. At the top, there are search and action fields. Below, the 'Location' tab is active, showing details for 'NP1QUAY002' and 'NPCT1 Quay Area 2'. The 'Type' is set to 'OPERATING'. To the right, the 'Site' is 'NPCT1' and the 'GL Account' is '281-0091-0-01'. A red box highlights the 'Status' field, which is currently 'ACTIVE', with a red circle containing the number '3' next to it. Below this, the 'Systems' section shows a table with one entry: 'PRIMARY' with description 'Primary System' and a 'Network?' checkbox.

System	Description	Network?
PRIMARY	Primary System	☐

2.4 Meters & Meter Groups Application

A meter record in EMS is virtual identifier used to record measurements. meter records is associated with assets and locations to record measurements, for example run hours, temperature, or pressure, that are used to track asset and location performance. **Meters** application is used to add, view, modify, and delete meter records.

Meters can have one of three types:

- Continuous - Meters where the readings increase continuously, for example odometers. These meters might be used to track miles, hours, engine starts, pieces produced, fuel consumed, and other continuous readings.
- Gauge - Meters where the readings may fluctuate, for example thermometers or pressure gauges. These meters might be used to track temperature, pressure, noise levels, oil levels, and other readings that fluctuate,
- Characteristic - Meters where an observed state is being tracked, for example a color change. These meters might be used to track Brick/Refractory Color (yellow, orange, white), or oil color (clear, turbid, dark).

Meter readings may be recorded for a variety of reasons, including:

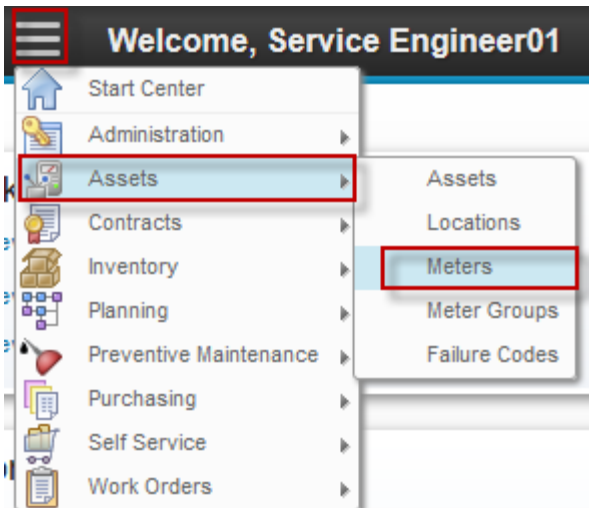
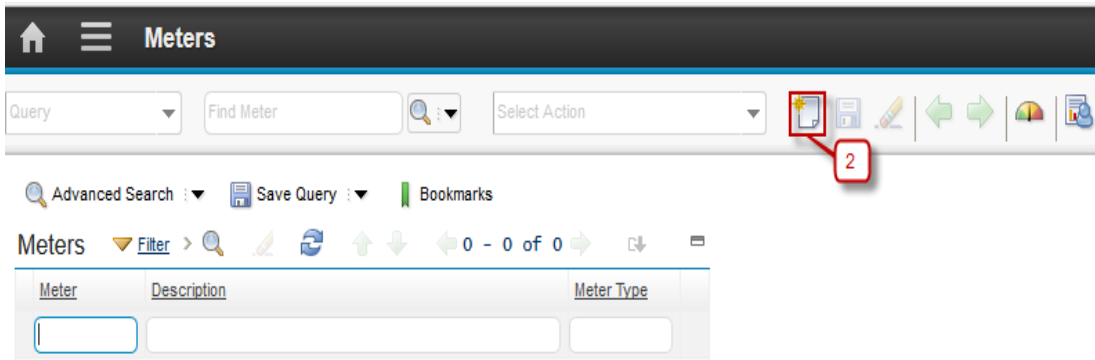
- determining the next time an asset is due for maintenance
- performing efficiency comparisons between similar asset performing similar work
- tracking production costs, for example tons per hour in the mining industry
- tracking service costs, for example odometer readings for a trucking fleet
- tracking efficiency indicators, for example miles per gallon
- conforming to regulatory requirements, for example maintaining cradle to grave details of time controlled components in the aviation industry

In **Meters** application, when user create a meter record, user can define the type of meter, the unit of measure used by the meter, and the type of reading (delta or actual) that the meter should record. Meter records can be associated with assets, inventory items, and locations, and can be used to drive condition monitoring and preventive maintenance. Meters can be applied individually, or as part of a meter group.

A meter group is two or more meters that are commonly used on the same types of assets or locations. **Meter Groups** application is used to create groups for meters that are commonly used together. For example all of the vehicles in a fleet use meters to track mileage, usage, fuel, and oil consumption. Rather than attach these four meters to a vehicle one at a time, user can define a meter group and use it to associate all four meters to the vehicle at the same time.

2.4.1 Create Meters

Below are step by step to Create Meters by *Service Engineer*:

Step	Action
1. Go to Meters application by click  (GoTo) – Assets – Meters.	
2. Click  New Row.	

3a. Input the Meter name.

3b. Input the Meter description.

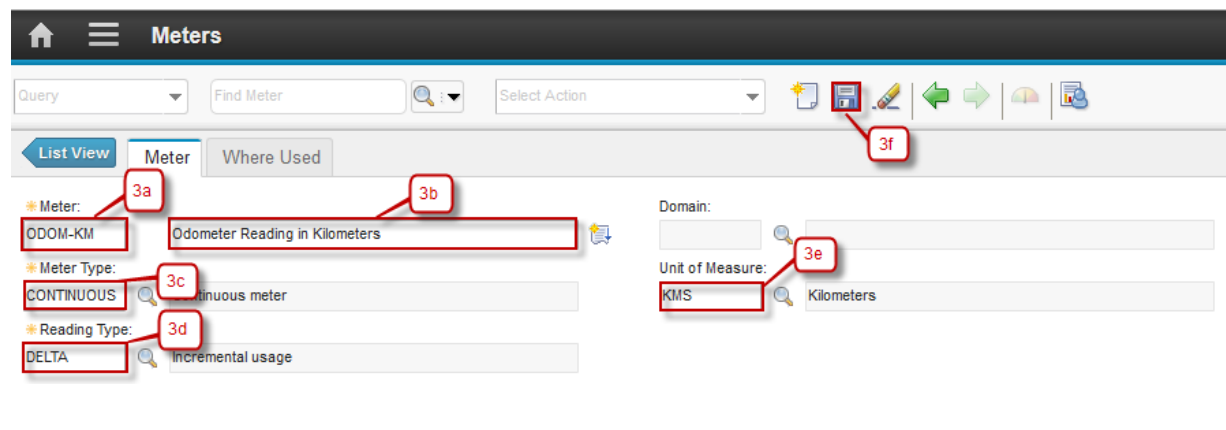
3c. Select the type of Meter.

3d. Input the Reading Type.

- If the type of Meter is 'CONTINUOUS', field Reading Type is mandatory.
- If the type of Meter is 'CHARACTERISTIC', field Domain is mandatory.

3e. Input the Unit of Measure

3f. Click **Save**.

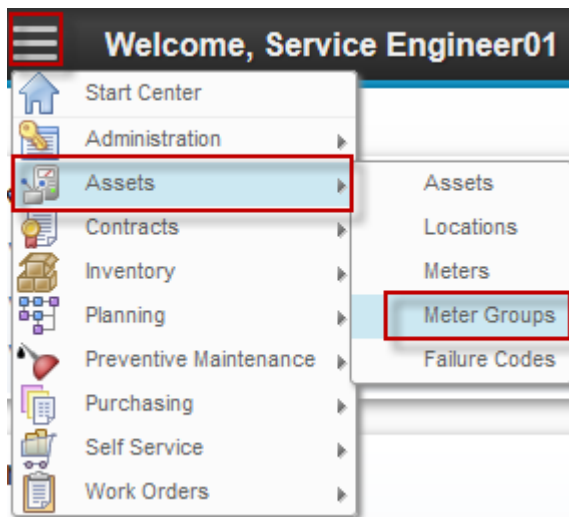


2.4.2 Create Meter Groups

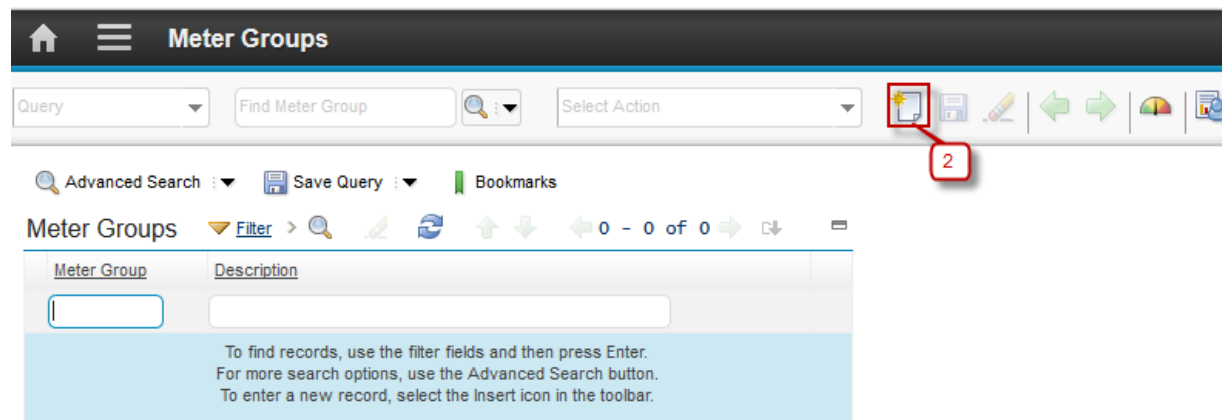
Below are step by step to Create Meter Groups by *Service Engineer*:

Step	Action
------	--------

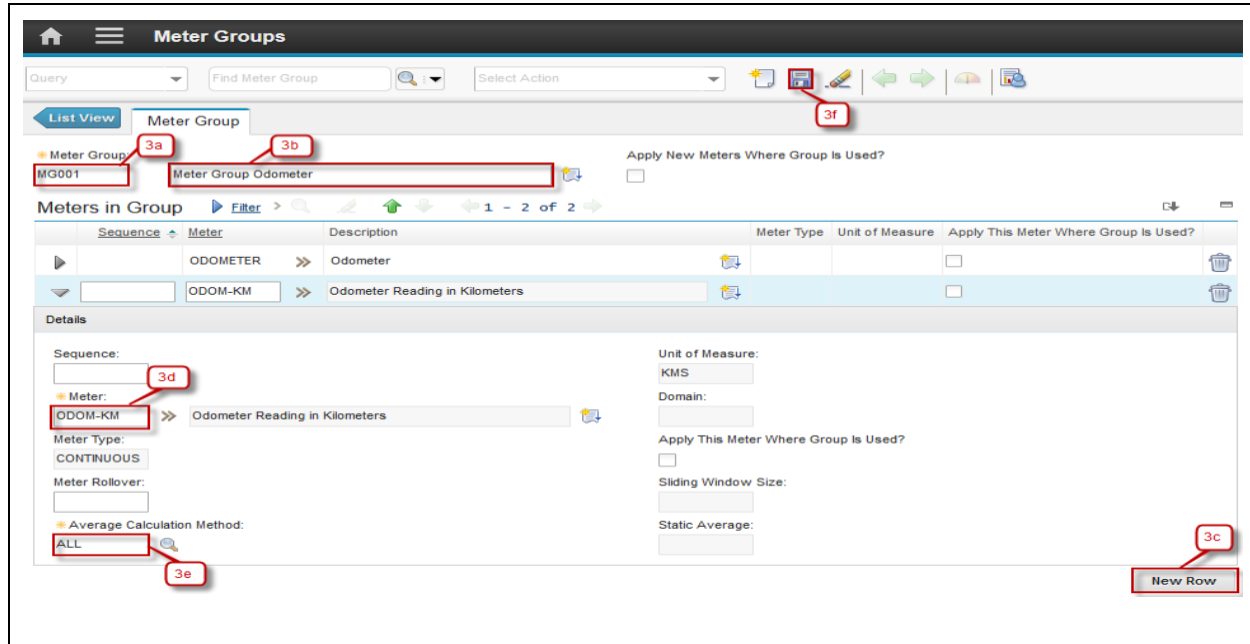
1. Go to Meters application by click  (GoTo) – Assets – Meter Groups.



2. Click  **New Row**.



- 3a. Input the Meter Group name.
- 3b. Input the Meter Group description.
- 3c. Click **New Row**.
- 3d. Select Meter using button .
- 3e. Select Average Calculation Method using button .
- 3f. Click **Save**.



The screenshot shows the 'Meter Groups' application interface. At the top, there is a navigation bar with a home icon, a menu icon, and the title 'Meter Groups'. Below this is a search bar with 'Find Meter Group' and a 'Select Action' dropdown. A toolbar contains icons for adding, editing, deleting, and other actions. The main content area is divided into two sections: 'List View' and 'Meter Group'. The 'List View' section shows a table of meter groups. The 'Meter Group' section shows the details for a selected meter group, 'MG001'. The details section includes fields for 'Sequence', 'Meter', 'Meter Type', 'Unit of Measure', 'Domain', 'Apply This Meter Where Group Is Used?', 'Sliding Window Size', 'Static Average', and 'Average Calculation Method'. Annotations 3a through 3f point to specific elements: 3a points to the 'Meter Group' dropdown, 3b points to the 'Meter Group Odometer' dropdown, 3c points to the 'New Row' button, 3d points to the 'Meter' dropdown, 3e points to the 'Average Calculation Method' dropdown, and 3f points to the 'Add' icon in the toolbar.

Meter Groups

Query: Find Meter Group Select Action

List View Meter Group

Meter Group MG001 Meter Group Odometer

Apply New Meters Where Group Is Used?

Meters in Group Filter 1 - 2 of 2

Sequence	Meter	Description	Meter Type	Unit of Measure	Apply This Meter Where Group Is Used?
	ODOMETER	Odometer			☐
	ODOM-KM	Odometer Reading in Kilometers			☐

Details

Sequence:

Meter: ODOM-KM Odometer Reading in Kilometers

Meter Type: CONTINUOUS

Meter Rollover:

Average Calculation Method: ALL

Unit of Measure: KMS

Domain:

Apply This Meter Where Group Is Used?

Sliding Window Size:

Static Average:

New Row

2.5 Assets Application

Asset application is used to create and store asset numbers and corresponding information, such as parent, location, vendor, up/down status and maintenance costs.

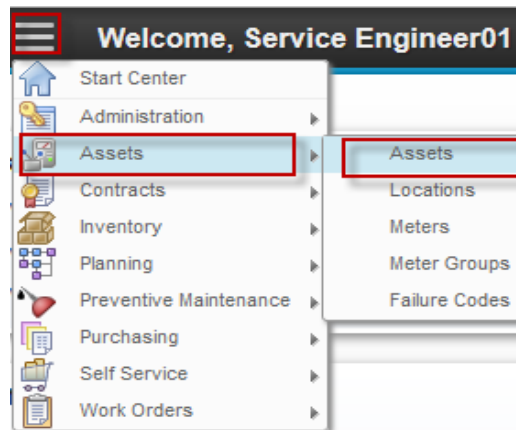
In **Asset** application user can build asset hierarchy, an arrangement of buildings, departments, assets and subassemblies. The asset hierarchy provides a convenient way to roll up maintenance costs so that user can check accumulated costs at any level, at any time. It also makes it easy to find a particular asset number.

2.5.1 Create Assets

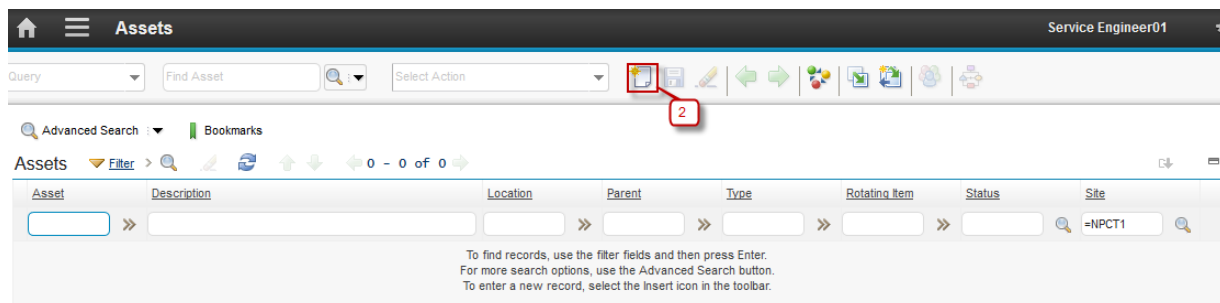
Asset creation can be done by *Service Engineer*. Below are step by step to Create Assets:

Step	Action
------	--------

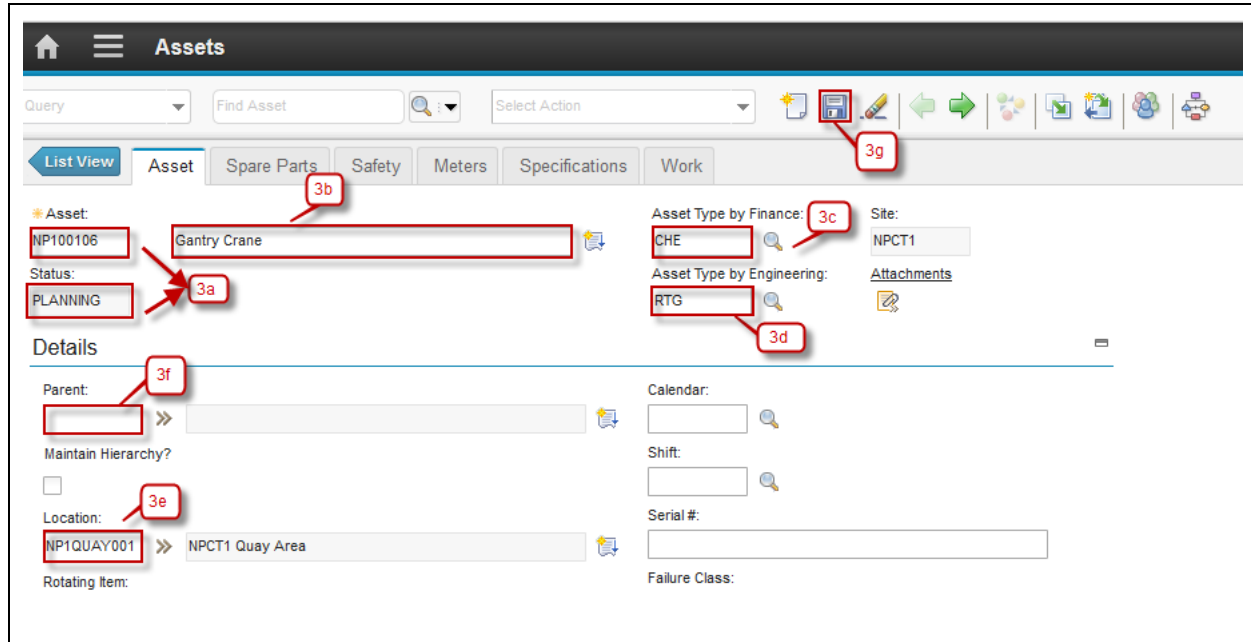
1. Go to Assets Application by click  (GoTo) – Assets – Assets.



2. Click  **New Row.**



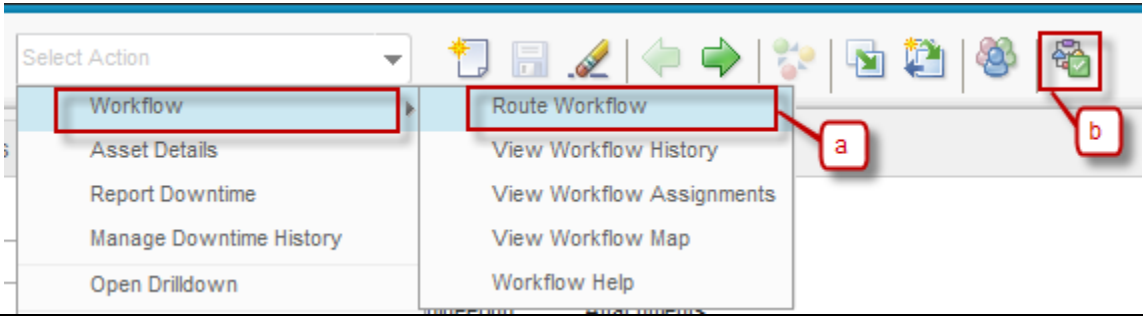
- 3a. Input the Asset name and make sure default status is 'PLANNING'.
- 3b. Input the Asset description.
- 3c. Select Asset Type by Finance using button .
- 3d. Select Asset Type by Engineering using button .
- 3e. Input Parent of Asset if the Asset have a parent.
- 3f. Input Location of Asset.
- 3g. Click **Save** and then after save the Asset button **Route Workflow** will be started automatically .



The screenshot shows the 'Assets' management interface. At the top, there's a 'Query' dropdown, a 'Find Asset' search bar, and a 'Select Action' dropdown. Below this is a tabbed interface with 'List View', 'Asset', 'Spare Parts', 'Safety', 'Meters', 'Specifications', and 'Work'. The 'Asset' tab is active. In the 'Asset' section, the 'Asset' field is 'NP100106' (callout 3a), 'Status' is 'PLANNING' (callout 3a), and 'Asset Type by Finance' is 'CHE' (callout 3c). The 'Asset Type by Engineering' is 'RTG' (callout 3d). The 'Site' is 'NPCT1'. In the 'Details' section, the 'Parent' field is empty (callout 3f), 'Maintain Hierarchy?' is unchecked, 'Location' is 'NP1QUAY001' (callout 3e), and 'NPCT1 Quay Area' is selected. The 'Rotating Item' field is empty. The 'Calendar' field is empty, 'Shift' is empty, 'Serial #' is empty, and 'Failure Class' is empty. A toolbar at the top right contains various icons, with a red box highlighting a specific icon (callout 3g).

2.5.2 Asset Creation Approval by CHE Engineer

Created Asset must be approved by *CHE Engineer*. Below are step by step to Asset Creation Approval:

Step	Action
1. Asset Creation Approval have two ways:	
a. From Select Action → Workflow → Route Workflow	
b. Click button 	
	 The screenshot shows the 'Select Action' dropdown menu. The 'Workflow' option is selected, and the 'Route Workflow' option is highlighted. Callout 'a' points to the 'Route Workflow' option, and callout 'b' points to the 'Route Workflow' button in the toolbar.
2. Click Route Workflow and dialog box will be appeared, see the figure below:	

Complete Workflow Assignment

Task:
Submit Asset NP100106 (Gantry Crane)

Action:
☒ Submit Asset
☐ Inactive

Memo:

Earlier Memos [Filter](#) >     0 - 0 of 0  

Memo	Person	Transaction Date
There are no rows to display.		

3a. Login as CHE Engineer 01.

3b. Assignment will be appeared in column **Inbox/Assignments** for approval Asset.

CHE Engineer 01 3a

Inbox / Assignments

Description	Last Memo	Start Date	Route
Next Assignment Due: 5/9/16 3:32 PM Refresh			
Need Approval for PR/ENG/10066/16/EMS (General)		2/15/16 2:07 PM	
Service Recovery 16000621 Needs to be Closed		2/23/16 9:04 PM	
Need Registration Approval for Asset NP100106 (Gantry Crane)		5/9/16 3:32 PM	 3b

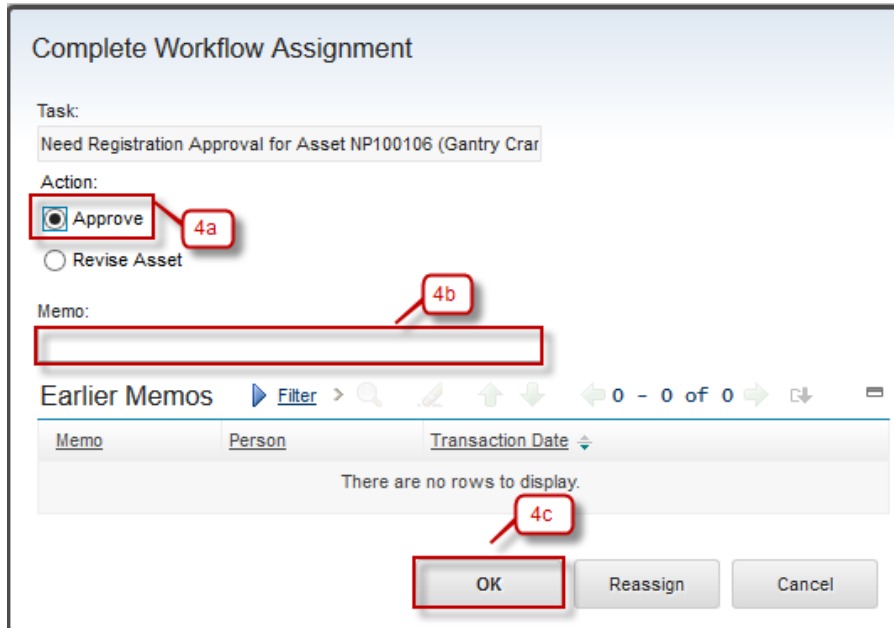
1 - 3 of 3

4. Click button **Route Workflow**  and dialog box will be appeared see the figure below:

4a. Select Action Approve

4b. Input Memo (if needed)

4c. Click **OK**

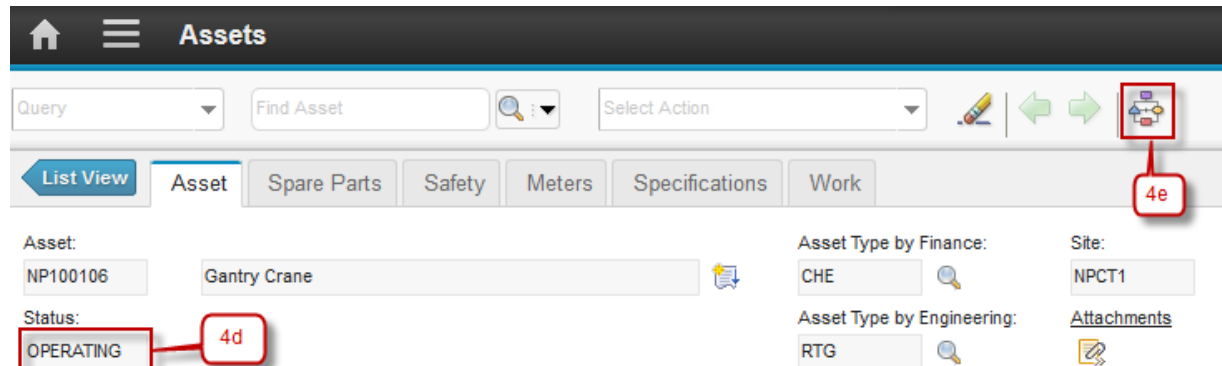


The dialog box titled "Complete Workflow Assignment" contains the following elements:

- Task:** Need Registration Approval for Asset NP100106 (Gantry Crar)
- Action:** Two radio buttons are present: "Approve" (selected, labeled 4a) and "Revise Asset".
- Memo:** A text input field (labeled 4b) is currently empty.
- Earlier Memos:** A table with columns "Memo", "Person", and "Transaction Date". It displays the message "There are no rows to display." (labeled 4c).
- Buttons:** "OK" (labeled 4c), "Reassign", and "Cancel".

4d. Status changed to 'OPERATING'.

4e. Workflow stopped.

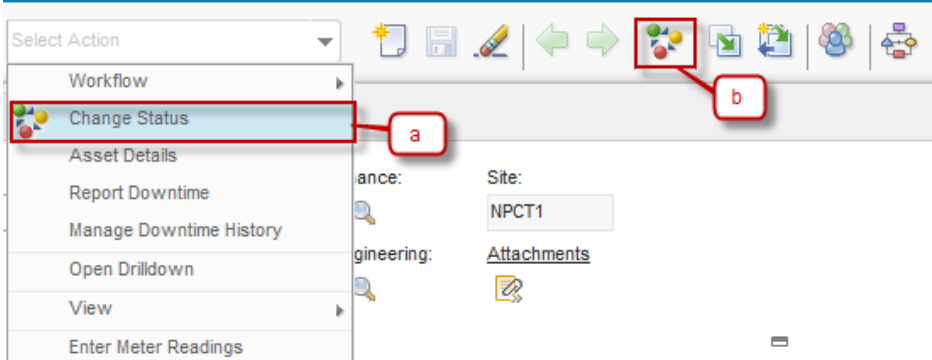
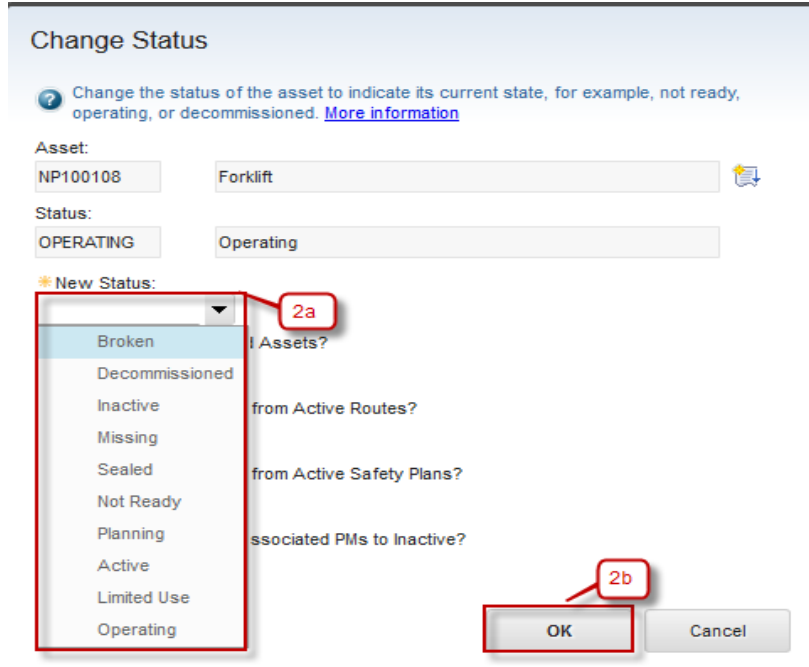


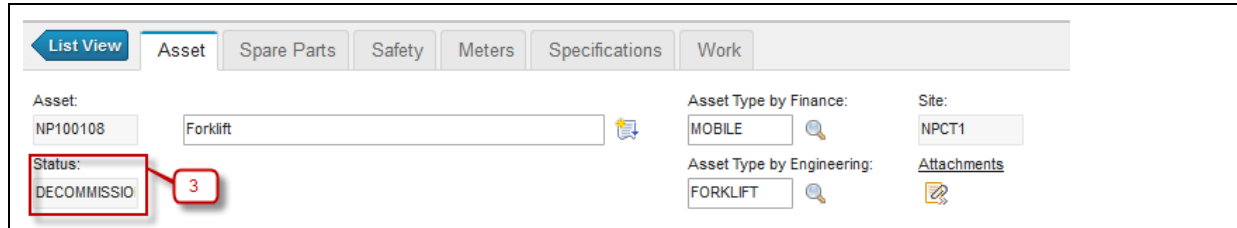
The "Assets" management interface shows the following details:

- Asset:** NP100106, Gantry Crane
- Status:** OPERATING (labeled 4d)
- Asset Type by Finance:** CHE
- Asset Type by Engineering:** RTG
- Site:** NPCT1
- Attachments:** A link to view attachments.
- Workflow:** A workflow icon in the top right (labeled 4e) indicates the workflow has stopped.

2.5.3 Change Status of an Asset

Sometimes, there is a needs to change the status of an asset. For example, when an asset is broken or decommissioned, user should change the status, to indicate that the asset is now in broken or decommissioned status. Below are step by step to Change Status of an Asset by *Service Engineer*:

Step	Action
1. Change Status of Person have two ways: a. From Select Action → Change Status b. Click button 	
2. Dialog box Change Status will be appeared, see the figure below : 2a. Select New Status of Asset. 2b. Click OK .	
3. Status changed to DECOMMISSIONED	



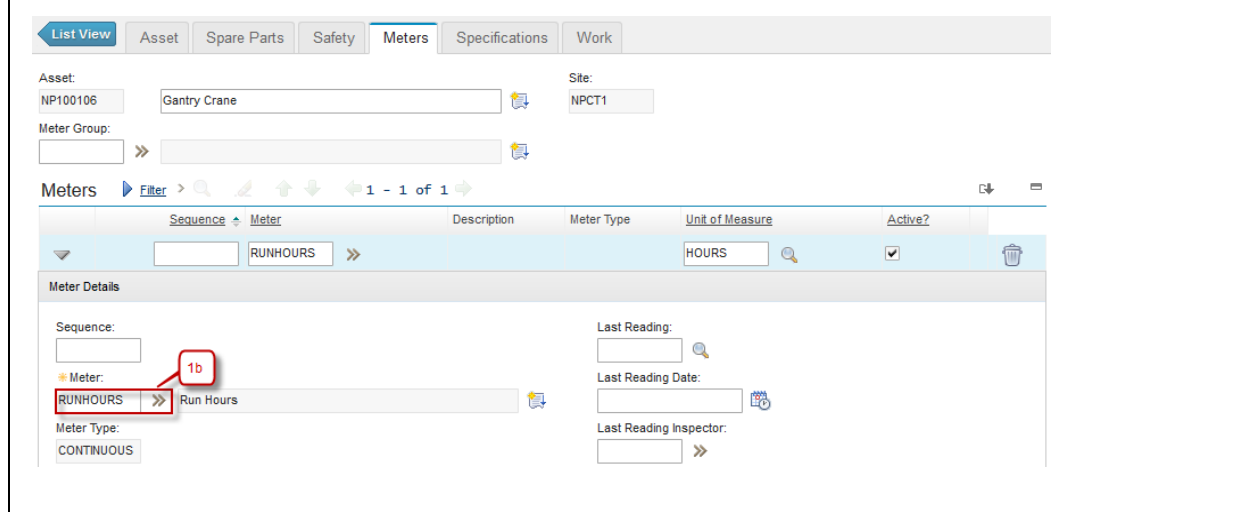
2.5.4 Assets Meters Registration & Meter Readings

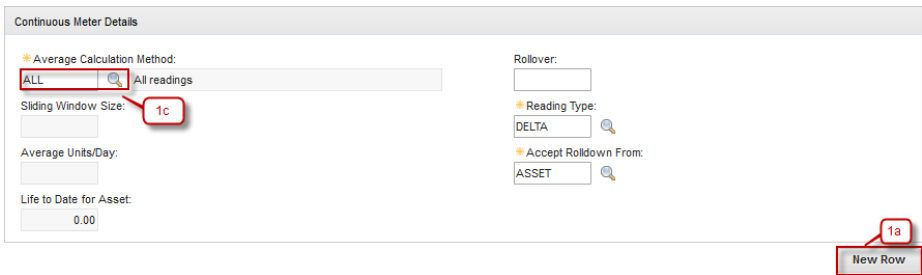
User can register all meters that apply to an asset in **Assets** application. Meters registration in an asset can be performed by registering a meter group that will automatically register all meters in meter group to the asset, or manually registering each meter to the asset.

In **Asset** application, user also can perform **Enter Meter Reading** action to report meter readings for the selected asset. This meter reading will be used as a measurement about the assets performance based the meters. This meter reading can be used to determine when a maintenance toward those asset should be performed. As additional notes, Enter Meter Reading action can be performed in **Work Order Tracking** application too (This application will be discussed later).

Below are step by step to register a meter to an asset, and performs **Enter Meter Reading** action, by *Service Engineer*:

Step	Action
1. Select one of Asset, go to Tab Meters .	
1a. Click New Row .	
1b. Select the Meter to use in Asset using button >> , field Meter Type and field Unit of Measure automatically fill.	
1c. If the type of Asset is 'CONTINUOUS' , field Average Calculation Method is mandatory.	





Continuous Meter Details

Average Calculation Method: **ALL** (1c) All readings

Sliding Window Size:

Average Units/Day:

Life to Date for Asset:

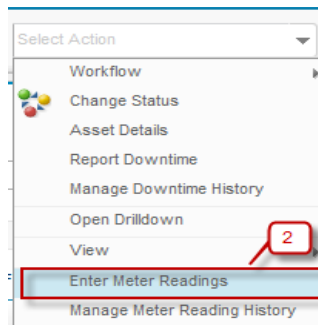
Rollover:

Reading Type: **DELTA**

Accept Rollover From: **ASSET**

1a New Row

2. To perform enter meter reading action, choose **Select Action** → **Enter Meter Readings**.



Select Action

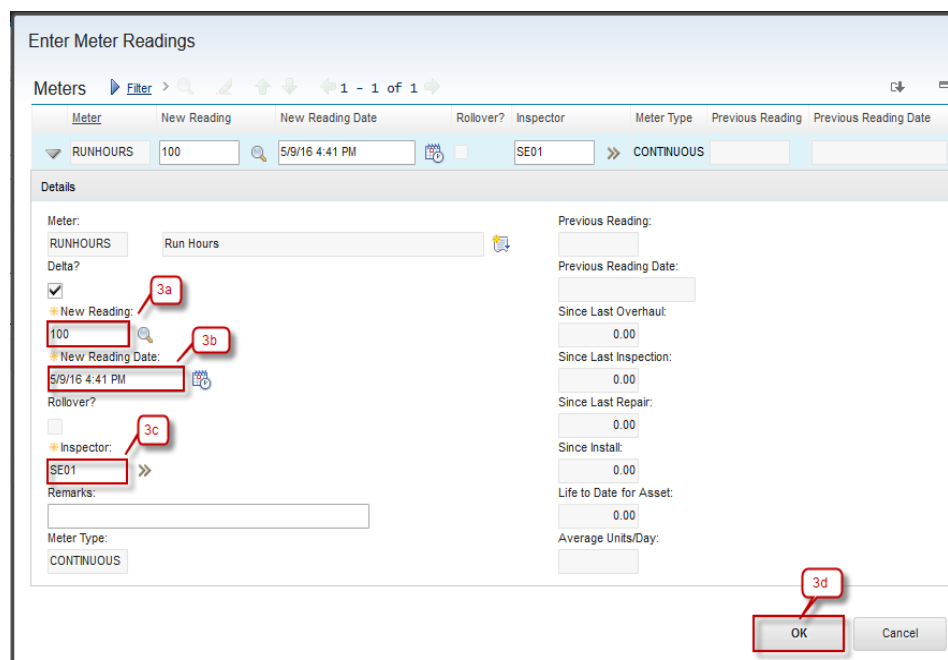
- Workflow
- Change Status
- Asset Details
- Report Downtime
- Manage Downtime History
- Open Drilldown
- View
- Enter Meter Readings** (2)
- Manage Meter Reading History

3a. Input value New Reading.

3b. New Reading Date automatically fill when input New Reading.

3c. Inspector automatically fill with user login.

3d. Click **OK**.



Enter Meter Readings

Meters **Filter** > 1 - 1 of 1

Meter	New Reading	New Reading Date	Rollover?	Inspector	Meter Type	Previous Reading	Previous Reading Date
RUNHOURS	100	5/9/16 4:41 PM	☐	SE01	CONTINUOUS		

Details

Meter: RUNHOURS Run Hours

Delta? ☒ (3a)

New Reading: 100 (3b)

New Reading Date: 5/9/16 4:41 PM (3b)

Rollover? ☐

Inspector: SE01 (3c)

Remarks:

Meter Type: CONTINUOUS

Previous Reading:

Previous Reading Date:

Since Last Overhaul: 0.00

Since Last Inspection: 0.00

Since Last Repair: 0.00

Since Install: 0.00

Life to Date for Asset: 0.00

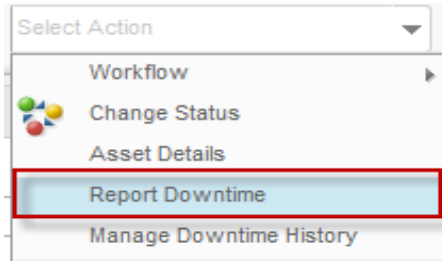
Average Units/Day:

3d OK Cancel

2.5.5 Downtime

Downtime is another feature that available in an asset. This feature can be used to determine and track how long an asset has been out of service. With this feature, user can be informed about how effective an asset is. A field of total accumulated downtime that ever happens to the asset is available in **Asset** application too.

Downtime is reported in **Assets** application using **Report Downtime** action. In this action, user should define how long a downtime occurs to the asset, the start time, and the end time. As a note, this report downtime feature also available in **Work Order Tracking** application. Below are step by step to report downtime of an asset:

Step	Action
1. Select one of Asset. Select Action → Report Downtime	

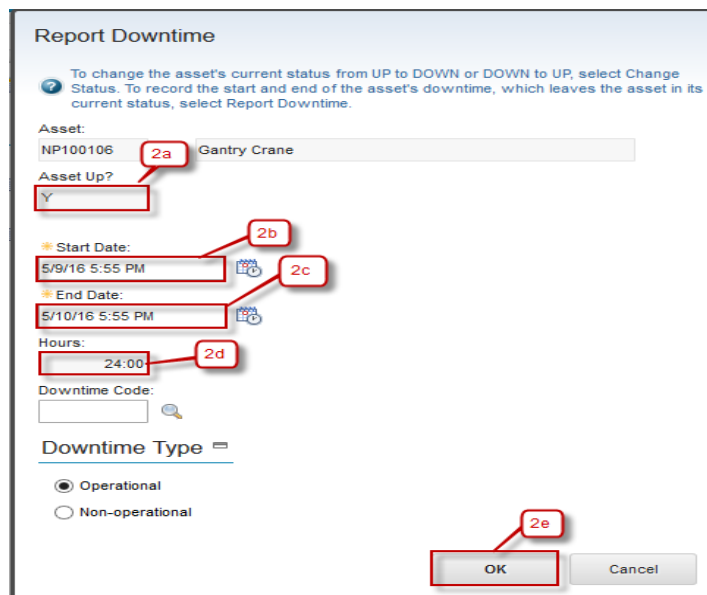
2a. Make sure default **Asset Up?** is **Y**.

2b. Start Date usually automatically fill with current date, but it's editable.

2c. Input End Date until when the asset will be up again.

2d. Value of Hours is interval between Start Date and End Date.

2e. Click **OK**.



Report Downtime

To change the asset's current status from UP to DOWN or DOWN to UP, select Change Status. To record the start and end of the asset's downtime, which leaves the asset in its current status, select Report Downtime.

Asset: NP100106 **2a** Gantry Crane

Asset Up? ☒ Y

Start Date: 5/9/16 5:55 PM **2b**

End Date: 5/10/16 5:55 PM **2c**

Hours: 24:00 **2d**

Downtime Code:

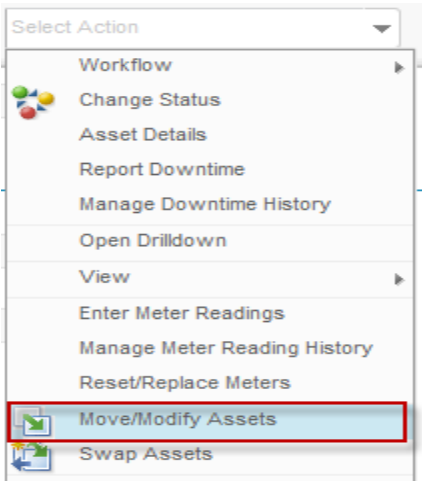
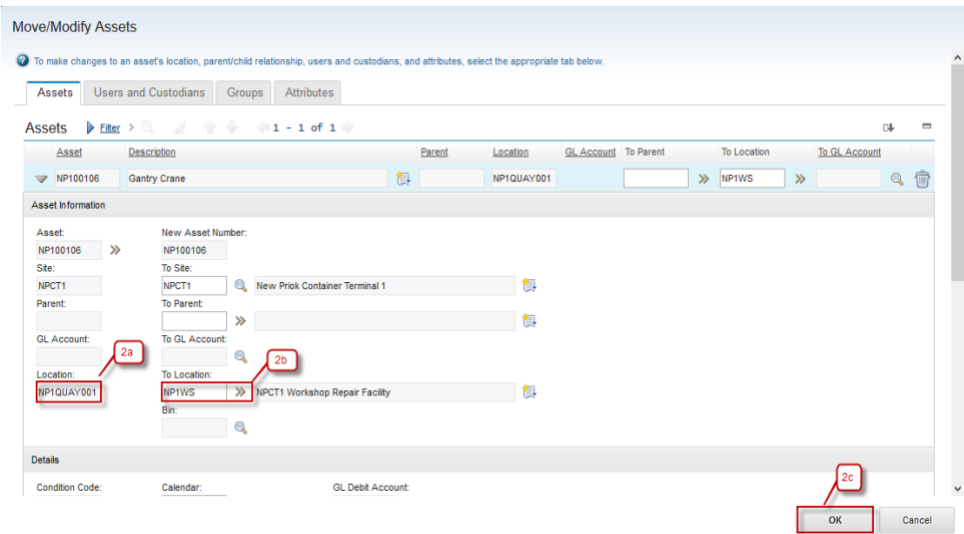
Downtime Type 

☒ Operational
☐ Non-operational

2e OK Cancel

2.5.6 Move Assets

When an asset need to be moved from a location to another location, for example, from operating location to repair location, user can do that from **Asset** application via **Move/Modify Assets** action. Besides locations, parent asset and condition code (for rotating asset) can be changed through this action too. Below are step by step to move location of an asset:

Step	Action
1. Select one of Asset. Select Action → Move/Modify Assets	
2a. Current Location before being moved. 2b. Field To Location where the Asset move. 2c. Click OK . 2d. Asset already move to New Location.	

List View

Asset

Spare Parts

Safety

Meters

Specifications

Work

Asset:

NP100106

Gantry Crane

Asset Type by Finance:

CHE

Site:

NPCT1

Status:

OPERATING

Asset Type by Engineering:

RTG

Attachments

Details

Parent:

>>

Calendar:

Maintain Hierarchy?

☐

Shift:

Location:

NP1WS

>>

NPCT1 Workshop Repair Facility

Serial #:

Rotating Item:

>>

Failure Class:

>>

Meter Group:

>>

2.6 Exercise Phase

3 Work Order

3.1 Work Order Overview

Work Orders are used to identify, plan, schedule, execute and track the work that is performed at Locations and/or on Assets throughout organization. Initially, a work order is a request for work to be performed. Once completed, a work order is a record of activities performed that make up the history of the Locations or Assets.

Work order record contains all information about a work, from work plan until work result. With a work order, a work can be tracked about its current status, all resource required, cost prediction, all resource used, and all task performed in the work.

3.2 Work Order Tracking Application

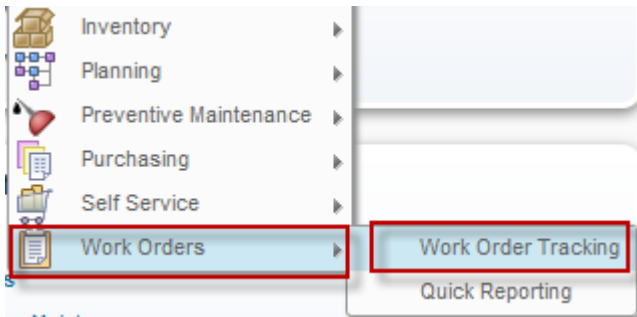
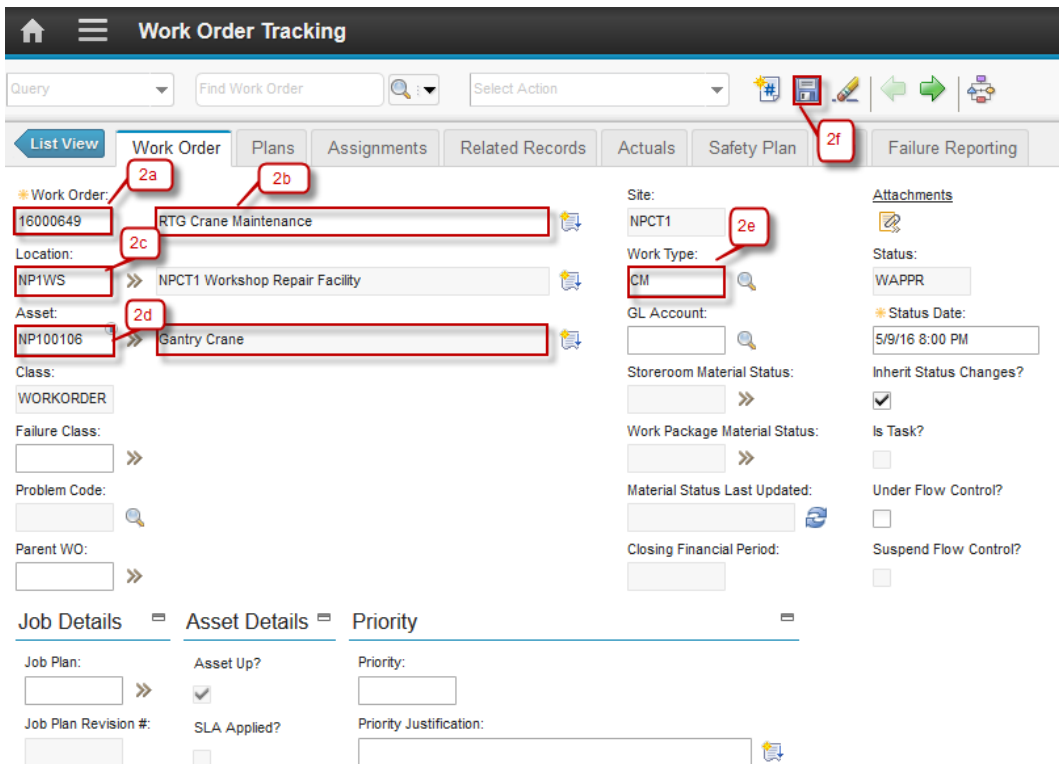
You use the **Work Order Tracking** application to plan, review and approve work orders for assets and locations. When you create a work order in Maximo, you initiate the maintenance process and create a historical record of work being performed. You can create work orders in several Maximo applications.

The **Work Order Tracking** application contains the following tabs:

- **List:** To search work orders.
- **Work Order:** To create, view and modify work order; view PM and scheduling information; see which job plan and safety plan are applied; view the origination work order for a follow-up work order; identify the failure hierarchy for the asset or location.
- **Plans:** To enter, view and modify job tasks and labor, material, services and tool requirements for the work plan.
- **Related Records:** to view, add and delete related work orders, and tickets; to view follow-up records for the current record.
- **Actuals:** to enter, view and modify actual work order start and finish times labor hours and costs, material quantities, locations, costs and tool quantities, hours and cost.
- **Safety Plan:** To enter, view and modify safety information on work order.
- **Failure Reporting:** to report asset and location failures to help identify breakdown patterns or trends.
- **Log:** to view and create work log and communication entries about the current record.

3.2.1 Materials Planning

Below are step by step to create Materials Planning:

Step	Action
1. Go to Work Orders Tracking Application by click  (GoTo) – Work Orders – Work Order Tracking.	
2a. Work Order number is autonumber. 2b. Input Work Order description. 2c. Location automatically fill when selecting asset. 2d. Select Asset. 2e. Select Work Type. 2f. Click Save .	

3. Move to Tab **Plan**. Go to Tab **Material**.

3a. Click **New Row**

3b. Select the Item and description automatically fill.

3c. Default Line Type is Item.

3d. Input Quantity, default value of Quantity is 1.00

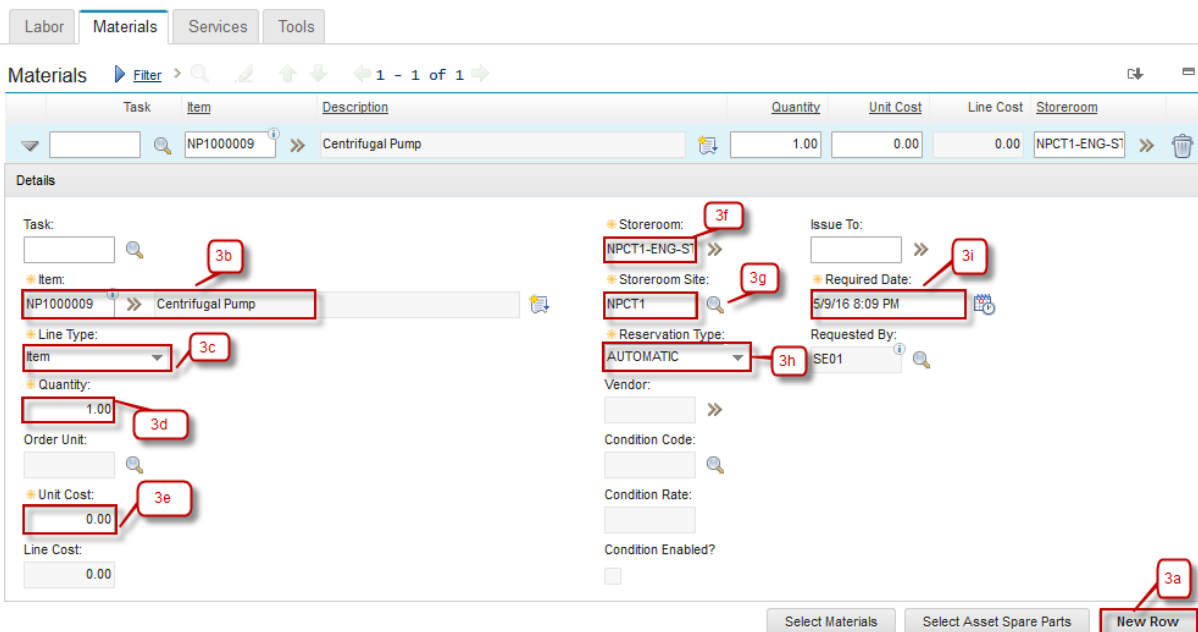
3e. Unit Cost Item, default value of unit cost is 0.00

3f. Select Storeroom where the Item is stored.

3g. Storeroom Site automatically fill when storeroom filled.

3h. Reservation Type automatically fill with default 'AUTOMATIC'.

3i. Required Date automatically with current date.

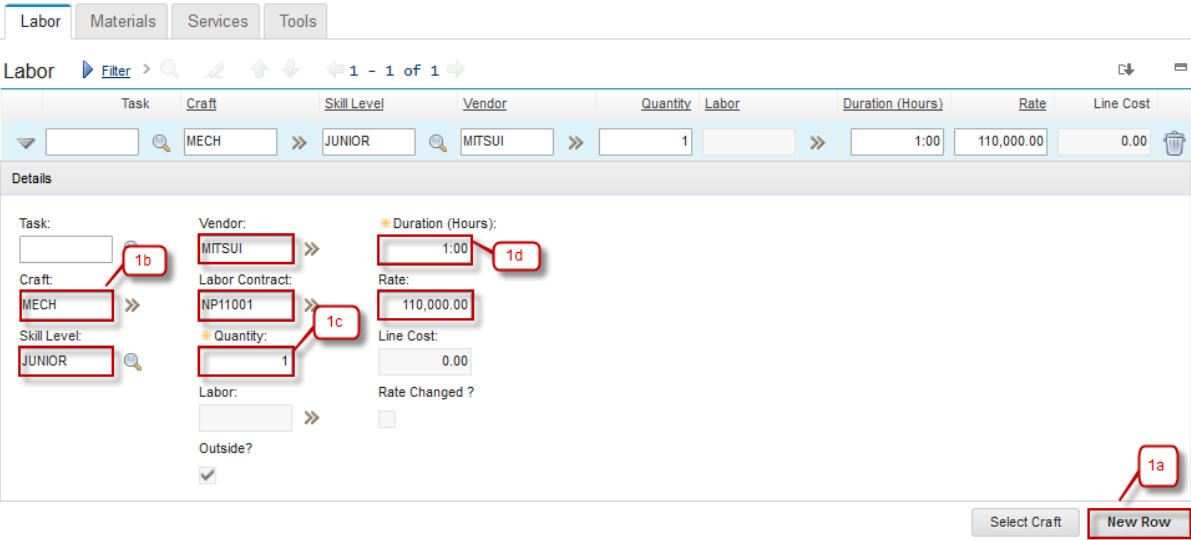


The screenshot shows the 'Materials' tab in the Agile software interface. At the top, there are tabs for 'Labor', 'Materials', 'Services', and 'Tools'. Below the tabs is a table with columns: Task, Item, Description, Quantity, Unit Cost, Line Cost, and Storeroom. The first row of the table is highlighted, showing 'NP1000009' as the Item, 'Centrifugal Pump' as the Description, '1.00' as Quantity, '0.00' as Unit Cost, '0.00' as Line Cost, and 'NPCT1-ENG-ST' as the Storeroom. Below the table is a 'Details' section with various fields. Callouts 3a through 3i point to specific fields: 3a points to the 'New Row' button at the bottom right; 3b points to the 'Task' field; 3c points to the 'Line Type' dropdown menu; 3d points to the 'Quantity' input field; 3e points to the 'Unit Cost' input field; 3f points to the 'Storeroom' dropdown menu; 3g points to the 'Storeroom Site' dropdown menu; 3h points to the 'Reservation Type' dropdown menu; and 3i points to the 'Required Date' input field. The 'Details' section also includes fields for 'Issue To', 'Requested By', 'Vendor', 'Condition Code', 'Condition Rate', and 'Condition Enabled'.

3.2.2 Labor Planning

Below are step by step to create Labor Planning:

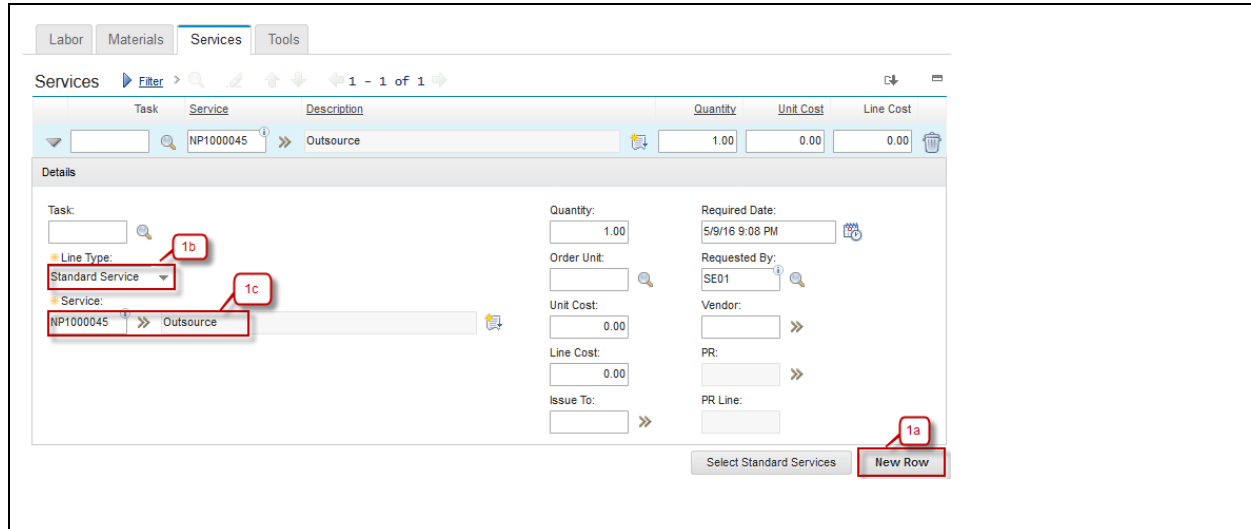
Step	Action
1. Move to Tab Plan . Go to Tab Labor .	
1a. Click New Row	
1b. If Labor from outside when choose the Craft, field Skill Level, Vendor, Labor Contract and Rate will be automatically fill in accordance with the data on the labor contract.	
1c. Input Quantity of Labor.	
1d. Input Duration of Labor.	



3.2.3 Services Planning

Below are step by step to create Services Planning:

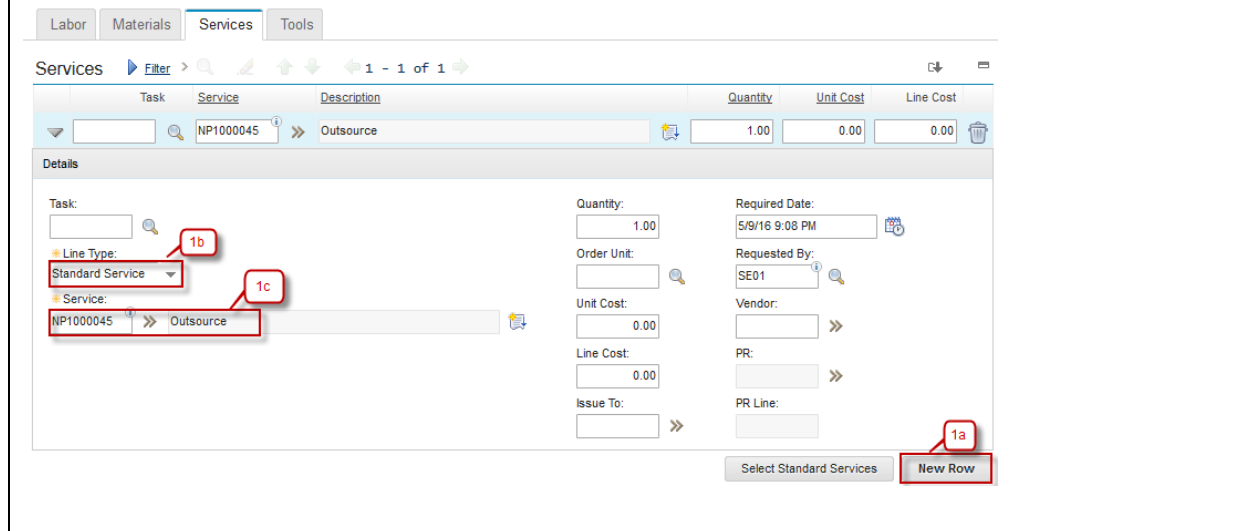
Step	Action
1. Move to Tab Plan . Go to Tab Services .	
1a. Click New Row	
1b. Select Line Type Service or Standard Service.	
- If Line Type is Service, field Description is required (mandatory). - If Line Type is Standard Service, choose Service using button »	
1c. Select Service will be used.	



3.2.4 Tools Planning

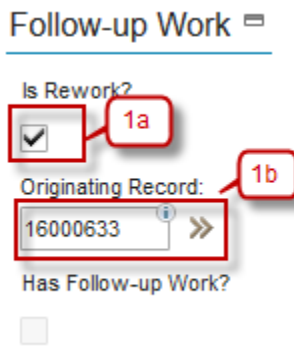
Below are step by step to create Tools Planning:

Step	Action
1. Move to Tab Plan . Go to Tab Tools .	
1a. Click New Row	
1b. Select Tool using button >>	
1c. Input Quantity of Tool.	
1d. Input Tool Hours.	
1e. Input Rate of Tool.	



3.2.5 Rework Work Order

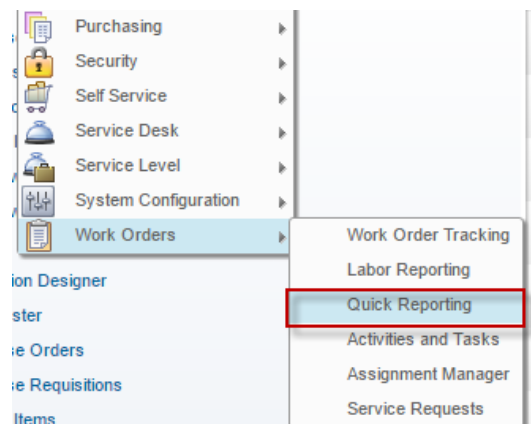
Sometimes a rework need to be performed to a work that already finished, maybe because there is an unfinished work or the result is not as expected. In EMS, the rework should be marked, to measure the maintenance performance. Below are steps to marks a rework work order:

Step	Action
1. Select one of Work Order to be set into a rework work order. 1a. Check box Is Rework checked 1b. Select the Originating Record that will be used as Work Order originator.	

3.3 Quick Reporting Application

Quick Reporting application is used to report a work that has been performed without any prior work plan. It is usually used for emergency work that should be performed immediately. So, when the work is finished, user should report the work result via **Quick Reporting** application. A work order will be created for the work result that has been reported. The different between creating a work order in **Work Order Tracking** application against **Quick Reporting** application is that work order created **Work Order Tracking** application contains planning information, while work order from **Quick Reporting** application is not.

Below is **Quick Reporting** application accessibility.

1	From Work Orders – Quick Reporting 
---	--

2 Show listed work order

Query Find Work Order Select Action

Advanced Search Save Query Bookmarks

Work Orders Filter 1 - 20 of 21

Work Order	Description	Location	Asset	Status	Scheduled Start	Priority	Site
1057	Sample Quick Reporting SR 2	NP1		COMP			NPCT1
1060	Sample Quick Reporting SR 0 Sekian	NPCT1-ENG-STORE		INPRG			NPCT1
1068	Sample Quick Reporting 01	NP1		COMP			NPCT1
1076		NP1		COMP			NPCT1
1077		NP1		COMP			NPCT1
1078	Sample SR 01	NP1		INPRG			NPCT1
1080	SR	NP1		COMP			NPCT1
1082	Sample RReport SR 03	NPCT1-ENG-STORE		COMP			NPCT1
1097	Quick Reporting SR Sample 01	NPCT1-ENG-STORE		INPRG			NPCT1
1102	Sample OR - CM 01	NPCT1		COMP			NPCT1
16000566		NP1BERTH	NP1PMMOVER	INPRG			NPCT1

2.a Filter work order by **Work Order** number, **Description**, **Location**, **Asset**, **Status**, **Scheduled Start**, **Priority** and **Site**

2.b Listed all work orders

3

Quick Reporting

Work Order: 1057

Location: NP1

Asset: New Priok Container Terminal One

Reported By: SE01

Reported Date: 1/30/16 8:31 PM

Supervisor: 3c

Status: COMP

Work Type: SR

Site: NPCT1

GL Account:

Attachments

Multiple Assets and Locations

Asset	Location	Target Description	Sequence	Mark Progress?	Site
There are no rows to display.					

Tasks

Task	Summary	Estimated Duration	Measurement Point	Measurement Value	Measurement Date	Route	Route Stop	Status
10	Sample Task 01	0:00						INPRG

Labor

Task	Labor	Name	Approved?	Start Date	Start Time	End Time	Duration (Hours)	Rate
	ELECT01	Electrical 01	✓	1/30/16			0:00	0.00
10	ELECT01	Electrical 01	✓	1/30/16			0:00	0.00

3.a **Quick Reporting** Tab, view update and detail of the work order.

3.b Information about work Order number

3.c Information about description of the work order.

3.d All Tasks listed in the work order.

3.e All Listed Labor, Material, Tools, Failure Reporting and Log in the quick reporting

4 Job Plan

4.1 Job Plan Overview

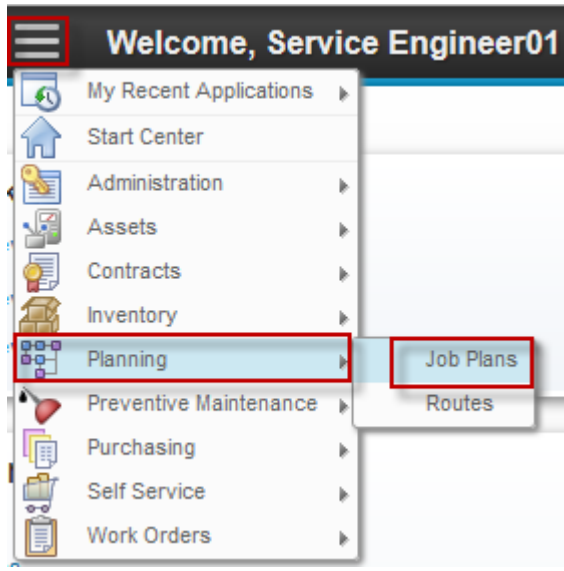
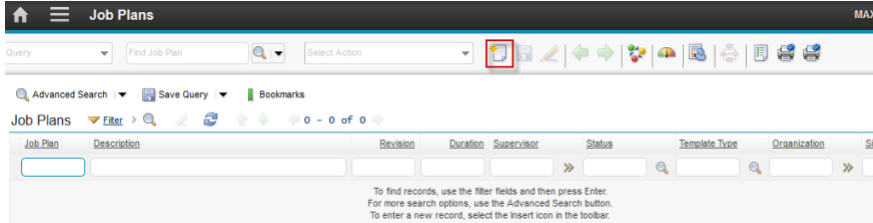
Job Plan is a work template that contains all planning information about a work. Job plans generally contain task (procedures), along with lists of estimated labor, labor hours, materials, services, and tools required for the work. Job plan can be applied to a work order, to automate planning resource and task of the work order.

Note: You also can create a job plan from a work plan that you already defined in a work order, using **Create Job Plan from Work Plan** action in **Work Order Tracking** application.

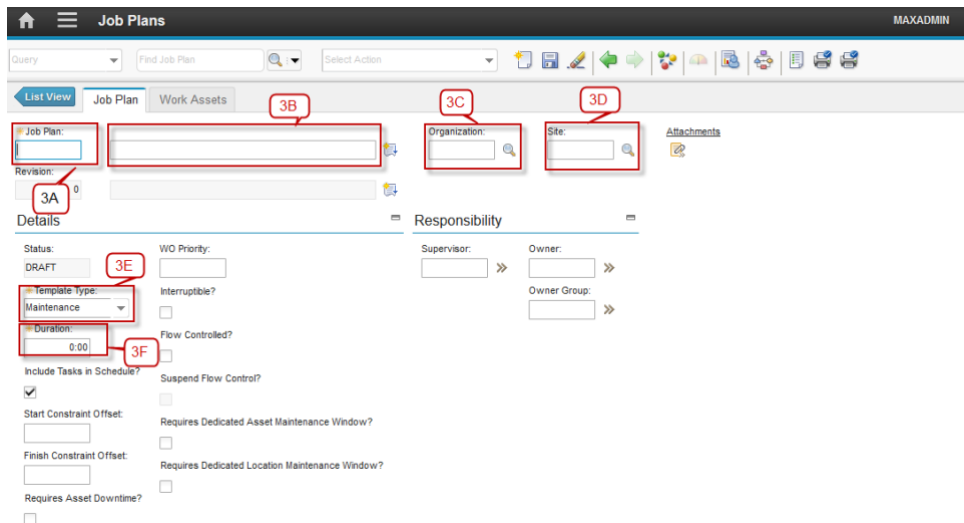
4.2 Job Plans Application

You use the **Job Plans** application to create, modify, or delete job plan records. A job plan is a detailed description of work to be performed for a work order. You can apply job plans to work orders to automate work planning definition. After a job plan becomes a work plan on a work order, you can modify the work plan without affecting the job plan.

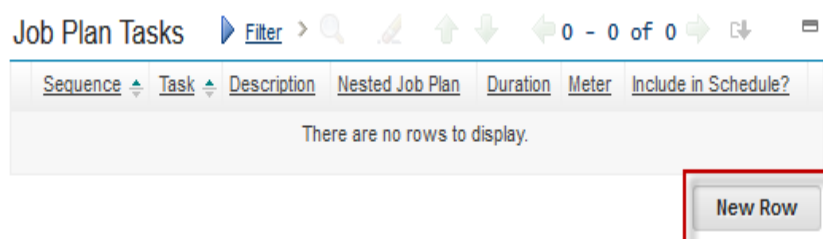
4.2.1 Create Job Plans

Step	Action
1. Go to Labor application by click  (GoTo) – Planning – Job Plans.	
2. Click  New Job Plan.	

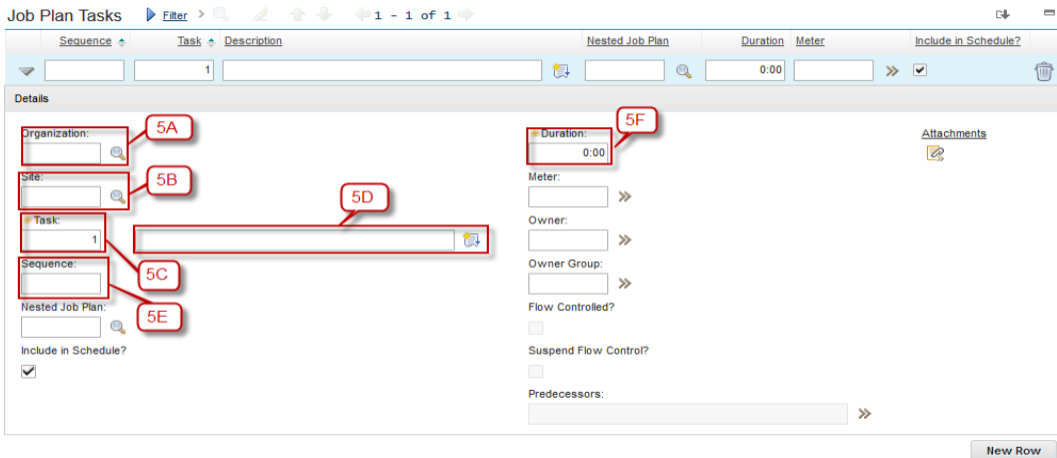
- 3A. Input Job Plan Number
- 3B. Input Description for Job Plan
- 3C. Click button  and select the Organization – COMP
- 3D. Click button  and choose the Sites – COMP1
- 3E. Select the Job Plan type, Activity / Maintenance / Process
- 3F. Input the duration of the Job Plan in hourly format



- 4. Click **New Row** on the Job Plan Tasks for creating new task



- 5A. Select the Organization with click button , then select COMP
- 5B. Choose the Site with click button , then select COMP1
- 5C. Input the number of Task
- 5D. Input the Task Description
- 5E. Input Sequence number
- 5F. Input the duration of this Task



The screenshot shows the 'Job Plan Tasks' form. Callouts point to the following fields:

- 5A: Organization dropdown
- 5B: Site dropdown
- 5C: Task dropdown
- 5D: Task description text field
- 5E: Sequence dropdown
- 5F: Duration field (0.00)

Other visible fields include: Meter, Owner, Owner Group, Flow Controlled?, Suspend Flow Control?, Predecessors, Attachments, and a 'New Row' button at the bottom right.

6A. Click on the Labor Tab

6B. Click New Row to add new data of Labor

6C. Click button , then select the Organization

6D. Click button , then select the Site

6E. Click button , then select Task

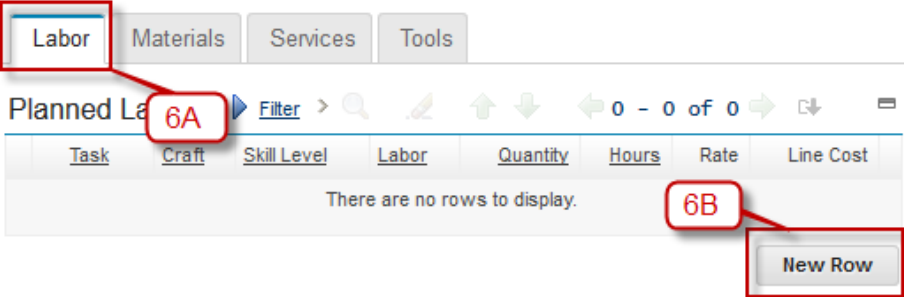
6F. Click button >> then click **"Select Value"** then select the Labor

6G. Click button >> then click **"Select Value"** then select the Craft

6H. Click button , then select the Skill Level

6I. Input Quantity of Labor

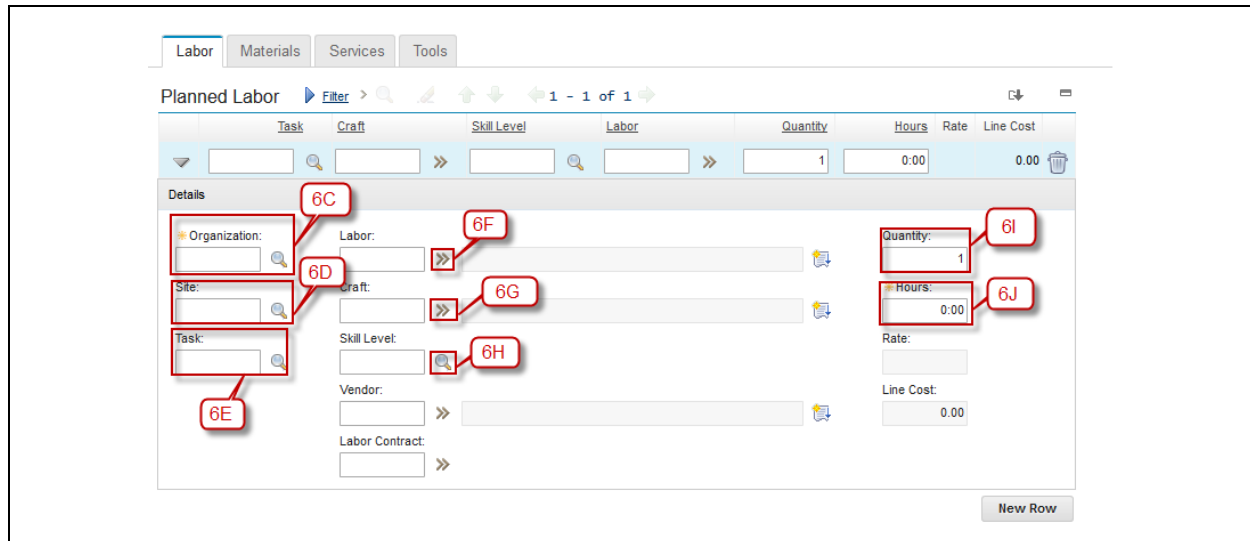
6J. Input work hours for Labor



The screenshot shows the 'Planned Labor' table. Callouts point to the following elements:

- 6A: Labor tab button
- 6B: New Row button

The table has columns: Task, Craft, Skill Level, Labor, Quantity, Hours, Rate, and Line Cost. The message 'There are no rows to display.' is shown below the table.



The screenshot shows the 'Planned Labor' form. At the top, there are tabs for 'Labor', 'Materials', 'Services', and 'Tools'. Below the tabs is a header bar with 'Planned Labor', a 'Filter' button, and navigation icons. The main form area is divided into two sections: 'Details' and a table. The 'Details' section contains fields for 'Organization' (6C), 'Site' (6D), 'Task' (6E), 'Labor' (6F), 'Craft' (6G), 'Skill Level' (6H), 'Vendor' (6I), and 'Labor Contract'. To the right of these fields are input fields for 'Quantity' (6I), 'Hours' (6J), 'Rate', and 'Line Cost'. A 'New Row' button is located at the bottom right of the form.

7A. Click on the Tab of Materials to select the Materials

7B. Click **New Row**

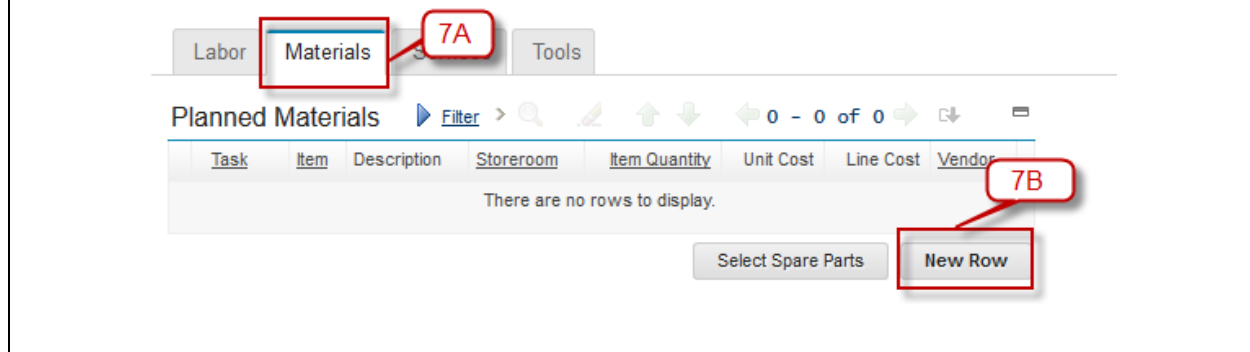
7C. Click on button  and select the Task

7D. Click button  and select the Item Set

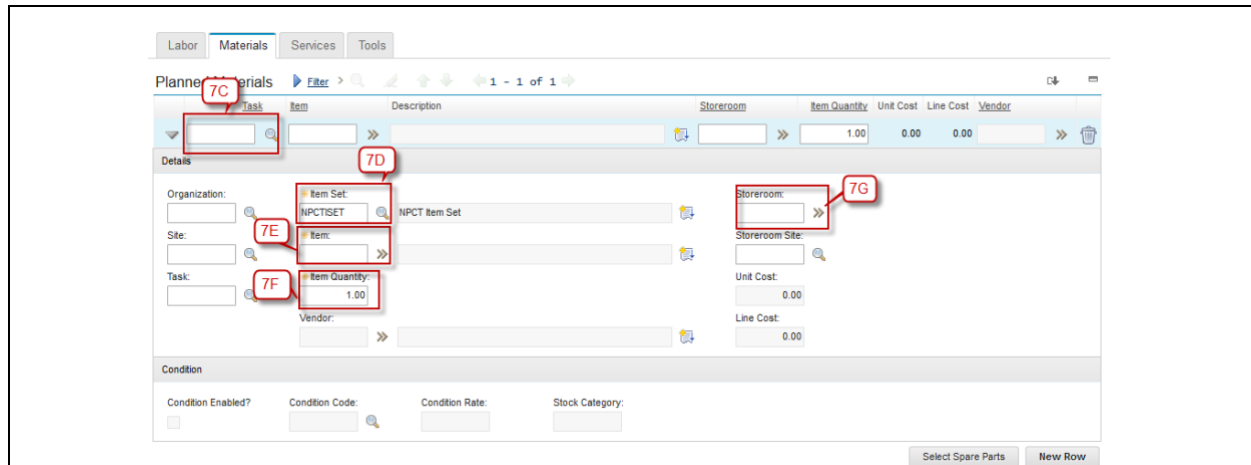
7E. Choose Item with click button >> then click magnifier icon (select icon)

7F. Then input the Quantity of Item

7G. Select the Storeroom with click >> icon



The screenshot shows the 'Planned Materials' form. At the top, there are tabs for 'Labor', 'Materials', 'Services', and 'Tools'. The 'Materials' tab is selected and highlighted with a red box and callout 7A. Below the tabs is a header bar with 'Planned Materials', a 'Filter' button, and navigation icons. The main form area contains a table with columns: 'Task', 'Item', 'Description', 'Storeroom', 'Item Quantity', 'Unit Cost', 'Line Cost', and 'Vendor'. The table is currently empty, with the text 'There are no rows to display.' below it. A 'Select Spare Parts' button is located at the bottom left, and a 'New Row' button is at the bottom right, highlighted with a red box and callout 7B.



The screenshot shows the 'Planned Materials' form. Callouts indicate the following fields and actions:

- 7C**: Task field
- 7D**: Item Set field (showing NPCTSET)
- 7E**: Item field
- 7F**: Item Quantity field (showing 1.00)
- 7G**: Storeroom field

Other visible fields include Organization, Site, Task, Vendor, Storeroom Site, Unit Cost, Line Cost, and Vendor. A 'Condition' section is at the bottom with fields for Condition Enabled?, Condition Code, Condition Rate, and Stock Category. Buttons for 'Select Spare Parts' and 'New Row' are at the bottom right.

8A. Click on the tab **Services** to create new services or see the services

8B. Click **New Row** to create new record of the services

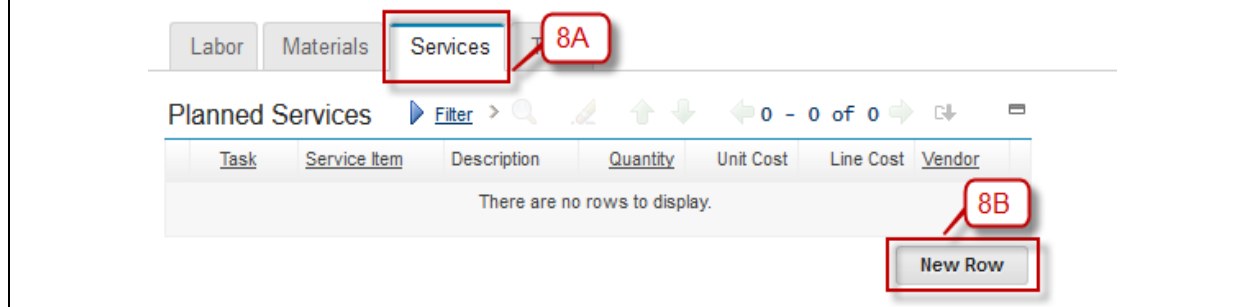
8C. Select the Task with click button 

8D. In Service Item field, you can select the Service Item already you have before. But if you want to create new Service Item, please ignore this field.

8E. Click magnifier icon to choose the Item Set, but on the default Item Set was fillin with default COMP1SET.

8F. Click >> for choose the Service Item

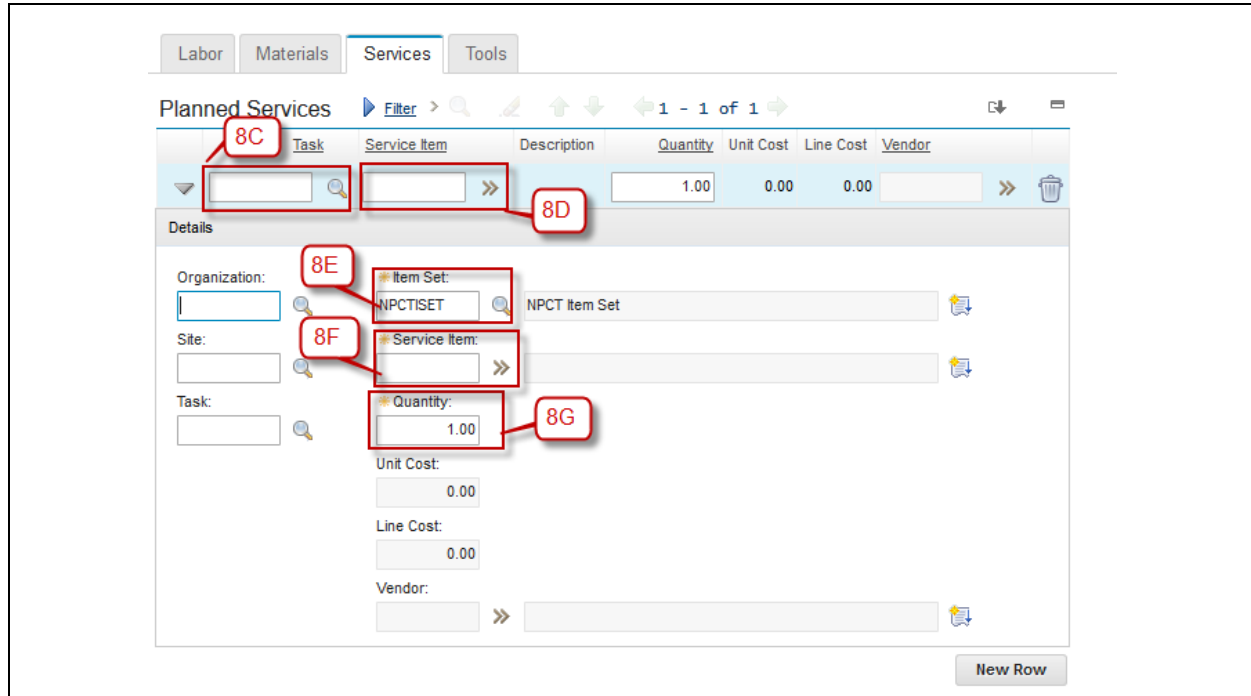
8G. Input the Quantity of Service item



The screenshot shows the 'Planned Services' form. Callouts indicate the following fields and actions:

- 8A**: Services tab
- 8B**: New Row button

The form has a table with columns: Task, Service Item, Description, Quantity, Unit Cost, Line Cost, and Vendor. The table is currently empty, displaying the message 'There are no rows to display.'



The screenshot shows the 'Planned Services' interface. At the top, there are tabs for 'Labor', 'Materials', 'Services', and 'Tools'. The 'Services' tab is active. Below the tabs, there is a table with columns: Task, Service Item, Description, Quantity, Unit Cost, Line Cost, and Vendor. A single row is visible with a quantity of 1.00. Callout 8C points to the 'Task' field, 8D to the 'Service Item' field, 8E to the 'Item Set' dropdown (showing 'NPCTSET'), 8F to the 'Service Item' dropdown, and 8G to the 'Quantity' field (showing 1.00). Below the table, there is a 'Details' section with fields for Organization, Site, Task, Item Set, Service Item, Quantity, Unit Cost, Line Cost, and Vendor. Callout 8E points to the 'Item Set' dropdown in the details section, 8F to the 'Service Item' dropdown, and 8G to the 'Quantity' field. A 'New Row' button is at the bottom right.

9A. Click on tab **Tools** to input what tools you used to finished this Job Plan.

9B. Click **New Row** to create a new record.

9C. Select Task for this tools

9D. In Tool field, you can select the Tool already you have before. But if you want to create new Tool, please ignore this field

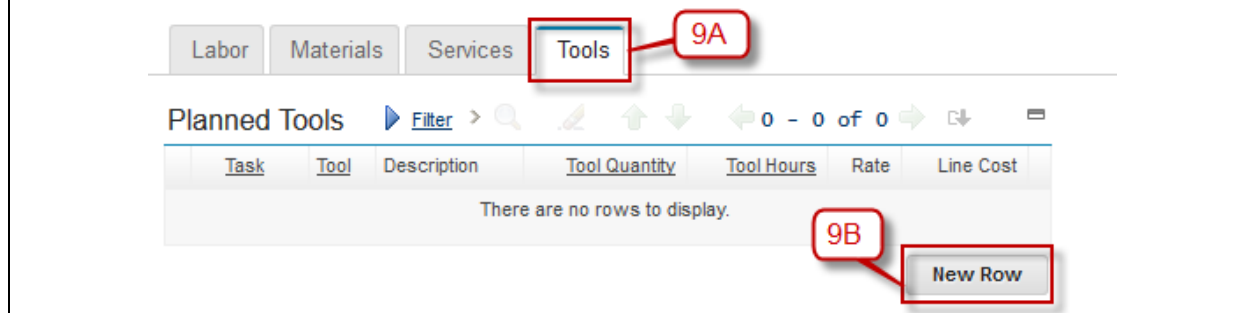
9E. Click button  to select the Item Set, but on the default Item Set was fillin with default COMP1SET

9F. Click >> icon to select and see list of tools and then choose

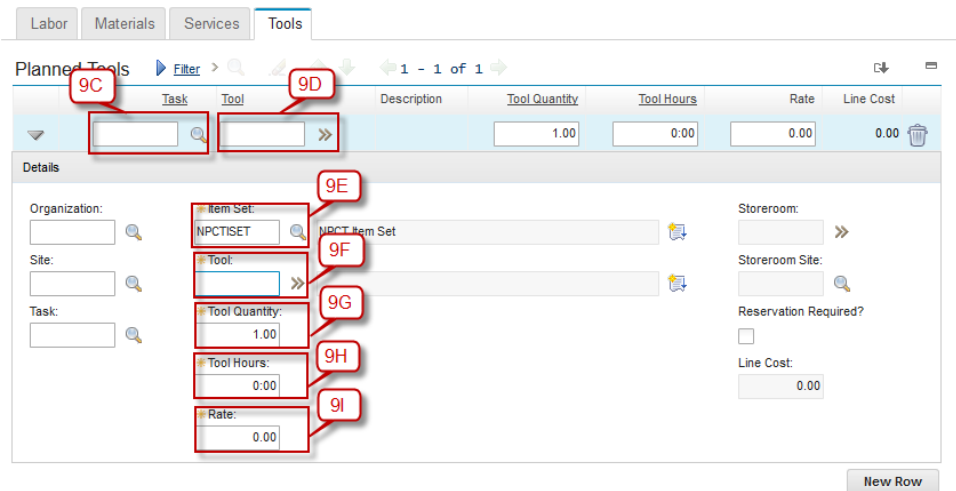
9G. Input Tool Quantity

9H. Input Total Hours for tool

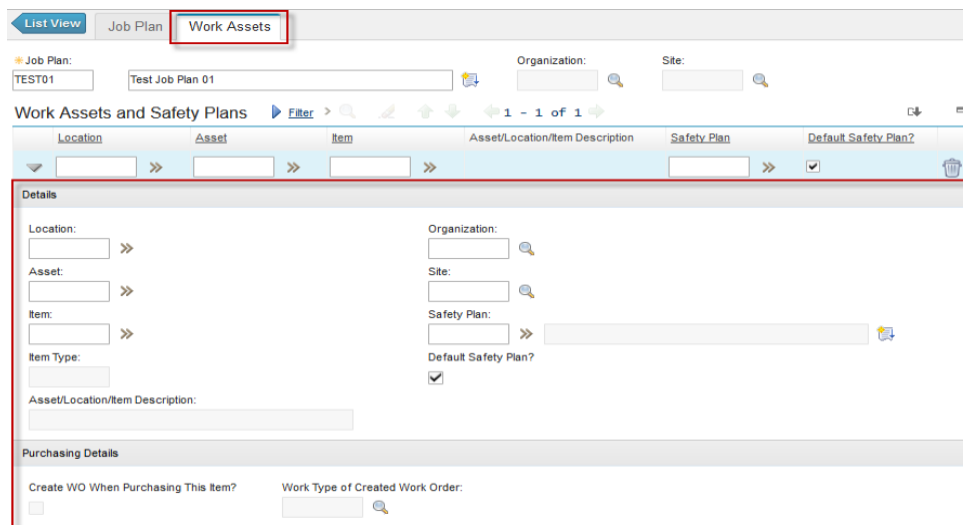
9I. Input Rate



The screenshot shows the 'Planned Tools' interface. At the top, there are tabs for 'Labor', 'Materials', 'Services', and 'Tools'. The 'Tools' tab is active. Below the tabs, there is a table with columns: Task, Tool, Description, Tool Quantity, Tool Hours, Rate, and Line Cost. The table is empty, and a message 'There are no rows to display.' is shown. Callout 9A points to the 'Tools' tab, and 9B points to the 'New Row' button at the bottom right.



10. Beside tab **Job Plan** is **Work Assets**, you can input detailed explanation for a specific asset and job plan. For the example, Job Plan TEST01 is only for ASS01.



4.2.2 Job Plan Approval

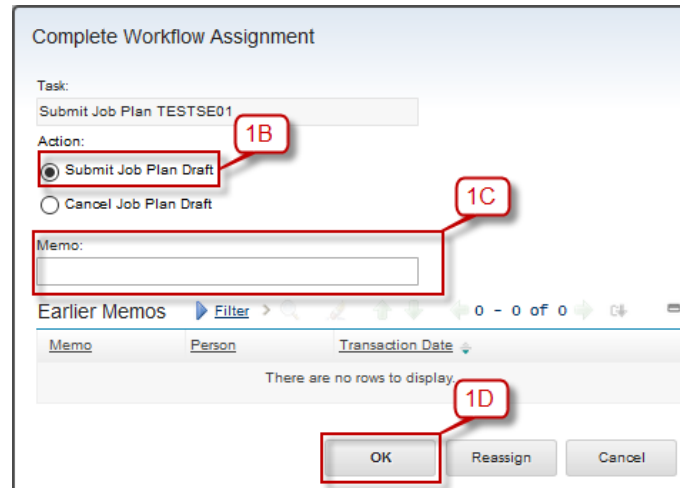
After *Service Engineer* create the Job Plan, Job Plan need to be approved by *CHE Engineer*. Below are step by step to Approval Job Plan:

1A. Click button Route Workflow  to start workflow and then after click the button changed to  its means the workflow is started.

1B. Select the option **“Submit Job Plan Draft”**

1C. Input Memo (if needed).

1D. Then click **OK** .



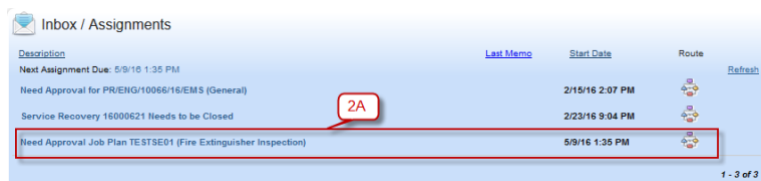
2. Login as CHE Engineer. User name : che01 and password : password.

2A. In CHE01 start center in portlet “Inbox / Assignment” CHE01 can see the Job Plan Need Approval, click on the list Need Approval. Then CHE01 will see the detail of Job Plan. After that click  to approval process

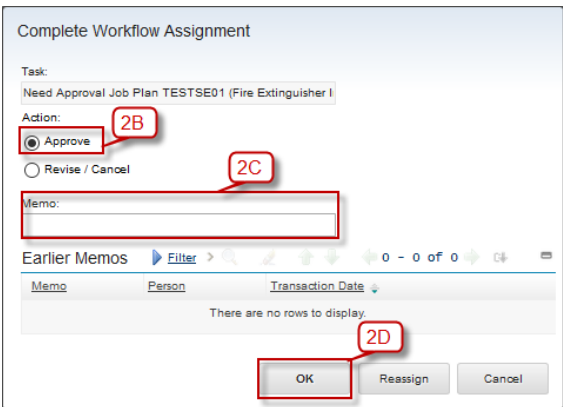
2B. Click on approve button

2C. CHE01 can give the memo if any notes for this Job Plan

2D. Click **OK**. Job Plan has been approved.



Description	Last Memo	Start Date	Route	Refresh
Need Approval Job Plan TESTSE01 (Fire Extinguisher Inspection)		5/9/16 1:35 PM		
Service Recovery 16000621 Needs to be Closed		2/23/16 9:04 PM		
Need Approval for PIRENG/10066/16/EMS (General)		2/15/16 2:07 PM		



Complete Workflow Assignment

Task:
Need Approval Job Plan TESTSE01 (Fire Extinguisher I)

Action:

☒ Approve (2B)

☐ Revise / Cancel (2C)

Memo: (2C)

Earlier Memos Filter > 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

(2D)

OK Reassign Cancel

3. CHE Engineer can **Revise / Cancel** a Job Plan was submitted by Service Engineer.

3A. To cancel Job Plan, CHE Engineer must click button **Revise / Cancel**.

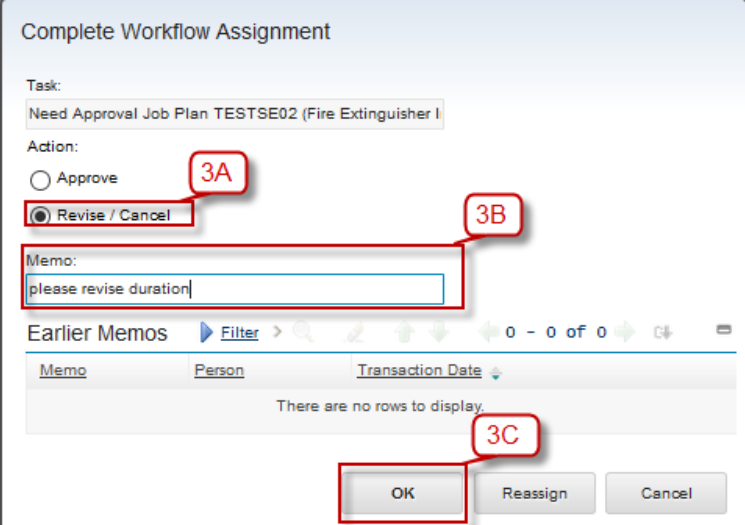
3B. CHE Engineer can give a memo

3C. Then click **OK** and dialog box manual input will be appeared.

3D. Click on **Cancel** button

3E. CHE Engineer can give a memo to cancel Job Plan

3F. Click **OK**



Complete Workflow Assignment

Task:
Need Approval Job Plan TESTSE02 (Fire Extinguisher I)

Action:

☐ Approve (3A)

☒ Revise / Cancel (3B)

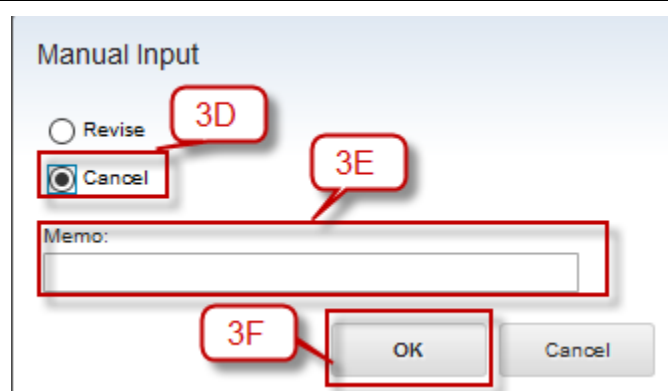
Memo: (3B)
please revise duration

Earlier Memos Filter > 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display.		

(3C)

OK Reassign Cancel



Manual Input

☐ Revise 3D

☒ Cancel 3E

Memo:

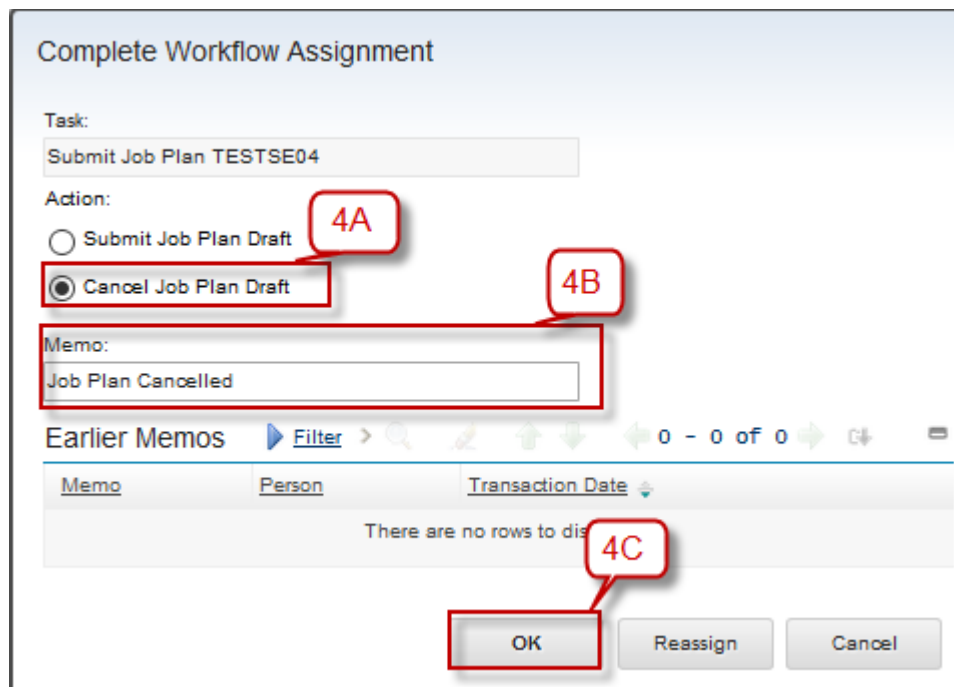
3F

4. Service Engineer can cancel Job Plan with this following step:

4A. Click button Route Workflow  to start workflow and then after click the button changed to  its means the workflow is started.

4B. Select the option “**Cancel Job Plant Draft**”

4C. Then click **OK**.



Complete Workflow Assignment

Task:
Submit Job Plan TESTSE04

Action:

☐ Submit Job Plan Draft 4A

☒ Cancel Job Plan Draft 4B

Memo:
Job Plan Cancelled

Earlier Memos Filter 0 - 0 of 0

Memo	Person	Transaction Date
There are no rows to display 4C		

4.2.3 Revise a Job Plan

Below are step by step to Revise a Job Plan:

1. Login as CHE Engineer. User name : che01 and password : password.

1A. In CHE Engineer start center in portlet “Inbox / Assignment” , CHE Engineer can see the Job Plan Need Approval, click on the list Need Approval. Then CHE Engineer will see the detail of Job Plan.

After that click  to approval process.

1B. Select **Revise / Cancel**

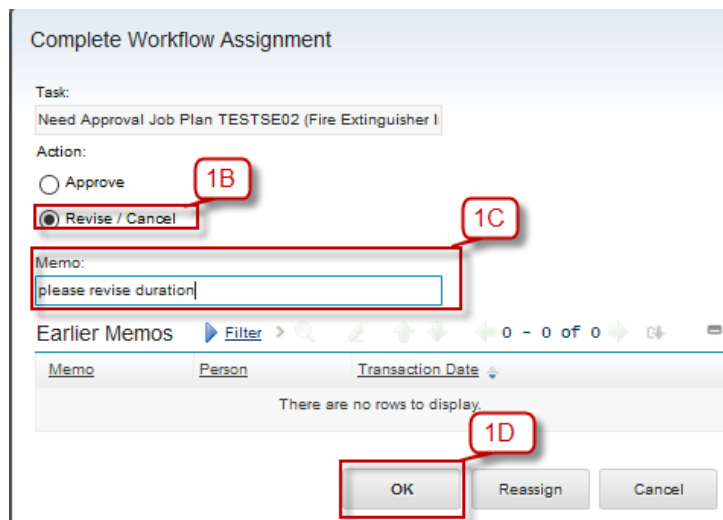
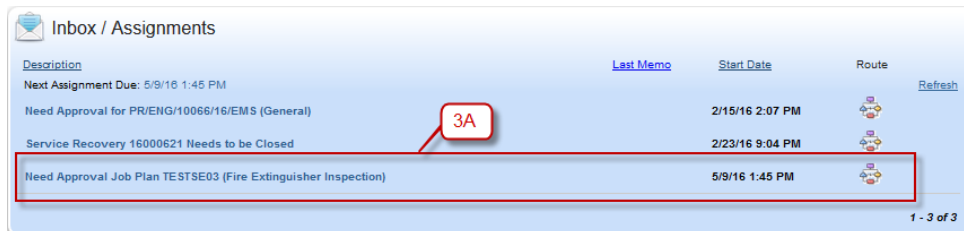
1C. CHE Engineer can give a memo

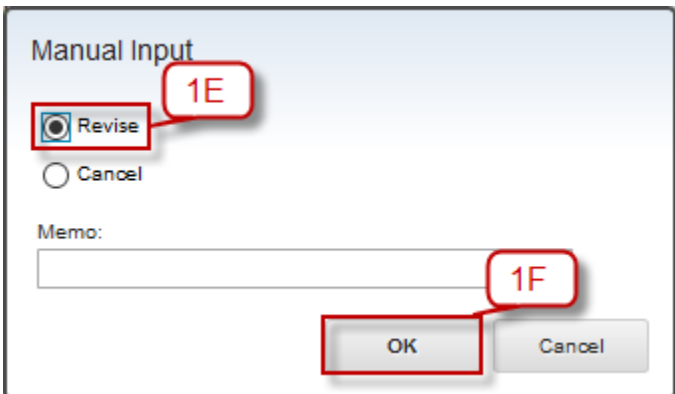
1D. Click **OK** and then dialog box manual input will be appeared.

1E. Click **Revise**

1F. Click **OK**

1G. Then the Job Plan was send back to Service Engineer to revised.





The image shows a 'Manual Input' dialog box. It contains a 'Revise' button with a camera icon, a 'Cancel' radio button, a 'Memo:' text field, and 'OK' and 'Cancel' buttons at the bottom. Red callouts '1E' and '1F' point to the 'Revise' button and the 'OK' button respectively.

2. Login as Service Engineer. User name : se01, password: password.

2A. In Start Center Service Engineer, in portlet Inbox / Assignment, Service Engineer can see the Job Plan need to Revision

2B. Click Need Revision Job Plan

2C. Click **Re-Submit Job Plan**

2D. Service Engineer can give a memo to this revision

2E. Click **OK**. Job Plan was send back to need approval to CHE Engineer.

Inbox / Assignments				
Description	Last Memo	Start Date	Route	Refresh
Next Assignment Due: 5/9/16 1:41 PM				
Special Maintenance 16000629 has to be Executed		3/3/16 9:58 AM		
Submit TESLAB02 for Approval		3/3/16 12:11 PM		
Submit LAB03 for Approval		3/3/16 3:08 PM		
Submit 16000633 for Approval		3/3/16 3:25 PM		
Submit LAB04 for Approval		3/3/16 3:29 PM		
Submit 16000643 for Approval		3/11/16 9:56 AM		
Service Recovery 16000644 Needs to be Completed		3/11/16 9:56 AM		
Need Revision Job Plan TESTSE02 (Fire Extinguisher Inspection)	please revise the Duration	5/9/16 1:41 PM		

Complete Workflow Assignment

Task:
Need Revision Job Plan TESTSE02 (Fire Extinguisher Ir

Action:

☒ Re-submit Job Plan **2C**

☐ Cancel Job Plan **2D**

Memo:

Earlier Memos [Filter](#) >  1 - 2 of 2 

Memo	Person	Transaction Date
please revise the Duration	CHE01	5/9/16 1:41 PM
please revise duration	CHE01	5/9/16 1:40 PM 2E

OK Reassign Cancel

3A. In Start Center of CHE Engineer, in portlet Inbox / Assignment, CHE Engineer can see the list to need approval CHE Engineer, click on the Need Approval Job Plan

3B. CHE Engineer will check the revision from Service Engineer

3C. After the revision is ok, then click  to approve the revision

3D. Click **Approve** button

3E. CHE Engineer can give a memo to this result of revision

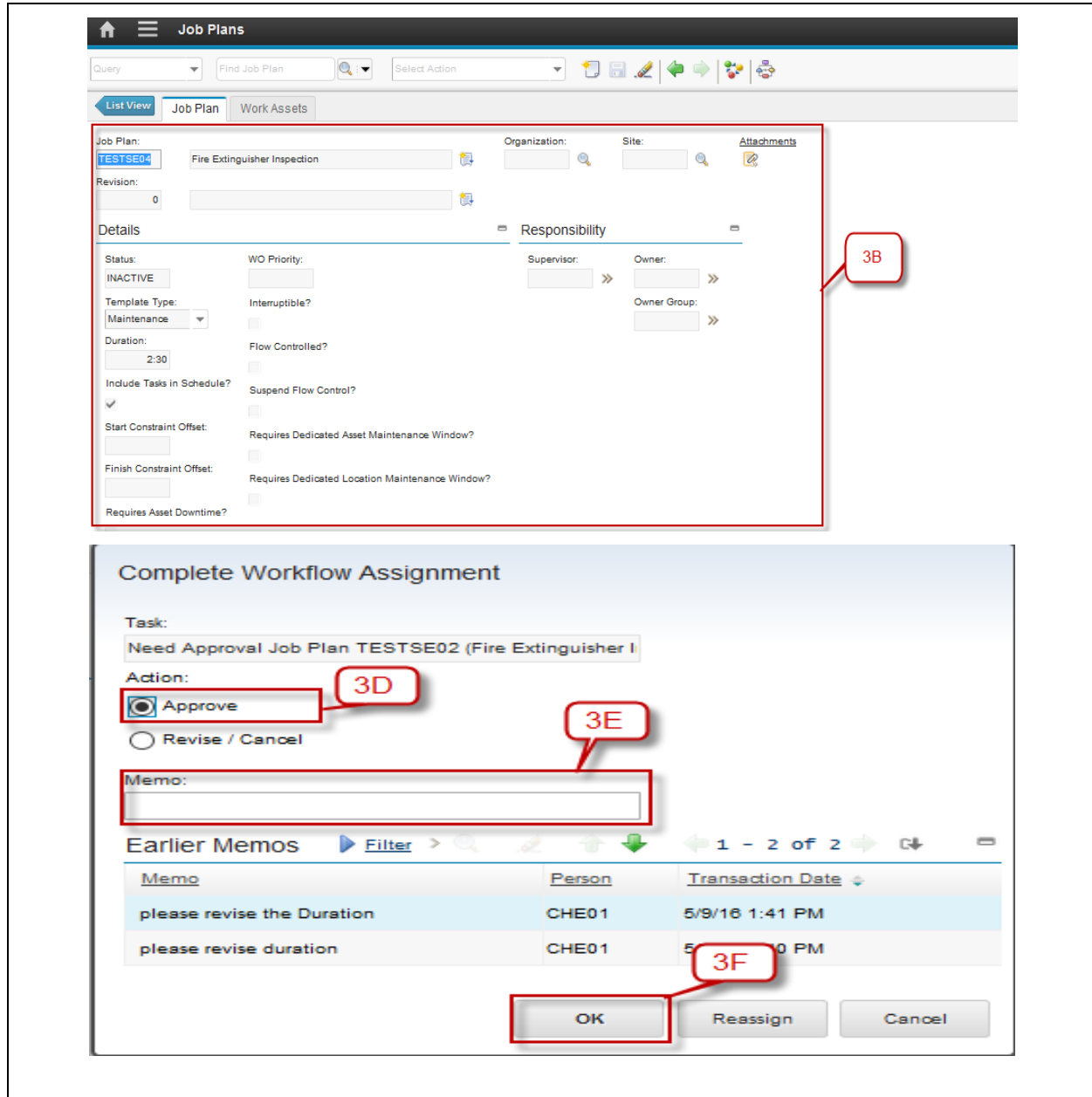
3F. Then click **OK**

 **Inbox / Assignments**

Description	Last Memo	Start Date	Route	Refresh
Next Assignment Due: 5/9/16 1:43 PM				
Need Approval for PR/ENG/10066/16/EMS (General)		2/15/16 2:07 PM		
Service Recovery 16000621 Needs to be Closed		2/23/16 9:04 PM		
Need Approval Job Plan TESTSE02 (Fire Extinguisher Inspection)	please revise the Duration	5/9/16 1:43 PM		

3A

1 - 3 of 3



The screenshot displays the 'Job Plans' interface with the 'Job Plan' tab selected. A red box labeled '3B' highlights the 'Job Plan' details section, including fields for Job Plan (TESTSE02), Fire Extinguisher Inspection, Organization, Site, Revision (0), and various checkboxes for details and responsibility. Below this, the 'Complete Workflow Assignment' dialog is shown. In this dialog, a red box labeled '3D' highlights the 'Approve' radio button, '3E' highlights the 'Memo' text area, and '3F' highlights the 'OK' button. The 'Task' field shows 'Need Approval Job Plan TESTSE02 (Fire Extinguisher I' and the 'Action' field shows 'Approve' selected. The 'Earlier Memos' table lists two entries: 'please revise the Duration' and 'please revise duration', both by user CHE01 on 5/9/16.

Memo	Person	Transaction Date
please revise the Duration	CHE01	5/9/16 1:41 PM
please revise duration	CHE01	5/9/16 1:41 PM

5 Preventive Maintenance

5.1 Preventive Maintenance Overview

Preventive maintenance is a maintenance schedule that is defined to plan a routine maintenance toward assets. Preventive maintenance schedule can be set based on time, or continuous meter that's the asset have.

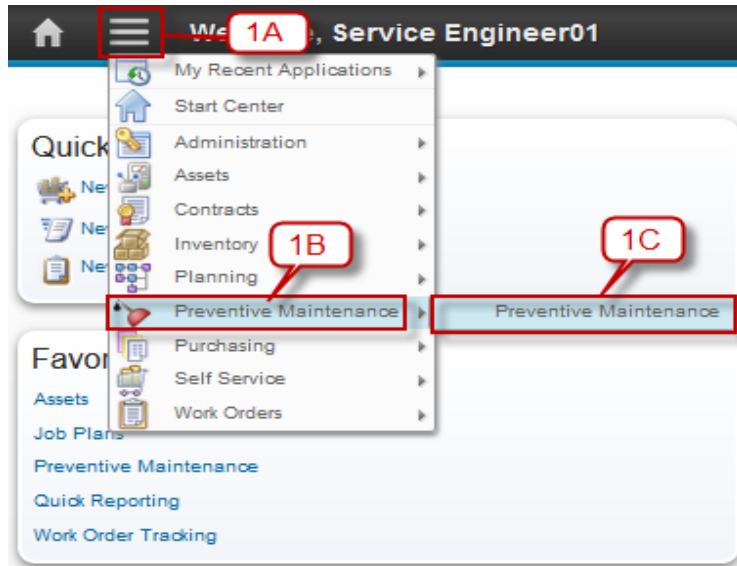
5.2 Preventive Maintenance Application

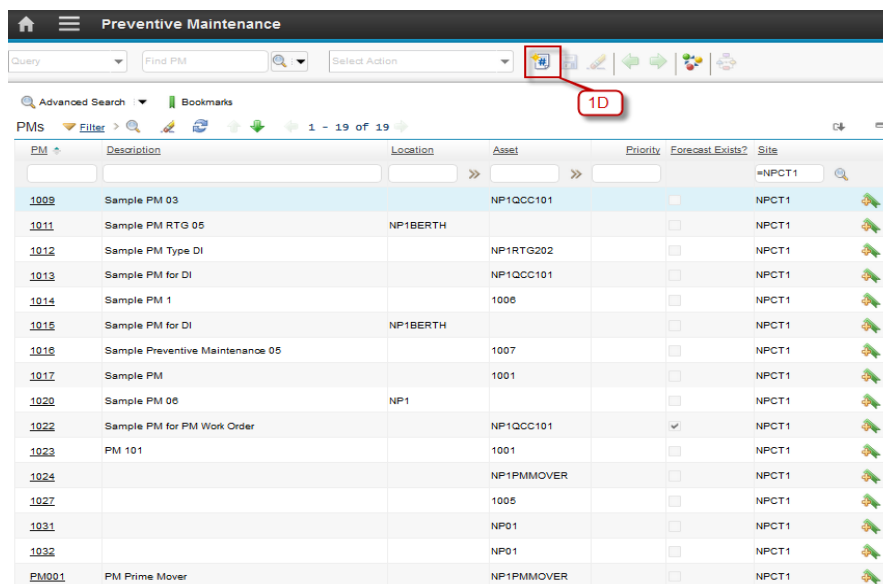
You use the **Preventive Maintenance** application to create, modify, and view preventive maintenance plans for work assets and locations. PM records are templates for work orders. On a PM record, you schedule job plans to be performed for preventive maintenance work. Maximo then generates work orders from the PM.

5.2.1 Time Based Preventive Maintenance

Below are step by step to create Time Based Preventive Maintenance:

Step	Action
1. Login as Service Engineer. User name : se01, password: password.	
1A. Click in icon menu ()	
1B. Click Preventive Maintenance	
1C. Click Preventive Maintenance	
1D. Click icon New ()	





PM	Description	Location	Asset	Priority	Forecast Exists?	Site
1009	Sample PM 03		NP1QCC101			NPCT1
1011	Sample PM RTG 05	NP1BERTH				NPCT1
1012	Sample PM Type DI		NP1RTG202			NPCT1
1013	Sample PM for DI		NP1QCC101			NPCT1
1014	Sample PM 1		1006			NPCT1
1015	Sample PM for DI	NP1BERTH				NPCT1
1016	Sample Preventive Maintenance 05		1007			NPCT1
1017	Sample PM		1001			NPCT1
1020	Sample PM 06	NP1				NPCT1
1022	Sample PM for PM Work Order		NP1QCC101			NPCT1
1023	PM 101		1001			NPCT1
1024			NP1PMMOVER			NPCT1
1027			1005			NPCT1
1031			NP01			NPCT1
1032			NP01			NPCT1
PM001	PM Prime Mover		NP1PMMOVER			NPCT1

2A. PM number is autonumber. Define description for Preventive Maintenance.

2B. Define Location or Asset or Route

2C. Click >> in **Job Plan** field, and select one Job Plan used for this Preventive Maintenance

2D. Click  in **Work Type** field and select **DI (Daily Inspection)**

2E. Click  in **Work Order Status** field and select **WEEXEC (Waiting Execute)**.

- If **Work Type** is **PM (Preventive Maintenance)** status in **Work Order Status** must be **APPR (Approve)**

2F. Click  **Save**.

2G. Input detail of Priority, Requires Asset Downtime, Storeroom, Storeroom Site

2H. Click on tab Frequency

2I. Click on Tab **Time Based Frequency**

2J. Define **Frequency** Value with 1

2K. Click  in **Frequency units** and select “**DAYS / MONTHS / WEEKS / YEARS**”

2L. Click  in **Estimated Next Due Date** and select current date

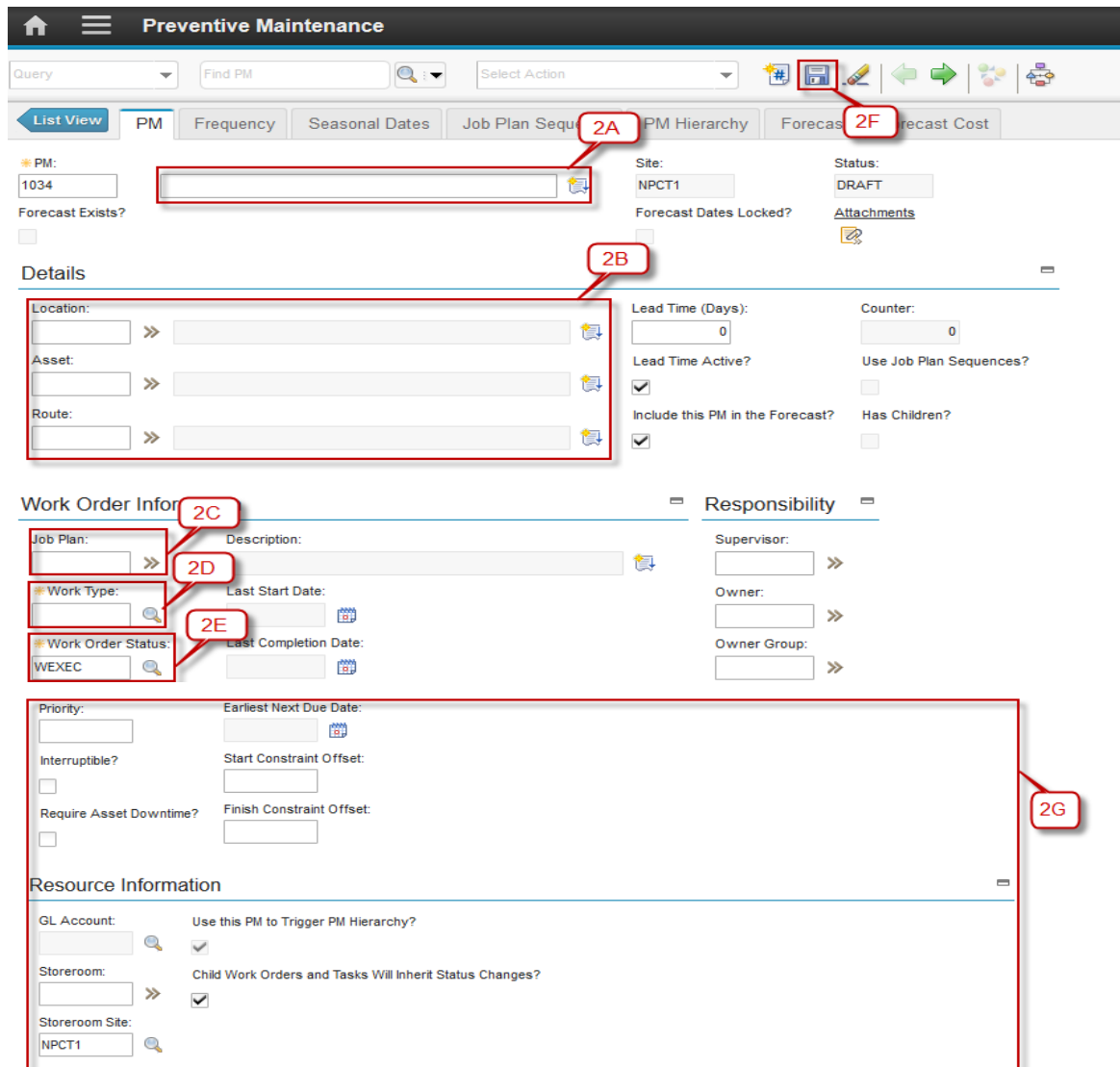
2M. Click  to Save PM

2N. Click  to Route Workflow.

2O. Click on **Submit PM**

2P. Input memo if needed

2Q. Click **OK**



Preventive Maintenance

Query: Find PM [] Select Action []

Buttons: List View, PM, Frequency, Seasonal Dates, Job Plan Sequences, PM Hierarchy, Forecast, Forecast Cost

PM: 1034 [] Site: NPCT1 Status: DRAFT

Forecast Exists? [] Forecast Dates Locked? [] Attachments []

Details

Location: [] >> []

Asset: [] >> []

Route: [] >> []

Lead Time (Days): [0] Counter: [0]

Lead Time Active? [x] Use Job Plan Sequences? []

Include this PM in the Forecast? [x] Has Children? []

Work Order Information

Job Plan: [] Description: []

Work Type: [] Last Start Date: []

Work Order Status: WEXEC Last Completion Date: []

Priority: [] Earliest Next Due Date: []

Interruptible? [] Start Constraint Offset: []

Require Asset Downtime? [] Finish Constraint Offset: []

Responsibility

Supervisor: [] >>

Owner: [] >>

Owner Group: [] >>

Resource Information

GL Account: [] Use this PM to Trigger PM Hierarchy? [x]

Storeroom: [] Child Work Orders and Tasks Will Inherit Status Changes? [x]

Storeroom Site: NPCT1

Callout boxes: 2A (Job Plan Sequences button), 2B (Details section header), 2C (Job Plan field), 2D (Job Plan dropdown), 2E (Work Type field), 2F (Forecast button), 2G (Resource Information section)

Agile – Proprietary/Confidential

Complete Workflow Assignment

Task:
Submit for PM 1039

Action:

☒ Submit PM 20

☐ Cancel

Memo: 2P

Earlier Memos [Filter](#) >     0 - 0 of 0 

Memo	Person	Transaction Date
There are no rows to display. 2Q		

3. Login as CHE Engineer. User name : che01, password : password.

3A. In CHE Engineer Dashboard in “Inbox / Assignment” click **Need Approval for PM**

3B. CHE Engineer will review detail of PM

3C. Click  (route workflow)

3D. Click **Approve**

3E. Input Memo if needed

3F. Click **OK**

Welcome, CHE Engineer 01

CHE Engineer 01

KPI List

Last Run: 2/3/16 8:05 PM [Update](#)

Status	KPI	Actual	Target	Variance
	Preventive Maintenance & Corrective Maintenance Ratio (%)	8.33	70	-61.67
	Year to date Labor Cost (in Million Rupiah)	1.47	90	-88.53
	Year to date Material Cost	0	100	-100
	Year to Date Service Cost (in Million Rupiah)	0	100	-100

Inbox / Assignments

Description	Last Memo	Start Date	Route
Next Assignment Due: 5/10/16 9:01 AM Refresh			
Need Approval for PRENG/10066/16/EMS (General)		2/15/16 2:07 PM	
Service Recovery 16000621 Needs to be Closed		2/23/16 9:04 PM	
Need Approval for PM 1038 (Sample PM 04 test)		5/10/16 8:54 AM	
Need Approval for PM 1039 (Sample PM 05 test)		5/10/16 9:01 AM	

1 - 4 of 4

List View
PM
Frequency
Seasonal Dates
Job Plan Sequence
PM Hierarchy
Forecast
Forecast Cost
3B

PM: 1039 Sample PM 05 test

Forecast Exists? ☐

Site: NPCT1

Forecast Dates Locked? ☐

Status: DRAFT

[Attachments](#)

Details

Location: >>

Asset: NP1QCC101 >> Quay Container Crane 101

Route: >>

Lead Time (Days): 0

Lead Time Active? ☒

Include this PM in the Forecast? ☒

Counter: 0

Use Job Plan Sequences? ☐

Has Children? ☐

Work Order Information

Job Plan: JP02 >>

Work Type: DI >>

Work Order Status: WEXEC >>

Description: Sample JP02

Last Start Date: >>

Last Completion Date: >>

Responsibility

Supervisor: >>

Owner: >>

Owner Group: >>

Home
Menu
Preventive Maintenance

Query
Find PM
Select Action

List View
PM
Frequency
Seasonal Dates
Job Plan Sequence
PM Hierarchy
Forecast
Forecast Cost
3C

PM: 1039 Sample PM 05 test

Forecast Exists? ☐

Site: NPCT1

Forecast Dates Locked? ☐

Status: DRAFT

[Attachments](#)

Complete Workflow Assignment

Task: Need Approval for PM 1039 (Sample PM 05 test)

Action: 3D ☒ Approve

☐ Revise / Cancel 3E

Memo:

Earlier Memos Filter >

Memo	Person	Transaction Date
There are no rows to display.		

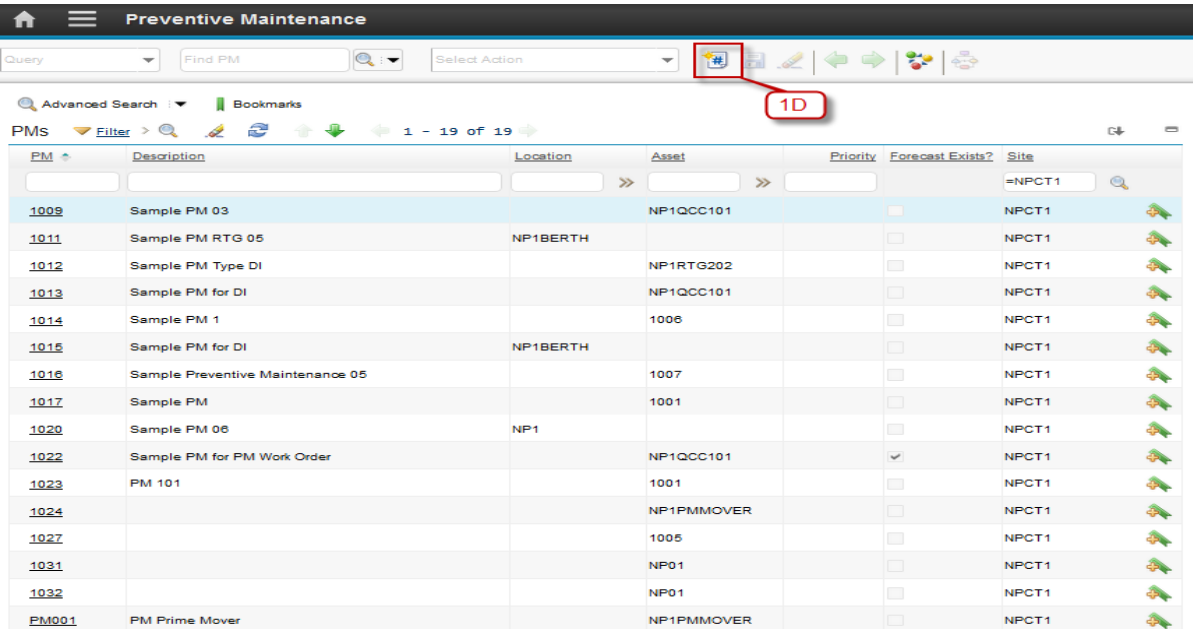
OK
Reassign
Cancel
3F

4. For Revised PM and Cancel PM please see the Revised and Cancel Job Plan, the step is same with Job Plan

5.2.2 Meter Based Preventive Maintenance

Below are step by step to create Meter Based Preventive Maintenance:

Step	Action
1. Login as Service Engineer. User name : se01, password: password.	
1A. Click in icon menu (☰)	
1B. Click Preventive Maintenance	
1C. Click Preventive Maintenance	
1D. Click icon New (✚)	

2A. PM Number is autonumber. Define description for Preventive Maintenance

2B. Define Location or Asset or Route

2C. Click >> in **Job Plan** field, and select one Job Plan used for this Preventive Maintenance

2D. Click  in **Work Type** field and select **DI (Daily Inspection)**

2E. Click  in **Work Order Status** field and select **WEXEC (Waiting Execute)**.

- If **Work Type** is **PM (Preventive Maintenance)** status in **Work Order Status** must be **APPR (Approve)**

2F. Click  to save Preventive Maintenance

2G. Input detail of Priority, Requires Asset Downtime, Storeroom, Storeroom Site

2H. Click on tab Frequency

2I. Click on Tab **Meter Based Frequency**

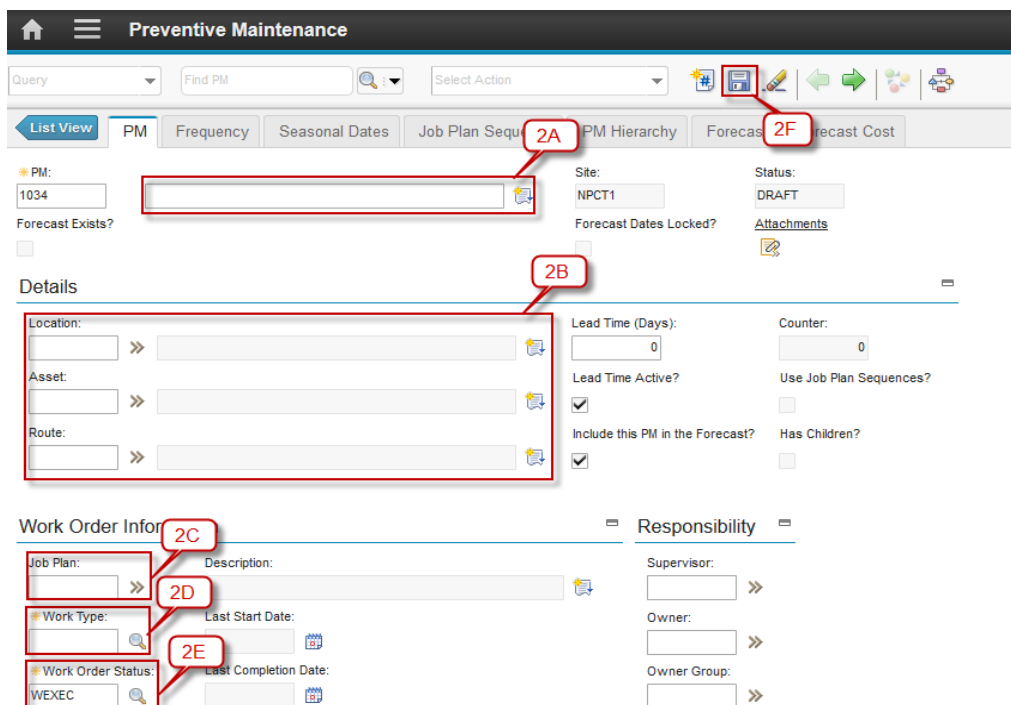
2J. Click **New Row**

2K. Click >> in **Meter** fields and choose one from available list

2L. Define **Frequency**

2M. If you want to add another meters repeat 3 step above from “New Row”

2N. Click  **Save**



Priority: Earliest Next Due Date: 

Interruptible? ☐ Start Constraint Offset:

Require Asset Downtime? ☐ Finish Constraint Offset:

Resource Information

GL Account: Use this PM to Trigger PM Hierarchy? ☒

Storeroom: Child Work Orders and Tasks Will Inherit Status Changes? ☒

Storeroom Site:

NPCT1

2G

  Preventive Maintenance

Query Find PM  Select Action

[List View](#) [PM](#) [Frequency](#) [Seasonal Dates](#) [Job Plan Sequence](#) [PM Hierarchy](#) [Forecast](#) [Forecast Cost](#)

PM: 1040 Sample PM 06 test Site: NPCT1 Status: DRAFT

Forecast Exists? ☐

Work Order Generation Information

Use Last Work Order's Start Date to Calculate Next Due Date? ☒ Generate Work Order Based on Meter Readings (Do Not Estimate)? ☐

Generate Work Order When Meter Frequency is Reached? ☐

Time Based Frequency [Meter Based Frequency](#)

Meter Based Frequency    0 - 0 of 0 

Meter	Description	Frequency	Units to Go	Generate WO Ahead By	Alert Lead
There are no rows to display.					

[New Row](#)

2H

2I

2J

Meter Based Frequency    1 - 1 of 1 

Meter	Description	Frequency	Units to Go	Generate WO Ahead By	Alert Lead

Details

Meter:  Average Units/Day:

Frequency: 0.00  Rollover:

Alert Lead:

Generate WO Ahead By:

Last Work Order Information

Meter Reading:

Meter Reading Date: 

Next Work Order Projections

Next Meter Reading:

Units to Go:

Estimated Next Due Date: 

[New Row](#)

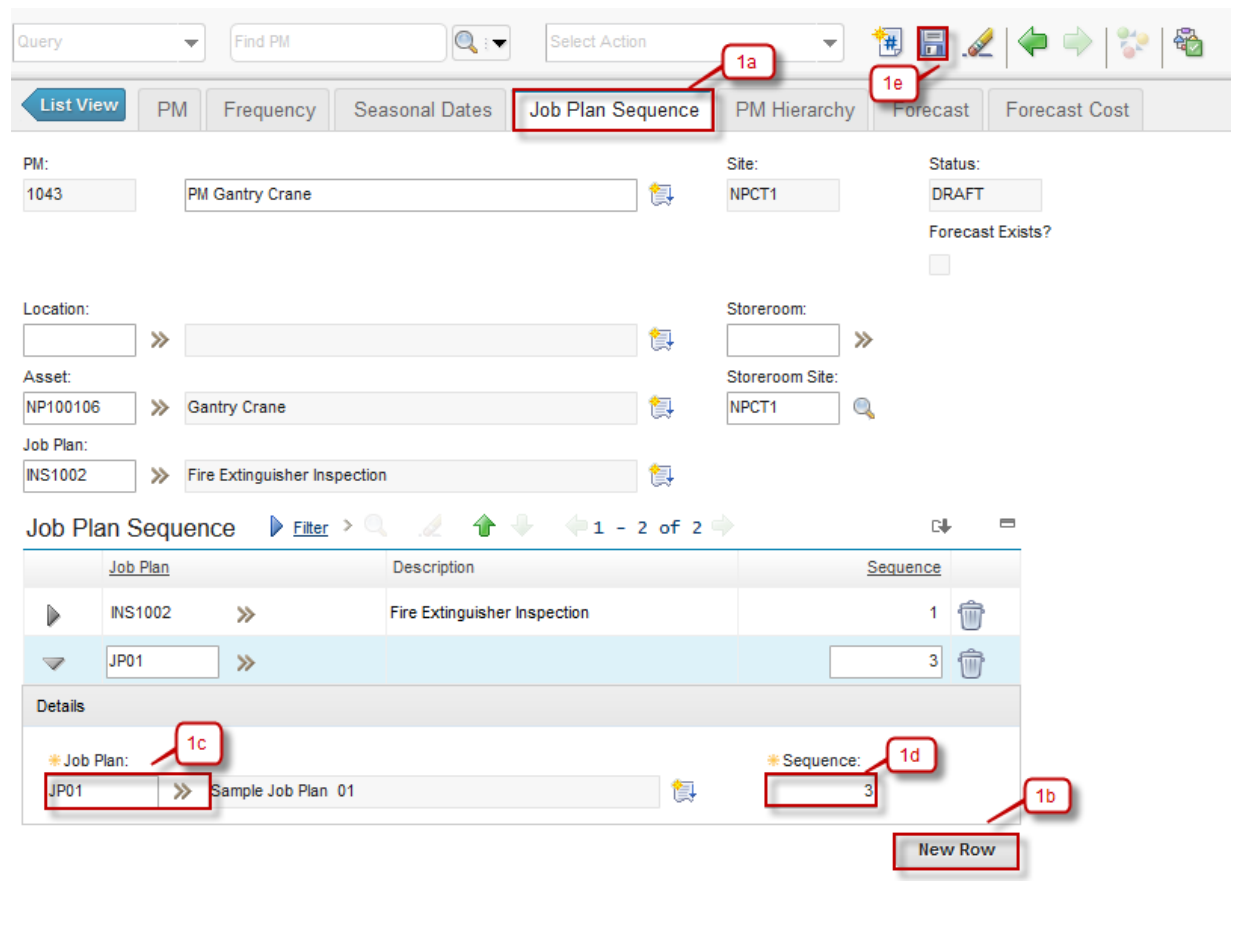
2K

2L

5.2.3 Multiple Job Plans for a Preventive Maintenance

Preventive Maintenance also have multiple Job Plans. Below are step by step to Multiple Job Plans for a Preventive Maintenance.

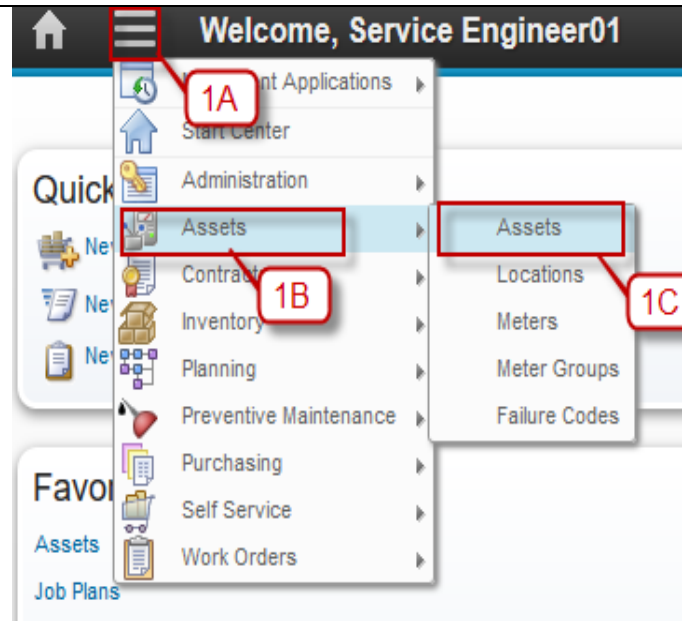
Step	Action
1a. Move to Tab Job Plan Sequence . Go to Job Plan Sequence Section on the page.	
1b. Click New Row	
1c. Select the required Job Plan using button »	
1d. Input Sequence of Job Plan. Sequence is interval the Job Plan is used.	
1e. Click Save .	



5.2.4 Forecasting – Meter Based

Below are step by step to create Forecasting Meter Based:

Step	Action
1A. Navigate to Go To Application	
1B. Navigate on Assets	
1C. Click on Assets menu	
1D. Search assets whose meter reading wants to be inputted. Then click Asset Number	
1E. Click Select Action with click on drilldown menu	
1F. Click on Enter Meter Reading menu	
1G. Input New Meter Reading Value	
1H. Input New Reading Date	
1I. Click OK	
1J. Navigate on Go To	
1K. Click on Preventive Maintenance	
1L. Click on Preventive maintenance	
1M. Search PM that wants to be processed. Click PM number	
1N. Click Select Action on drilldown icon	
1O. Choose Generate Forecast	
1P. Define forecast until or forecast for (days) field. (Defining 1 field will fill the other field automatically)	
1Q. Click OK	
1R. Click on Forecast Tab	
1S. Click on Calculation Cost	
1F. Click on Select Action on drilldown icon	
1U. Click Generate Work Orders	
1V. Define Generate Wos Due Today Plus This Number of Days	
1W. Click OK	



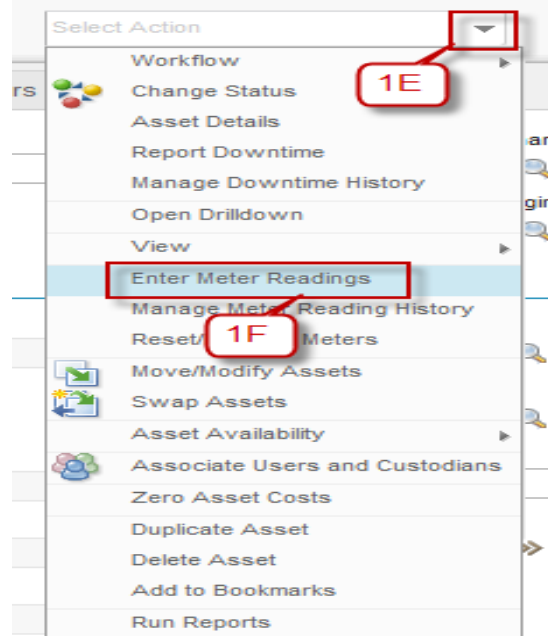
Assets Service Engineer01

Query Find Asset Select Action

Advanced Search Bookmarks

Assets Filter 1 - 20 of 76

Asset	Description	Location	Parent	Type	Rotating Item	Status	Site
NP1QCC101	Quay Container Crane 101	NP1WS		QCC		OPERATING	NPCT1
NP1QCC102	Quay Container Crane 102	NP1BERTH		QCC		OPERATING	NPCT1
NP1RTG201	Rubber Tire Gantry 201	NP1BERTH		RTG		OPERATING	NPCT1
NP1RTG202	RTG 202	NP1BERTH		RTG		PLANNING	NPCT1
NP1RTG203	Rubber Tire Gantry 203	NP1BERTH		RTG		OPERATING	NPCT1
NP1PMMOVER	Prime Mover	NP1BERTH		PRIMEOVER		OPERATING	NPCT1
NP1PMMOVER02	Prime Mover 02	NP1BERTH		PRIMEOVER		OPERATING	NPCT1
1010	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	PLANNING	NPCT1
1009	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1008	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1007	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1006	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1005	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1004	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1003	Rotating Item 01	NP1QUAY001			ITM0001	PLANNING	NPCT1

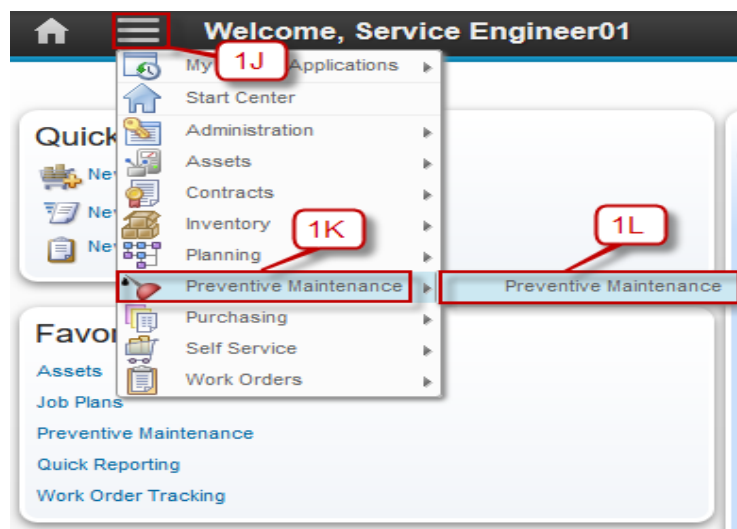


Enter Meter Readings

Meters Filter > 1G 1 - 10 1H

Meter	New Reading	New Reading Date	Rollover?	Inspector	Meter Type	Previous Reading	Previous Reading Date
RUNHOURS	10 (1G)	5/10/16 1:38 PM (1H)	☐	SE01	CONTINUOUS	8.00 (1I)	3/16 12:01 PM

OK Cancel



Preventive Maintenance

Query: Find PM Select Action

Advanced Search Bookmarks

PMS Filter 1 - 20 of 21

PM	Description	Location	Asset	Priority	Forecast Exists?	Site
1009	Sample PM 03		NP1QCC101			NPCT1
1011	Sample PM RTG 05	NP1BERTH				NPCT1
1012	Sample PM Type DI		NP1RTG202			NPCT1
1013	Sample PM for DI		NP1QCC101			NPCT1
1014	Sample PM 1		1006			NPCT1
1015	Sample PM for DI	NP1BERTH				NPCT1
1016	Sample Preventive Maintenance 05		1007			NPCT1
1017	Sample PM		1001			NPCT1
1020	Sample PM 06	NP1				NPCT1
1022	Sample PM for PM Work Order		NP1QCC101		☒	NPCT1
1023	PM 101		1001			NPCT1
1024			NP1PMMOVER			NPCT1
1027			1005			NPCT1
1031			NP01			NPCT1

Select Action

- Workflow
- View Sequence
- Generate Work Orders
- Set PM Counter
- Set Reading At Last WO
- Generate Forecast**
- Delete Forecast
- Clear Pending Reforecast
- Lock Forecast Dates
- Unlock Forecast Dates
- Duplicate PM
- Delete PM
- Add to Bookmarks
- Run Reports

1M

1N

1O

Generate Forecast

 Enter the period of time for which to forecast preventive maintenance records. The forecast can be run in the background. Forecasted dates will appear on the Forecast tab.

Last Forecast Date:

1P

 Forecast Until:



 Forecast For (Days):

Run Forecast Generation in the Background?

☐

Notification E-mail for Forecast Generation:

NOTE: If reforecast is pending, forecast dates will be adjusted appropriately when this action is executed.

1Q

OK

Cancel

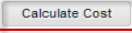
  **Preventive Maintenance** BMXAA4205I - Record has been saved.

Query Find PM  Select Action       

List View PM Frequency Seasonal Dates Job Plan Sequence PM Hierarchy Forecast **Forecast Cost**

PM: 1009 Sample PM 03 Site: NPCT1

Last Calculated Date: 5/10/16 11:34 AM Grand Total Cost: 200.00

 **Calculate Cost**

Forecast Details        1 - 1 of 1

Forecast Date	Job Plan	Nested Job Plan?	Route	Route Stop	Total Labor Hours	Total Labor Cost	Total Material Cost	Total Tool Cost	Total Service Cost	Total Cost
5/12/16	JP02	☐			0.00	0.00	200.00	0.00	0.00	200.00

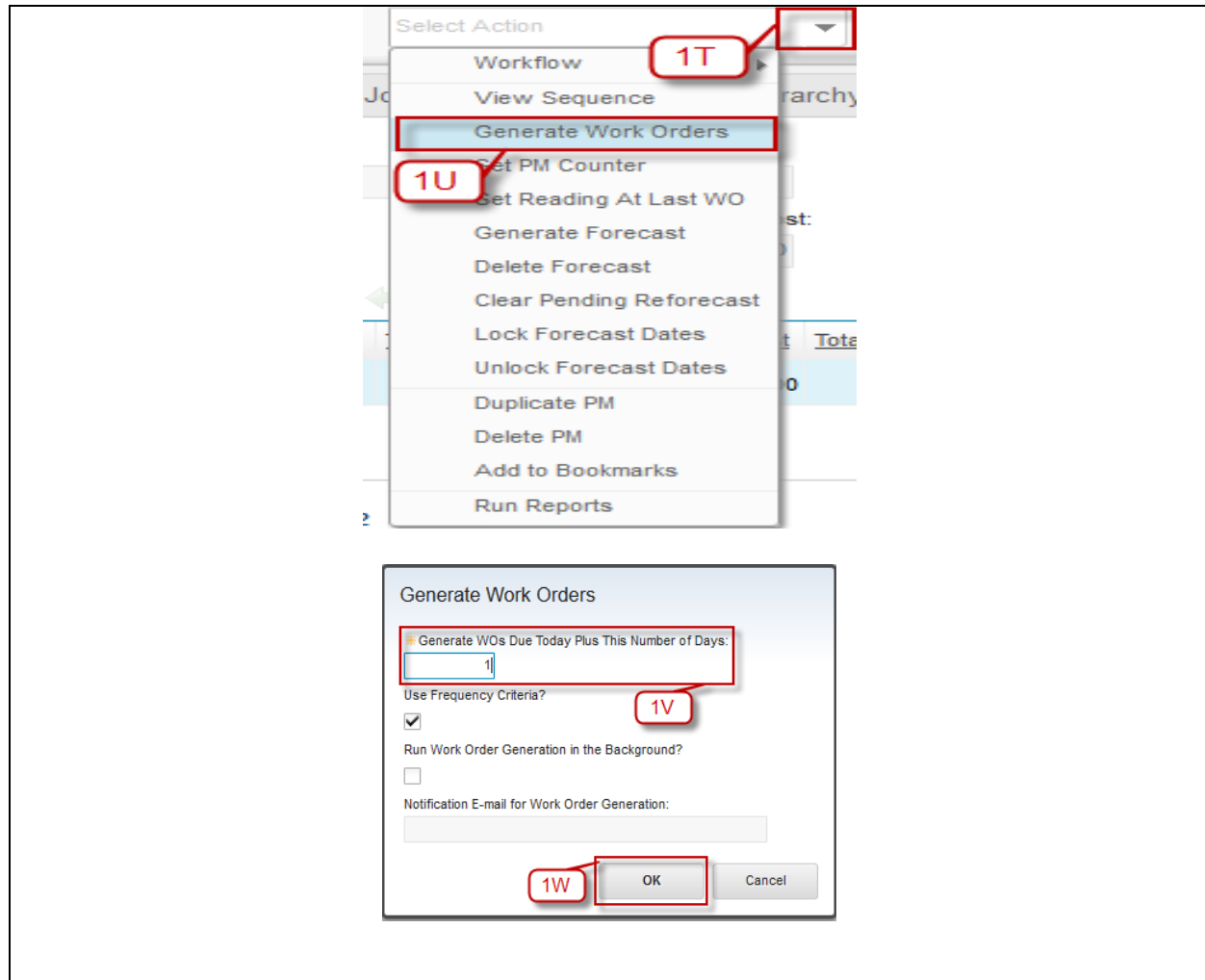
LABOR MATERIAL TOOL **SERVICE**

Service        1 - 2 of 2

Service Item	Quantity	Service Cost
SV001	1.00	0.00
SV001	1.00	0.00

1R

1S



5.2.5 Forecasting – Time Based

Below are step by step to create Forecasting Time Based:

Step	Action
1A. Click Go To	
1B. Navigate to Preventive Maintenance	
1C. Click on Preventive Maintenance	
1D. Search PM that wants to be processed. Click PM number	
1E. Click Select Action in drilldown icon	
1F. Choose Generate Forecast	
1G. Define Forecast until or Forecast For (Days) field. (Defining 1 field will fill the other field)	

automatically)

1H. Click **OK**

1I. Click **Forecast** Tab

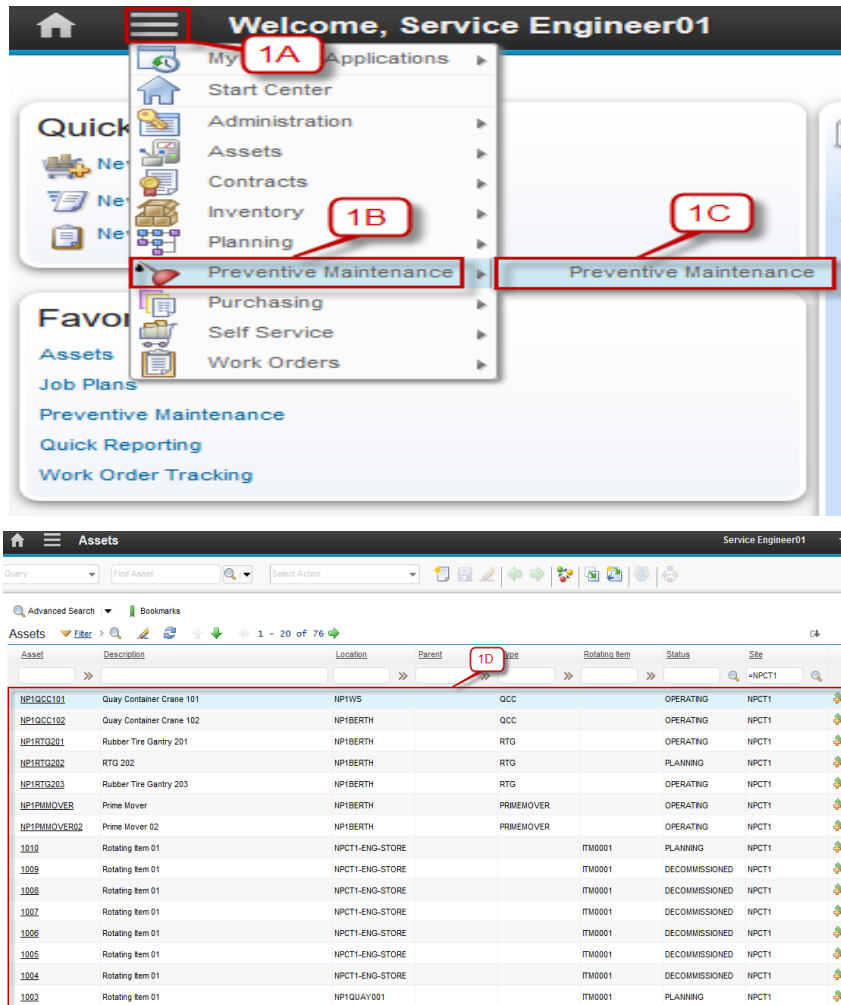
1J. Click **Calculate Cost** Button

1K. Click **Select Action** in drilldown menu

1L. Choose **Generate Work Orders**

1M. Define “Generate Wos Due Today Plus This Number of Days”

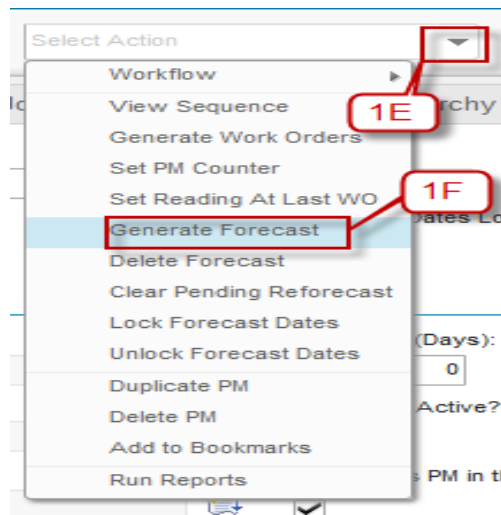
1N. Click **OK**



The screenshot displays the Agile Service Engineer01 interface. The top navigation bar includes a home icon, a menu icon, and the text "Welcome, Service Engineer01". A dropdown menu is open, showing options: My Applications, Start Center, Administration, Assets, Contracts, Inventory, Planning, Preventive Maintenance, Purchasing, Self Service, and Work Orders. Red callouts 1A, 1B, and 1C point to the menu icon, the Preventive Maintenance option in the dropdown, and the Preventive Maintenance link in the main content area, respectively.

Below the navigation bar, the "Assets" section is visible. It includes a search bar, a "Find Asset" button, and a "Select Action" dropdown. A table of assets is displayed, with columns: Asset, Description, Location, Parent, Type, Rotating Item, Status, and Site. Red callout 1D points to the "Type" column header.

Asset	Description	Location	Parent	Type	Rotating Item	Status	Site
NP1GCC101	Quay Container Crane 101	NP1WS		GCC		OPERATING	NPCT1
NP1GCC102	Quay Container Crane 102	NP1BERTH		GCC		OPERATING	NPCT1
NP1RTG201	Rubber Tire Gantry 201	NP1BERTH		RTG		OPERATING	NPCT1
NP1RTG202	RTG 202	NP1BERTH		RTG		PLANNING	NPCT1
NP1RTG203	Rubber Tire Gantry 203	NP1BERTH		RTG		OPERATING	NPCT1
NP1PMOVER	Prime Mover	NP1BERTH		PRIMEOVER		OPERATING	NPCT1
NP1PMOVER02	Prime Mover 02	NP1BERTH		PRIMEOVER		OPERATING	NPCT1
1010	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	PLANNING	NPCT1
1009	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1008	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1007	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1006	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1005	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1004	Rotating Item 01	NPCT1-ENG-STORE			ITM0001	DECOMMISSIONED	NPCT1
1003	Rotating Item 01	NP1QUAY001			ITM0001	PLANNING	NPCT1



Generate Forecast

 Enter the period of time for which to forecast preventive maintenance records. The forecast can be run in the background. Forecasted dates will appear on the Forecast tab.

Last Forecast Date:

Forecast Until:
 

Forecast For (Days):

Run Forecast Generation in the Background?
☐

Notification E-mail for Forecast Generation:

NOTE: If reforecast is pending, forecast dates will be adjusted appropriately when this is executed.

Preventive Maintenance BMXAA42051 - Record has been saved.

Query: Find PM Select Action

PM: 1009 Sample PM 03 Site: NPCT1

Last Calculated Date: 5/10/16 11:34 AM Grand Total Cost: 200.00

Forecast Details 1 - 1 of 1

Forecast Date	Job Plan	Nested Job Plan?	Route	Route Stop	Total Labor Hours	Total Labor Cost	Total Material Cost	Total Tool Cost	Total Service Cost	Total Cost
5/12/16	JP02	☐			0.00	0.00	200.00	0.00	0.00	200.00

Service 1 - 2 of 2

Service Item	Quantity	Service Cost
SV001	1.00	0.00
SV001	1.00	0.00

Select Action

- Workflow
- View Sequence
-
- Set PM Counter
- Set Reading At Last WO
- Generate Forecast
- Delete Forecast
- Clear Pending Reforecast
- Lock Forecast Dates
- Unlock Forecast Dates
- Duplicate PM
- Delete PM
- Add to Bookmarks
- Run Reports

Generate Work Orders

1

Use Frequency Criteria? ☒

Run Work Order Generation in the Background? ☐

Notification E-mail for Work Order Generation:

5.3 Routes Application

A route is a list of related work assets, which are considered "stops" along the route. These route stops represent assets or locations. Route simplifies building hierarchies of work orders for inspections. You can use a route in the following ways:

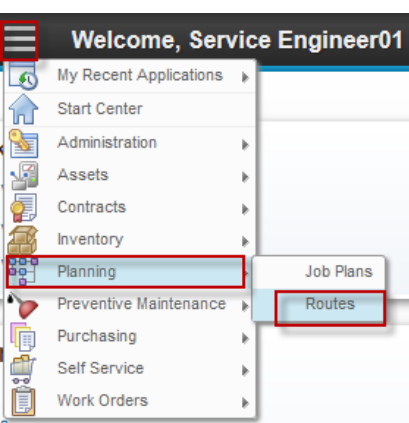
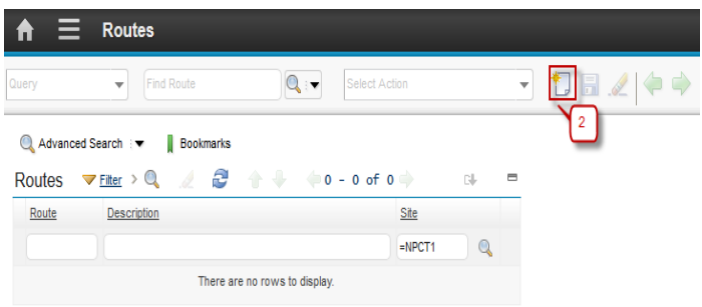
- Apply the route to a preventive maintenance record to generate work orders for all work assets listed as stops on the route.
- Apply the route to a work order and generate child work orders for each work asset listed as a stop on the route.

A route is a list of related asset or location work assets. You can create a route in the **Routes** application of the **Plans** module. You can create a route that lists all assets of the same type, or all assets in a certain location, or both. For example, you might create an inspection route that lists all fire extinguishers in a plant, or all safety equipment in a building.

When you create the route, you can specify a job plan for some or all route stops. Maximo copies these job plans to work orders generated for the route stop. If a safety plan is required when you use this job plan on the selected work asset, Maximo also copies that information to the work order.

5.3.1 Create Routes

Below are step by step to create Routes:

Step	Action
1. Go to Routes Application. Click button  (GoTo) – Planning - Routes	
2. Click  New Row	

3a. Input the Route Number.

3b. Input the Route Description.

3c. If you want the status of all child work orders to update simultaneously, select the **Route Stop Inherit Change Status?** checkbox. Otherwise, clear the checkbox.

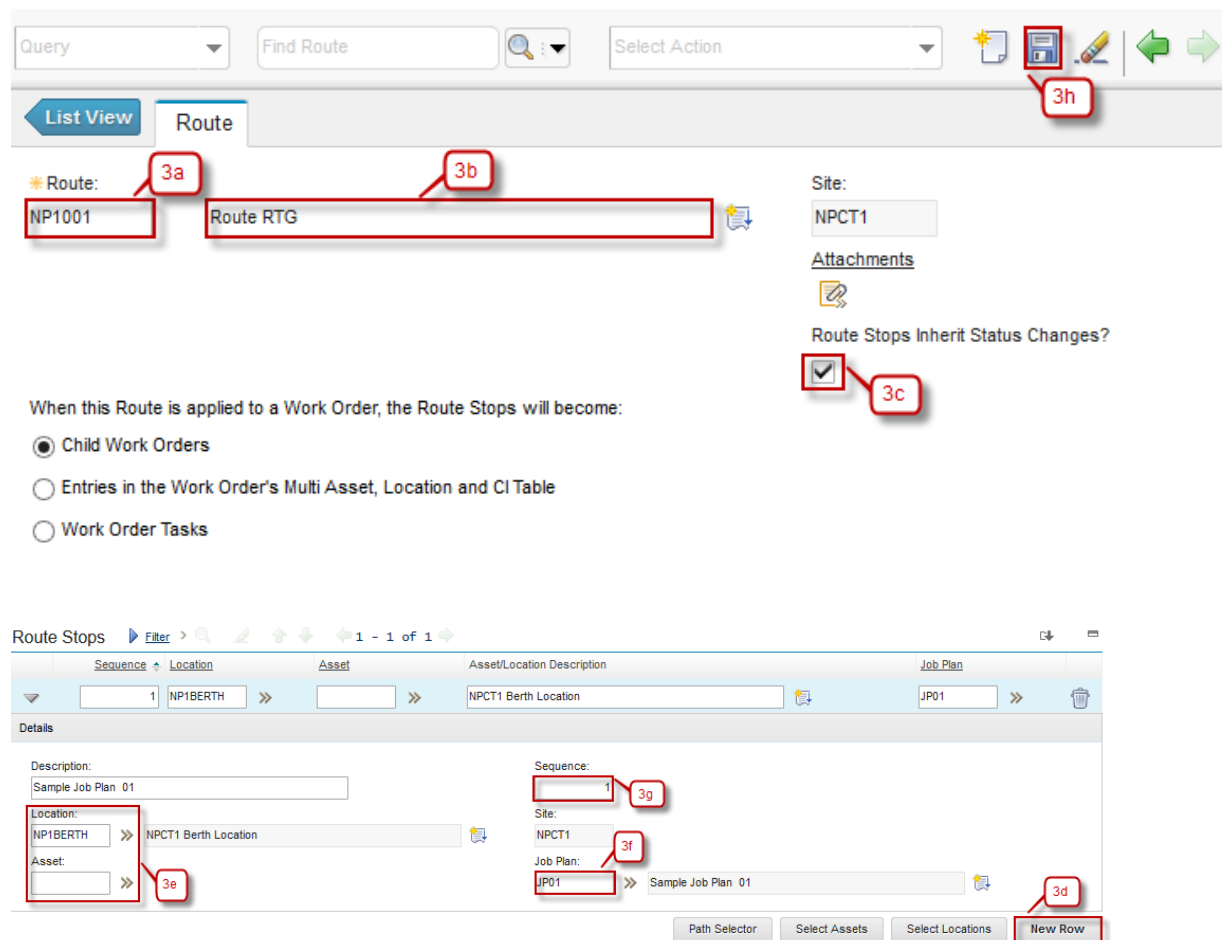
3d. At the bottom of the **Route Stops** table window, click **New Row**.

3e. In the **Location** or **Asset** field, enter a value or click » (details menu) and select an option.

3f. In the **Job Plan** field, enter a value or click » (details menu) and select an option.

3g. (Optional) In the **Sequence** field, enter a sequence number to correspond to each identifier, or route stop.

3h. Click **Save**.



The screenshot displays the Agile Route Management interface. At the top, there is a toolbar with a 'Query' dropdown, a 'Find Route' search bar, a 'Select Action' dropdown, and icons for adding, saving, editing, and navigating. Below the toolbar, the 'Route' tab is active, showing a form for creating or editing a route. The form includes fields for 'Route' (annotated 3a) with the value 'NP1001', 'Route Description' (annotated 3b) with the value 'Route RTG', and 'Site' with the value 'NPCT1'. There is an 'Attachments' section and a checkbox for 'Route Stops Inherit Status Changes?' (annotated 3c) which is checked. Below this, a section titled 'When this Route is applied to a Work Order, the Route Stops will become:' contains three radio button options: 'Child Work Orders' (selected), 'Entries in the Work Order's Multi Asset, Location and CI Table', and 'Work Order Tasks'. The bottom section is titled 'Route Stops' and shows a table with one row. The table has columns for 'Sequence', 'Location', 'Asset', 'Asset/Location Description', and 'Job Plan'. The first row has '1' in the Sequence column, 'NP1BERTH' in the Location column, and 'NPCT1 Berth Location' in the Asset/Location Description column. Below the table, there is a 'Details' section for the selected row. It includes fields for 'Description' (Sample Job Plan 01), 'Sequence' (annotated 3g) with the value '1', 'Site' (NPCT1), 'Location' (NP1BERTH, annotated 3e), 'Asset' (annotated 3e), 'Job Plan' (JP01, annotated 3f), and 'Sample Job Plan 01' (annotated 3d). At the bottom right of the details section, there is a 'New Row' button (annotated 3d) and buttons for 'Path Selector', 'Select Assets', and 'Select Locations'.